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Nasals and Nasalization in Spanish and Portuguese

Perception, phonetics and phonology

C. Elizabeth Goodin-Mayeda

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Nasals and Nasalization in Spanish and Portuguese

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Volume 9

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Perception, phonetics and phonology
by C. Elizabeth Goodin-Mayeda

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Perception, phonetics and phonology

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To my parents – all of them –,
without whose examples of persistence and bravery
I would not have found my own courage to succeed

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CHAPTER 1

Introduction

Many commonly attested phonological processes occur in contexts that involve nasals, including nasal assimilation, post nasal voicing and/or devoicing, and vowel nasalization, just to name a few. For reasons that will be explored in this book, nasality, whether expressed on consonants or vowels, has certain phonetic and phonological characteristics that cause outcomes seen time and time again in languages across the globe regardless of whether these share common ancestry.

One could approach the study of nasals and nasalization from a variety of phonetic and/or phonological perspectives, and indeed many linguists have. Almost since the founding of modern phonology in the early 20th century, debate over the relationship between phonetics and phonology has abounded. As the two fields became more autonomous in subsequent decades, the recurring question of how phonetics and phonology interact has persisted among phoneticians and phonologists. Some linguists (Beddor, 1983; Hayes, 1999; Ohala, 1990a, 1990b and others) have argued that phonological rules often have natural explanations in the physiological nature of speech production and perception. For example, phonological models often consider the crosslinguistic preference for consonants to be syllabified in onset rather than coda position as something universal. This is most likely rooted in the fact that consonantal cues for place of articulation (among others) are more easily perceived in the onset than in coda. This is also in part what makes codas so susceptible to weakening; because the cues are less perceptible, codas undergo more mutation, including deletion, than onsets. Thus, coda-licensing restrictions seen in many generative models of phonology reflect a perceptibility issue.

Other linguists oppose the notion of phonology being constrained in any way by phonetics (Buckley, 2000; Hale & Reiss, 1998; Kaye, 1989 and others). Hale and Reiss (1998, p. 6–7), for example, state that "the substance of phonological entities is never relevant to how they are treated by the computational system, except in *arbitrary, stipulative* ways" (their emphasis). Hyman (2001) takes a somewhat more centrist approach to address the question "how phonetic is phonology?" by proposing that phonology is the intersection of phonetics and grammar, which he schematizes as in Figure 1.

As such, the distinction between phonetics and phonology is phonology's relationship to the mental grammar. In Hyman's view, phonetics is only relevant to

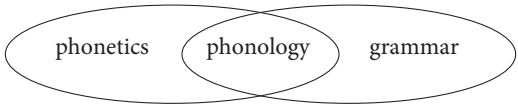

Figure 1. Hyman's (2001) conception of the relationship of phonetics and phonology

phonology insofar as it may condition changes that later become phonologized: "universal phonetics determines in large part what will become a language-specific phonetic property, which ultimately can be phonologized to become a structured, rule-governed part of the grammar" (Hyman, 2001, p. 153). Ohala (1990a), on the other hand, disagrees with the entire notion of an "interface" between phonetics and phonology, because this implies that these are two independent systems. "The term "interface" is inappropriate for phonology and phonetics because it deliberately under-represents the overlap in the domains covered by these disciplines as well as their interests, methods, and data" (Ohala, 1990a, p. 154–155).

Some theories of phonology integrate phonetic concepts such as articulatory efficiency and perceptual distinctiveness, most notable among these being Articulatory Phonology (Browman & Goldstein, 1986, 1992) and Optimality Theory (Prince & Smolensky, 1993). Articulatory Phonology conceives phonology as a set of gestural commands for the articulators in the vocal tract. In this model, the basic units of phonological contrast are not phonemes or features but gestures, which are abstract representations of articulatory events. As we will see in Section 3.1, this conception of phonology lends itself quite well to accounting for coarticulated speech.

Optimality Theory proposes that the grammar is made up of a universal set of violable constraints, and that the relative ranking of these constraints is what accounts for phonological differences among languages. For example, instead of a rule that states that nasals must assimilate to the place of articulation of an adjacent consonant, Optimality Theory might present a markedness constraint such as Agree (place), which states that adjacent consonants must have the same place of articulation. If this constraint is ranked higher than the faithfulness constraint mediating against changes to the input, the result will be nasal assimilation; if faithfulness is ranked higher than Agree (place), then nasals will not assimilate. Markedness constraints are grounded in phonetic notions such as articulatory efficiency, and there are even those who posit Optimality Theoretic grammars for perception (Boersma 1997, 1998, 1999 and others). The advantage of phonetically driven constraints is that this allows the phonology to not only describe and explain regularities in a particular grammar, but also to predict language typologies. By taking into account articulatory efficiency and perceptual distinctiveness,

Optimality Theory predicts what types of phenomena are expected – and perhaps more importantly not expected – to occur crosslinguistically.

The role of perception is not only important to synchronic accounts of phonology, but also to diachronic changes that occur in a language's sound system. Ohala (1981, 2012) claims that crosslinguistically common changes may begin as listener misperceptions, which later become the societal norm. For example, the development of distinctive nasal vowels is often the result of listeners erroneously considering nasalization as part of the speaker's intended pronunciation of the vowel instead of attributing it to a nasal consonant. This in turn can lead to the loss of the conditioning environment, namely the nasal consonant. When that happens, nasal vowels are subject to changes in height due to the acoustic consequences of nasalization in the spectral region associated with vowel height. According to Ohala, failure to perceptually factor out the spectral effects of nasal coupling has lead to vowel height changes in many languages. In French, for instance, high nasal vowels lowered sometime in the 16th century, e.g., ĩ > ẽ, ỹ > œ̃, ũ > õ (Ruhlen, 1979). Another shift is underway in Modern French whereby mid-front nasal vowels have lowered, as in ɛ̃ > æ̃ and œ̃ > æ̃, the latter of which is not complete but very advanced (Ruhlen, 1979).

Among synchronic and diachronic analyses of nasals and nasalization in Romance languages, French has always been afforded a position of distinction due in part to the fact that it comes from a well-known parent language (i.e., Latin) and to the existence of extensive, uninterrupted French documentation from the 9th century to the present (Ruhlen, 1979). Certainly, French has the longest history of diachronic variation related to nasals and nasalization, but Spanish and Portuguese offer insights into synchronic and diachronic phenomena related to nasals and nasalization as well. Both of these languages are widely spoken throughout the world, and as such, there is considerable dialectal variation between varieties of each, and yet they are so closely related that they are often mutually intelligible. As we will see in more detail in Chapter 4, some Spanish dialects tend to preserve features of coda nasals and have very little nasal overlap with preceding vowels, while other Spanish dialects show greater degrees of nasalization and possible final nasal deletion. Portuguese, on the other hand, has phonemic nasal vowels that are historically derived from VN sequences, e.g., *lã* 'wool' (from Latin *lanam*) vs. *lá* 'there'. Due to their particularly close relationship, Spanish and Portuguese can been seen as representing a continuum of nasal lenition and vowel nasalization, from the more conservative dialects of Spanish, to the more radical dialects of Spanish, to Portuguese. A similar argument has been made for Spanish, Galician and Portuguese with regard to nasal lenition and nasalization (Colina & Simonet, 2014).

The goal of this book is to examine nasals and nasalization in Spanish and Portuguese from the perspective of perception in order to provide insight on how perception can inform models of phonetics, phonology and language change. Specifically, the approach here will integrate phonetic and phonological models that often attempt to account for the same data but have developed largely independently of each other. A further research objective is to investigate the role of the listener's native language in perception of nasal place of articulation and nasal coarticulation with vowels, which will contribute to our understanding of synchronic and diachronic variation of nasals and nasalization in Spanish and Portuguese, and, perhaps more generally, in Romance languages.

This book is organized into six chapters, beginning with discussion of prior studies and models of perception and coarticulation. Chapter 2 presents previous studies in infant speech perception and models of perceptual development as well as studies in adult perception and models of L2 speech perception in order to frame the question of the role of language experience in perception. Chapter 3 focuses on the importance of coarticulation and phonetic context in speech perception, which is relevant to understanding the effects of nasal coarticulation with the vowel. In Chapter 4, the relevant language data for the varieties of Spanish and Portuguese under consideration will be discussed. Chapter 5 presents data for two perception experiments: Experiment 1 examines the role of native language experience and syllable position on the perception of nasal consonant place of articulation, and Experiment 2 investigates the role of native language experience and context on the perception of nasal vowel height. Finally, Chapter 6 discusses the results obtained with respect to the models presented in Chapters 2 and 3 and the implications for synchronic and diachronic shifts in Spanish and Portuguese.

CHAPTER 2

From citizens of the world to language specialists

Infant and adult speech perception

Adults are often said to perceive non-native speech input through the "filter" of their native language (Sapir, 1921). Indeed, a multitude of studies examine the challenges adults face when listening to a language that is not their L1. The pervasive difficulties adults encounter in perceiving non-native speech contrasts are made only more striking by the relative ease with which infants perceive such contrasts. Regardless of the language community in which an infant is born, for the first several months of her life she will discriminate contrasts that her parents cannot. For example, an infant born in Japan can discriminate English /r/ and /l/, a contrast with which Japanese-speaking adults notoriously struggle (Goto, 1971; Miyawaki et al., 1981). However, an infant's amazing perceptual acuity diminishes during the first year of life, as she becomes specialized in perceiving the contrasts of her native language (Aslin, Pisoni, & Jusczyk, 1983; Jusczyk, 1993, 1994; Kuhl, 1991, 1993; Werker & Tees, 1984a, 1884b).

2.1 Studies in infant speech perception

Infants face several hurdles when presented with the task of decoding speech. First, speech sounds vary across speakers due to differences in the size and shape of vocal tracts as well as across speech styles for any given speaker. In addition, articulatory gestures tend to interact; thus, the same phonetic segments have different acoustic patterns depending on their position within a word or utterance. Finally, speech is continuous: although pauses may occur at some syntactic boundaries, speech can continue for long intervals without interruption. Thus, language learners must identify meaningful segments (i.e. words or morphemes) within a continuous stream of speech. In spite of these hurdles, infants demonstrate extraordinary perceptual abilities as early as the first month of life.

Most studies on infant perception have focused on sensitivity to prosodic cues, segmental cues and phonotactic patterns. In each domain, infants become less responsive to non-native cues and patterns while in turn becoming more sensitive to native ones. Some of the earliest linguistic preferences that infants

demonstrate are those related to prosodic cues. Christophe, Dupoux, Bertoncini, and Mehler (1994), for example, showed that 3-day-old French infants were able to discriminate between sequences that contained a word boundary and those that did not based on differences in stress cues. In order to test such young infants, Christophe et al. (1994) and many other researchers have used variations of a High Amplitude Sucking (HAS) technique, originally developed by Siqueland and DeLucia (1969). In this technique, a baseline rate of high-amplitude, non-nutritive sucking is established. The nipple is connected to a positive pressure transducer that records changes in sucking rate and pressure. When new stimuli are presented, in this case new auditory stimuli, there is an increase in sucking as compared to the baseline rate. Eventually, the sucking rate will slow as the infant becomes habituated to the initial stimulus. At that point, a second stimulus is introduced, and an increase in the rate of sucking after the presentation of the second stimulus indicates that the infant has detected a change, which is taken as inferential evidence that the infants perceive the two stimuli as different.

To test infants' discrimination of words with different stress patterns, Christophe et al. (1994) extracted bisyllabic stimuli from natural speech samples in French. The stimuli were extracted from within words (e.g., **mati** in mathé**mati**cien) and between words (e.g., **mati** in panora**ma ty**pique). Because French has word-final stress, the between-word stimuli bore stress on the first syllable and the within-word stimuli bore no stress. The results of this study indicated that 3-day-old infants were able to distinguish between the within-word stimuli and the between-word stimuli. The authors propose that, since newborns seem to be sensitive to cues that correlate with word boundaries, such as stress, it is plausible that they might use this information during lexical acquisition.

Older infants seem to be sensitive to prosodic marking of major phrasal units, which may also help them begin to organize the speech signal (Jusczyk et al., 1992). Jusczyk et al. (1992) tested 9- and 6-month-old infants' preference for listening to speech with pauses at phrasal boundaries or within a phrase. To do this, they used a Head Turn Preference Procedure in which the infant sat on her mother's lap in the center of a test booth that had speakers on either side. After orienting the infant toward the front, a blinking red light over one of the speakers began to flash. When the infant turned to that side, speech from one of the conditions (pauses at phrasal boundaries or pauses within phrases) would begin playing. If the infant turned away, the speech playing from that speaker would stop. The authors found that 9-month-old infants listened longer to speech that was interrupted at phrasal boundaries than to speech that was interrupted within phrases. This pattern was not found for 6-month-olds, who showed no listening preference for either type of interruption. They conclude that discovering how phrases are marked may be one way that infants begin to organize speech input,

especially considering that this sensitivity emerges just before first words start to appear.

Jusczyk and his colleagues have also found that as infants get older, they begin to show preference for prosodic patterns present in their native language. Jusczyk, Cutler, and Redanz (1993) found that 9-month-old infants show preference for the stress patterns of their native language. This study found that American 9-month-olds, but not 6-month-olds, listened longer to words with the prevalent (strong/weak) stress pattern of English words. This was tested by low-pass filtering the words – that is, by reducing the amplitude of higher frequencies in the speech signal but not the low frequencies, which essentially results in removing the segmental information from the signal while leaving the rhythmic and intonational features intact. The 9-month-old group, but not the 6-month-old group, still demonstrated preference for the strong/weak stress pattern even after the words were low-pass filtered, indicating that they were responding to prosodic rather than lexical cues. The fact that American 9-month-old infants are sensitive to the predominant stress pattern of English suggests that they are in a position to use such information in segmenting words from the speech stream (Jusczyk, Cutler, & Redanz, 1993).

A multitude of studies on infant speech perception have focused on the variety of segmental contrasts that infants can discriminate. The speech cues that have received the most attention are those related to voicing distinctions, in particular Voice Onset Time (VOT). VOT studies have been a hotbed for debate on claims regarding the innateness of voicing distinctions and the notion of linguistic vs. non-linguistic modes of speech perception.

Lisker and Abramson (1964) investigated VOT in the production of initial stop consonants in eleven languages and found that productions across all languages fell into three relatively restricted ranges of VOT: long voicing lag in which voicing began about 75 msec after the release of the stop closure (e.g., English [p^h]), short voicing lag in which voicing started about 10 msec after the release of the stop (e.g., English [b]), and long voicing lead in which voicing preceded the stop release by about 100 msec (e.g., prevoiced consonants in Thai). Based on these results, Eimas et al. (1971) set out to examine whether infants were sensitive to the voicing contrasts in their native language. Using a non-nutritive sucking technique, they tested American 1- and 4-month-olds' ability to discriminate voiced and voiceless stop consonants /b/ and /p/. Synthetic stimuli /ba/ and /pa/ were synthesized with VOT values of –20, 0, +20, +40, +60, +80 msec (a negative number indicates that voicing begins before the release of the consonant). The pairs were chosen such that all pairs differed by only 20 msec, but some pairs constituted within-category pairs for native listeners of English (e.g., –20 msec [ba] vs. 0 msec [ba] or +60 msec [pa] vs. +80 msec [pa]), and other pairs represented between-category values (e.g., +20 msec [ba] vs. +40 msec [pa]). The results of

Eimas et al. (1971) showed that both the 1-month and 4-month-old English-learning infants discriminated the between-category pairs but neither group reliably discriminated the within-category pairs.

Eimas et al. (1971) drew three main conclusions from their results: (1) infants perceive VOT categorically; (2) underlying perceptual mechanisms are innate; and (3) that the findings suggest perception in a linguistic mode. To further examine the possibility that perceptual mechanisms such as those for perceiving VOT boundaries are innate, Eimas (1975b, 1985) tested 2- and 3-month-old infants from an English environment on their perception of the prevoiced-voiced distinction, which does not exist in English. Eimas reported that the infants were able to discriminate the prevoiced-voiced contrast; however, his results have been criticized for several reasons, among them the fact that the VOT pairs differed by 80 msec, a much larger difference than the 20 msec that was used to show the voiced-voiceless distinction (Jusczyk, 1981).

Lasky, Syrdal-Lasky, and Klein (1975) attempted to further examine the role of native language experience in the development of VOT perception. They tested Guatemalan 4- and 6-month-old infants on their perception of VOT pairs –60 msec/–20 msec, –20 msec/+20 msec, and +20/+60 msec. The first distinction was designed to span the Thai boundary as described by Eimas (1975b). The second pair was designed to span the (adult) Spanish boundary found by Abramson and Lisker (1970, 1972) and the third pair straddled the English boundary. Infants were tested by habituation and dishabituation measured by cardiac deceleration. The results indicated that infants were able to discriminate the –60 msec/–20 msec pair and the +20/+60 pair, but not the –20/+20 pair. These results are significant for two reasons: first, in this experiment, infants were shown to reliably discriminate the prevoiced-voiced distinction as well as the voiced-voiceless; second, infants in a monolingual Spanish-speaking environment discriminated VOT boundaries at different values than Spanish-speaking adults. Streeter (1976) found similar results in Kikuyu infants. Kikuyu is a Bantu language spoken in Kenya that has prevoiced and voiced stops (Streeter, 1976). Kikuyu infants were shown to discriminate the same categories as the Guatemalan infants in Lasky et al. (1975), in spite of the fact that the voiced-voiceless distinction does not occur in Kikuyu.

The claim by Eimas et al. (1971) that there is a linguistic mode for perceiving language as opposed to other auditory stimuli is probably the most contentious (Jusczyk, 1981). The notion of perception in a linguistic mode stems from the observation that infants seem to perceive voicing contrasts categorically and the assumption that categorical perception is unique to language. However, subsequent studies have shown categorical perception of non-speech sounds (Miller,

Wier, Pastore, Kelly, & Dooling, 1976 and others) as well as categorical perception of VOT by chinchillas (Kuhl & Miller, 1978).

Miller et al. (1976) attempted to ascertain whether categorical perception is a unique perceptual mode used by humans when listening to speech or whether it may be a general property of sensory behavior. Categorical perception is typically tested by corroborating identification and discrimination results such that the most easily discriminable pairs are those that are labeled differently in the ID task (/t/ vs. /d/, for example) while pairs that are labeled within the same category are more difficult to distinguish (two tokens that were both labeled /t/, for example). Subjects were tested on their perception of noise-buzz sequences using identification and discrimination tasks. The noise lead-time in the sequences ranged from –10 to +80 msec in 10 msec steps, and the noise and buzz terminated simultaneously in each stimulus. The results supported that the noise-buzz sequences were perceived categorically. Interestingly, the data for noise-buzz sequences appears to approach chance levels at the same rate and lead-time value as data from VOT studies. In other words, the category boundary for the noise-buzz sequences are close to the category boundary found for VOT. Miller et al. (1976) take this as support that the perception of noise-buzz sequences heard as nonspeech sounds can be similar to those heard as speech.

Kuhl and Miller (1978) address the same question of whether discrimination of VOT differences is accomplished by means of mechanisms specialized for the perception of speech in a different way. They tested chinchillas on discrimination of synthetic VOT pairs for bilabial, alveolar and velar consonants. Not only did they find that the chinchillas perceived VOT differences categorically, but the location of the VOT boundaries coincided with those found for humans. This has lead many to conclude that any underlying mechanisms for perceiving VOT discriminations likely reflect general constraints imposed upon the mammalian auditory system, and that human phonological systems take advantage of this (Pisoni, 1977).

Besides voicing contrasts, infants have been shown to be sensitive to other acoustic cues in discriminating segments based on place and manner of articulation (Eimas, 1974; Eimas & Miller, 1980; Moffit, 1971). Moffitt (1971) was the first to extensively study infant perception of distinctions in place of articulation. The principle acoustic cue signaling differences in place of articulation of stop consonants is formant transitions, especially second formant (*F*2) transitions. This study used synthesized stimuli that varied only in *F*2 values to test infants on their discrimination of the syllables [ba] and [ga] by measuring changes in heart rate. Based on these results, young infants are able to distinguish stop consonant place of articulation based on differences in *F*2 transitions.

Both Moffitt (1971) and Eimas et al. (1971) speculated that the discriminatory abilities that they observed in infants are present at birth, but since their subjects were at least one month old, the possibility that language experience played a role in their subjects' perceptual performance could not be entirely ruled out. For this reason, Trehub (1976) tested English-speaking infants that were 5–17 days old on foreign speech contrasts. Using foreign speech contrasts, Trehub could ensure that her subjects had no previous experience with the distinctions under investigation. The two contrasts examined were oral vowel vs. nasal vowel [pa]–[pã], which is phonemic in French and Polish, and [za] vs. [řa], which differ in stridency, an important contrast in Czech (Trehub, 1976). The infants in this study, who were from English-speaking environments, were able to discriminate both of these contrasts, which revealed that very young infants are able to discriminate contrasts along a range of acoustic cues, regardless of native language environment.

It is clear that infants' ability to perceive segmental contrasts that may or may not be phonemic in their language is remarkable; however, sometime between infancy and adulthood, this ability declines. In the first study of its kind, Werker, Gilbert, Humphrey, and Tees (1981) compared 7-month-old infants and adults on the ability to discriminate three contrasts: English /ba/–/da/, Hindi /ta/–/ʈa/ (voiceless, unaspirated, dental stop vs. retroflex stop, respectively) and Hindi /t^h/–/d^h/ (voiceless, aspirated, dental stop vs. the breathy voiced dental stop, respectively). Previous studies had been concerned with either infant perception or adult perception, but not both. For this reason, Werker et al. used similar procedures to test infants and adults on the same stimuli.

Four groups of subjects were tested: infants from English-speaking environments, English-speaking adults with no prior experience or training on the Hindi contrasts, another group of English speaking adults also with no prior training on the Hindi contrasts who were given training on the discrimination paradigm, and Hindi-speaking adults (the control group). A visually reinforced infant speech discrimination (VRISD) paradigm was used to test the infants. In this procedure, an infant is conditioned to turn her head away from an experimental assistant and toward a loudspeaker within a certain period of time after hearing a change in the auditory stimulus. In this study, correct head turns were reinforced with the illumination of an electronically activated toy inside a Plexiglas box, and false positives were not reinforced. Adults were tested with a modified version of this procedure: they pressed a button to signal a change in stimulus, and correct responses were reinforced with illumination of a smoked Plexiglas box. The results showed that for both Hindi contrasts, there was no statistical difference between the performance of the adult Hindi group and the infants on one hand, and the trained and naïve adult English groups on the other hand. Regardless of training,

the adult English groups did not attain the pre-established criterion on the Hindi pairs, while the 7-month-olds did.

After directly testing and finding discrepancies in the perceptual abilities of infants and adults tested on the same stimuli, Werker and Tees (1983) examined the perception of non-native speech contrasts across childhood in order to determine whether the decline of such abilities corresponded with puberty, as suggested by Lenneburg (1967), or earlier in childhood. 4-, 8- and 12-year-old English-listening children were tested on the same English and Hindi contrasts described above. Surprisingly, not only was there no decline in performance across the three age groups, but all three groups performed as poorly as English-listening adults. Not one of the 4-year-olds came within reach of the 8 out of 10 criterion on either Hindi contrast, in spite of the fact that they could all discriminate English /ba/–/da/, demonstrating that they were able to perform the task.

Given the results of their two previous studies, Werker and Tees (1984a, 1984b) sought to identify the timeline during which perceptual abilities in infants progressively decline. In order to determine if the results found in previous research were generalizable to other non-native contrasts, the first experiment tested infants and adults on a place of articulation contrast in Thompson, an Interior Salish (Native Indian) language spoken in south central British Columbia. The results of this echoed the results of their previous work: English-listening adults were unable to discriminate glottalized velar and glottalized uvular sounds /k̓i/–/q̓i/, while 6- to 7-month-old infants acquiring English and Salish-listening adults were able to do so. A second experiment set out to test infants of different ages on both the Salish /k̓i/–/q̓i/ and Hindi /ta/–/ʈa/ contrasts. They found that 6–8 month-old English-acquiring infants' performance did not significantly differ from that of infants in a Hindi-speaking environment with regards to perception of the Hindi contrast or from Salish infants on the Salish contrast. However, by 8–10 months, the infants exposed to English had depressed rates of differentiation of the non-English contrasts, and by the age of 10–12 months, they no longer made the non-English distinctions. Both cross-sectional and longitudinal data supported these findings. They concluded that "selective tuning of initial sensitivities in accordance with specific phonology … occurs at about the age that the child is beginning to understand and possibly produce sounds appropriate to his/her language" (Werker & Tees, 1984b, p. 62).

Given the timeline of perceptual development in monolingual infants, a logical question would be whether the timeline is the same for infants exposed to two languages. Sundara, Polka, and Molnar (2008) set out to examine just that. This study tested monolingual French, monolingual English and French-English bilingual infants at 6–8 months and 10–12 months of age on their perception of voiced alveolar versus dental stops ([d] vs. [d̪]). Coronal stops in English are

alveolar, while in French, they are dental. Results from previous research on adults showed that English listeners were able to discriminate between alveolar and dental /d/, perhaps due to the perceptual similarity between French /d/ and English /ð/ or their experience with dental-alveolar place differences in fricatives, while French listeners were not (Sundara & Polka, 2008). The results of the study with infants demonstrated that all infants in the 6- to 8-month-old groups discriminated French and English /d/. However, by 10- to 12-months-old, only the English monolingual and French-English bilingual groups discriminated dental and alveolar /d/; the French monolingual infants did not. Thus, the timeline for perceptual development in the French-English bilinguals was consistent with their monolingual peers. It should be noted that other studies have found differences between monolingual and bilingual phonological development (Bosch & Sebastián-Gallés, 2003a, 2003b; Sundara, Polka, & Genesee, 2006). However, unlike the Sundara et al. (2008) study, which compares two different phonemes in two different languages, these other studies involve contrasts that are phonemic in one language but not the other, meaning that one phoneme exists in both languages, which may result in statistical irregularities across the two languages (Bosch & Sebastián-Gallés, 2003a).

As older infants lose the ability to discriminate contrasts that are not phonemic in their L1, they also become more sensitive to phonetic and phonotactic patterns in their native language. Besides identifying the inventory of sounds that are used in the language, a child must also determine the organization of these sounds in her language (i.e. the phonotactics). For example, a child acquiring English must learn that while [h] is a permissible component of an English word, it cannot occur syllable finally, in contrast to aspirating dialects of Spanish (*estos* 'these' ['eh-t̪oh]). In a series of experiments by Jusczyk, Friederici, Wessels, Svenkerud, and Jusczyk (1993), Dutch- and English-acquiring infants were shown to prefer phonotactic patterns of their native language at 9 months of age, but not at 6 months. Infants listened to low-frequency, abstract words in Dutch and English. The words that were chosen in each language contained sequences that were impermissible in the other language. For example, unlike English, Dutch does not allow [d] in syllable-final position, and unlike Dutch, English does not permit sequences like [kn] or [zw] syllable initially. Dutch and American 9-month-olds, but not 6-month-olds, listened longer to the words from their native language. Furthermore, when the speech was low-pass filtered, there was no listening preference by the 9-month-olds, due to the similar prosodic characteristics of the two languages, indicating that it was indeed the phonetic, and not prosodic, properties of the speech to which the infants responded (Jusczyk, Friederici, Wessels et al., 1993).

Older infants are also sensitive to frequency effects of phonetic and phonotactic patterns. Jusczyk, Luce, and Charles-Luce (1994) tested sensitivity to phonotactic patterns *within* the child's native language by comparing English-acquiring infants' attention to non-words that had high probability phonotactic patterns for English (e.g., 'riss' [ɹɪs]) versus low-probability patterns (e.g., 'shawch' [ʃɔtʃ]). American 9-month-olds, but not 6-month-olds, listened significantly longer to words from the high probability list than to words from the low probability list, even when vowel quality was equated across the two lists. Thus, not only do older infants show strong preference for words that observe the phonotactic constraints of their language (Jusczyk, Friederici, Wessels et al., 1993), they are also sensitive to the distribution of these patterns in the input (Jusczyk et al., 1994).

2.2 Models of perceptual development

As we have seen, during the first year of life, and especially during the latter half of that year, infants' sensitivity to non-native speech contrasts is diminishing. Interestingly, this period is not defined merely by the loss of perceptual abilities that are unnecessary for the language they are acquiring, but also by the simultaneous addition of perceptual acuity in relevant contrasts, as their sensitivity to properties of their native language is becoming more finely tuned (Kuhl et al., 1992).

Early models of speech perception (e.g., Eimas, 1982; Liberman & Mattingly, 1985) claimed that the fact that infants distinguish speech contrasts from such an early age, that they tend to do so categorically, and that some contrasts, such as VOT, are crosslinguistically similar is evidence for innate perceptual mechanisms specialized for perceiving speech. The Motor Theory of speech perception (Liberman & Mattingly, 1985), for example, claimed that speech is perceived by a biologically specified system, or 'module,' which specializes in detecting the *intended gestures* of the speaker, and that these gestures are represented in the brain as invariant motor commands (Liberman & Mattingly, 1985). Accordingly, listeners do not hear the speech signal as ordinary sounds; rather, through a biologically based link between perception and production, they are able to "use the systematic, yet special, relation between signal and gesture to perceive the gesture" (Liberman & Mattingly, 1985, p. 6). Thus, the link between perception and production is innately specified, not learned through association.

One of the problems with this model is that it is not clear how listener misperception and sound change is accounted for if the listener directly perceives the intended gestures of the speaker. In other words, according to this model, speech perception should not be influenced by the effects of coarticulation. As we will see in the next chapter, coarticulation plays a significant role in perception – in

particular *mis*perception – which can ultimately lead to sound change (Ohala, 1981, 1990a, 1990b, 1993, 2012). In addition, as we have seen earlier in this section, there are some problems with the claim that speech perception is unique and different from the perception of other auditory stimuli, including the fact that categorical perception is not unique to humans (Kuhl & Miller, 1978) and evidence that humans have shown categorical perception of non-speech contrasts (Miller et al., 1976). For these reasons, later models, such as the Direct Realist Theory (Fowler, 1986), Work Recognition and Phonetic Structure Acquisition or WRAPSA (Jusczyk, 1993) and Native Language Magnet (NLM) (Kuhl, 1991, 1992) do away with the notion of language specific modes of perception.

Similar to the Motor Theory of speech perception, the Direct Realist view assumes that the perceptual primitives in speech perception are articulatory gestures (Best, 1995). However, the perceptual primitives in the Direct Realist view are the speaker's *actual* gestures, not intended gestures. This model draws a parallel between speech perception and other types of perception, such as vision and smell. In this way, acoustic information is seen as an informational medium (similar to reflected light in vision) that informs the perceiver about "distal events" (i.e., real-world events), in this case the moving vocal tract. Thus, just as reflected light allows a perceiver to ascertain physical properties of an object or event, the acoustic signal relates information about the speaker's articulatory gestures. Accordingly, there is no need for a perceptual mode especially for speech, because the speaker's gestural information is available via the acoustic structure, which is shaped by principles of acoustic physics. In addition, this gestural information is obtained directly due to the lawful physical relationship between the acoustic signal and the shape of the vocal tract. In other words, it is not obtained by perceiving acoustic cues and then "translating" them into the gestures that produced them.

The WRAPSA and NLM models focus on the role of language experience during early infancy and how this experience shapes an infant's perception. While these two theories differ in several key aspects, they do share some similarities. For example, both begin with the premise that the innate abilities to perceive auditory stimuli observed in infants are not due to mechanisms specific to speech, but to more general auditory and cognitive mechanisms. In addition, both models claim that infants' native language experience alters their perception of speech. The models differ, however, in how they account for the emergence of phonetic contrasts during the first year.

According to the WRAPSA Model, phonemic contrasts emerge not due to an infant's focus on the sounds themselves, but as one result of a larger goal, which is language acquisition.

> It is important not to lose sight of the fact that speech perception abilities develop within the context of language acquisition as a whole, and that the driving force that motivates the child is to establish a means of effectively communicating his or her wants, needs, thought and desires. Viewed from this perspective, it is unlikely that the child intentionally starts out with a goal of learning the set of phonemic contrasts that hold in his or her native language. Rather, those contrasts emerge as a consequence of success in pursuing various communicative goals, such as trying to learn new words. (Jusczyk, 1993, p. 4)

Thus, according to this model, infants are not analyzing phonetic segments, but bigger chunks, such as syllables or words, which are cued by prosodic and phonotactic regularities in their native language. In this way, WRAPSA attempts to account for the evolution of structural features of a language in a way that is efficient for dealing with online speech processing, and the beginning stages of word recognition involve exploiting prosodic cues to segment words. An additional component of this model is the development of a weighting scheme that discerns meaningful distinctions between words.

> At the preliminary level of analysis, then, the description that emerges is neutral with respect to the language that is spoken. This is the type of description that is attained by the infant during the first few months of life. However, once a language has been acquired, then the output of the auditory analyzers is weighted to give prominence to those features that are most critical to making meaningful distinctions between words in the language. The weighting scheme basically amounts to focusing attention on the critical features and ignoring the other ones. (Jusczyk, 1993, p. 6)

Thus the emphasis of WRAPSA is on developing a system that can be used in online speech processing, which will allow an infant to segment words, and distinguish the most important acoustic details for making lexical distinctions.

Kuhl (1993) criticizes this point, arguing that while linguistic contrast may be important for phonological development later, 6-month-olds are not developmentally in a position to use this information yet. Instead, Kuhl's NLM suggests that infants' earliest language-specific perceptual abilities are the result of attraction or assimilation. Kuhl has shown that infants demonstrate a "perceptual magnet effect" for phonetic categories. In a series of experiments, infants and adults were tested on perception of ideal members of phonetic categories (aka prototypes), both vocalic and consonantal. To give a concrete example, adult listeners were asked to rate the category goodness of many computer synthesized instances of the vowel /i/. The results showed that there was a "hotspot" in the vowel space in which vowels received better ratings. The authors then chose an exemplar from the hotspot, which they called the prototype, and an outlying exemplar that did

not get very good ratings, which they called the non-prototype. They synthesized 32 variants of each vowel: eight vowels that varied in formant frequencies for each of four rings orbiting the vowel, which were a controlled distance from the center vowel, either the prototype or the non-prototype. Adults, 6-month-old infants and rhesus macaques were tested on the discrimination of these pairs, and the results were that adults and infants, but not monkeys, perceived vowels close to the prototype as more similar than vowels equally similar close to the non-prototype. Kuhl takes this as evidence that "the prototype perceptually assimilates surrounding stimuli to a greater extent than is the case for the non-prototype" (Kuhl, 1993, p. 127), and calls the prototype a perceptual magnet. Interestingly, in spite of other studies that show similarities in categorical perception in humans and monkeys (Kuhl & Miller, 1978), rhesus macaques exhibited no magnet effect in Kuhl's studies. This suggests that the perceptual magnet effect and categorical perception are somehow different in significant respects (Kuhl, 1993).

NLM claims that based on their language experience, infants develop stored representations of speech by the age of six months, and that these representations of native language sounds make up the initial stages of language-specific speech perception. This is another major difference between WRAPSA and NLM. Whereas WRAPSA assumes that infants do not analyze phonetic segments but rather higher-level units, such as syllables, NLM holds that there is not one single unit of analysis. That is, the representations that infants form can be based on any or many levels of detail, from phonetic features to phonetic segments, prosodic cues for intonation and/or stress, etc.

2.3 Studies in adult speech perception

The way that adults perceive speech sounds has been said to depend on properties of their native language (Polivanov, 1931; Sapir, 1921 and others). A number of studies have demonstrated adults' difficulty in perceiving contrasts that are not phonemic in their first language (Best, 1991; Dupoux, Peperkamp & Sebastián-Gallés, 2001; Escudero & Williams, 2012, 2014; Peperkamp, 2005; Werker & Tees, 1984b and others). Such difficulties have been found in the perception of segments that are not phonemic in the L1, as well as with suprasegmental properties of language, such as contrastive stress.

There is no shortage of research demonstrating L2 learners' difficulty with consonantal contrasts that are not phonemic in their L1. The classic example is that of Japanese speakers mapping English /ɹ/ and /l/, which is not a native contrast in Japanese, onto a single Japanese phoneme /ɾ/ (Brown, 1998; Goto, 1971; Miyawaki et al., 1981; Yamada & Tohkura, 1992 and others). Brown (1998) found

that Japanese speakers could not reliably perceive the difference between American English (AE) /ɹ/ and /l/ in onsets or clusters. This study employed natural speech recordings in a discrimination task and a picture identification task. In the discrimination task, subjects heard pairs like 'rake' and 'lake' and judged whether the words were the same or different. Interestingly, although performance in onsets and clusters was below chance, Japanese speakers correctly distinguished between AE /ɹ/ and /l/ in coda position 99.3% of the time. In the picture identification task, the subjects heard a word such as 'lake' and made a forced choice between two pictures (in this example, a picture of a rake and another of a lake). Again, the Japanese speakers scored at or below chance when the /ɹ/–/l/ contrast was in onset position or in clusters, but performed substantially better when the phone was in coda position.

In a similar study, Yamada and Tohkura (1992) used both natural and synthesized speech to examine Japanese speakers' perception of AE /ɹ/ and /l/. They employed a variety of experiments consisting of identification tasks using natural and synthesized speech stimuli as well as an ABX discrimination task using synthesized stimuli. An ABX discrimination task involves hearing three stimuli, the first two of which are different (A and B) and deciding whether the last stimulus (X) sounds more like the first or second. It is thought that the ABX paradigm is more effective, compared to a simple AX paradigm, at preventing subjects from relying on acoustic memory traces. Yamada and Tohkura (1992) examined the existence of a /w/ category in the synthesized /ɹ/–/l/ continuum, the effect of stimulus range on perception of this contrast and the perceptual cue(s) involved in identifying /ɹ/, /l/ and /w/. The results showed considerable variability in Japanese speakers' perception of the /ɹ/–/l/ contrast, but generally speaking, the subjects did not perceive this contrast categorically. In addition, some Japanese listeners identified some of the synthetic stimuli in the /ɹ/–/l/ continuum as /w/. The results suggest that while American English speakers use *F*3 frequency as the main perceptual cue when distinguishing /r/ and /l/, Japanese speakers rely not only on *F*3 but also *F*2 frequency. Direct evidence for this can be found in Iverson et al. (2003), who mapped the underlying perceptual space of /ɹ/ and /l/ for German, Japanese and American adults. They found that Japanese speakers were more sensitive to *F*2 as an acoustic cue, which is irrelevant in English /ɹ/–/l/ categorization.

In a study on L2 perception of nasal place of articulation, L1 Spanish speakers showed difficulty in perceiving certain nasal consonants in final position in English (Marckwardt, 1946). Spanish has three nasal phonemes: /m/, /n/ and /ɲ/. Unlike English, the velar nasal [ŋ] is not a phoneme in Spanish and occurs in most dialects only as an allophone of /n/ before velar consonants, although some dialects (e.g., in Andalucia, Galicia and *tierras bajas* or 'low lands' regions) velarize final nasals syllable and/or word finally. /m/ and /ɲ/ are often said to be "defective

phonemes" in Spanish because, with the possible exception of loanwords, they do not occur in absolute final position.[1] In the case of loanwords, the orthographic final <m> is almost always pronounced as [n] (e.g., *álbum* [álβun] 'album').

Marckwardt tested Spanish speakers' perception of final [m], [n] and [ŋ] in English words. The subjects demonstrated an overall rate of error of 15.8%. The fewest mistakes were made with English final /m/, which showed errors favoring [n]. The most common errors were bidirectional confusion in misidentifying final [n] as [ŋ] and final [ŋ] as [n]. Marckwardt takes these results to suggest that the more complete distribution of a defective L1 phoneme (in this case, final [m]) is not as problematic for language learners as an L1 allophone that is phonemic in the L2 (in this case, English /ŋ/) (Marckwardt, 1946). However, Malécot (1956) found similar results for English speakers listening to natural speech tokens of CV and VC non-words. The bilabial nasal /m/ was the most perceptible whether it appeared syllable initially, syllable finally or as an isolated resonance, that is, only the nasal murmur. Accuracy of identification judgments in these contexts was best for /m/, followed by /n/, and lastly /ŋ/. Thus, it is possible that the perceptual patterns found for Spanish speakers are due to inherent acoustic properties of the nasals themselves, resulting in some PA features being more perceptible than others, and not to native language background *per se*. The role of native language experience on the perception of nasal place of articulation is examined further in Chapter 5.

Unlike consonantal contrasts, which tend to be perceived categorically, vowel contrasts tend to be perceived in a more continuous manner, with less sharp distinctions between labeling categories and better within category discrimination than what is normally seen with categorical perception (Liberman, Cooper, Shankweiler, & Studdert-Kennedy, 1967; Stevens, Liberman, Studdert-Kennedy, & Öhman, 1969; Vinegrad, 1972). It is thought that differences in perceiving consonants and vowels are due at least in part to the dynamic nature of vowels; unlike most consonants, vowels have robust acoustic signatures, a fact which results in many more cues available to the listener (Strange, 1987, 2011). In spite of this, difficulties in perceiving non-native vowels can be pervasive and ongoing.

Gottfried (1984) found significant L1 effects on the perception of French vowels by naïve English listeners as well as L1 English L2 learners of French. This study presented natural speech tokens produced by several Parisian speakers in identification and discrimination tasks. The results indicated that both naïve as well as L2 French listeners made perceptual errors more often than native French

1. Note that [m] does occur in absolute final position in some dialects, such as Yucatan Spanish. Also, [m] from /n/ occurs word finally due to place assimilation with a following bilabial consonant (e.g., /un pan/ [um ˈpan]).

speakers, although the non-French speakers fared worst of all. Not surprisingly, the English listeners had the most difficulty with French contrasts that do not occur in English, front rounded vowels [y]–[ø], and front and back rounded vowels [y]–[u].

Interestingly, L1 background does not always affect L2 perception in a predictable fashion. In a 1990 study, Bohn and Flege tested German, Spanish and Mandarin listeners on their ability to perceive the English contrasts [i]–[ɪ] and [ɛ]–[æ]. Both of these contrasts differ in terms of both vowel height and duration. The *F*1 of [i] and [ɛ] are lower than [ɪ] and [æ], respectively. In addition, [i] is longer than [ɪ] and [æ] is longer than [ɛ].

Bohn & Flege sought to discover which acoustic cues linguistically naïve listeners of German, Spanish and Mandarin would use to identify stimuli along [i]–[ɪ] and [ɛ]–[æ] continua. Since the German vowel system makes use of height differences as well as durational differences to distinguish vowel pairs, it was expected that they might use both of these cues to distinguish the English pairs. Spanish, on the other hand, does not use duration contrasts to distinguish vowels. Likewise, Mandarin does not use duration to signal segmental contrasts either; however, duration is a factor in distinguishing between rising and dipping tones. Therefore, while Spanish speakers were not hypothesized to be sensitive to durational differences, German speakers were, and the possibility was left open for Mandarin speakers.

Bohn & Flege found that Mandarin and Spanish listeners relied heavily on durational differences, in spite of the fact that Spanish does not make vocalic distinctions based on duration. This suggests that duration cues in vowel perception may be easy to access whether or not listeners use such cues in the L1 and thus, native language experience does not completely determine how non-native vowel contrasts are perceived (Bohn, 1995).

Studies of adult early bilinguals demonstrate that even early and extensive exposure to a second language does not guarantee fully native-like phonological competence (Flege, Yeni-Komshian, & Liu, 1999; Pallier, Bosch, & Sebastián-Gallés, 1997). For example, Pallier et al. (1997) examined the perception of the /e/–/ɛ/ contrast in bilingual listeners of Spanish and Catalan in Catalonia, which is a highly bilingual region in northeastern Spain. In kindergarten, Catalan is the dominant language and in public elementary schools, students begin to write Catalan before Spanish. In addition, most lectures at university are given in Catalan (Pallier et al., 1997). Thus, most young people are highly proficient, relatively balanced bilinguals.

While Spanish and Catalan are closely related languages, their phonological systems are quite different. Catalan has two contrasting phonemes /e/ and /ɛ/ (e.g., [te] 'take' vs. [tɛ] 'tea'), while Spanish has only one phoneme /e/, which is slightly

less tense than Catalan /e/. Pallier et al. (1997) examined the perception of the /e/–/ɛ/ contrast in forty listeners, half of whom had Spanish-speaking parents and the other half of whom had Catalan-speaking parents. A series of seven vowel stimuli were synthesized along a continuum from [e] to [ɛ], and subjects were tested using an identification task and an AX discrimination task. The Spanish group failed to consistently differentiate [e] and [ɛ] in the words *Pere* [ˈpeɾə] 'Peter' and *pera* [ˈpɛɾə] 'pear'. Some Spanish listeners performed more or less like Catalan listeners while others performed at chance levels. In the discrimination task, Catalan listeners, unlike Spanish listeners, showed a peak in their discrimination at the phonemic boundary. This suggests that while Spanish listeners are not totally unaware of the existence of two different vowels in Catalan, only those subjects that have been exposed to Catalan since birth have two phonemic categories for /e/ and /ɛ/ (Pallier et al., 1997). Pallier et al. conclude that even with very early and intense exposure to a second language, many learners do not converge on a native-like phonological system.

In another study on early bilinguals, Højen and Flege (2006) compared discrimination of English vowels by monolingual English listeners, native Spanish listeners who had learned L2 English as children (i.e. early learners) and monolingual Spanish listeners. The authors include both Spanish and English monolingual groups, aptly pointing out that "identical perception by groups of early learners and L2 native speakers should probably not be used as a litmus test for whether the speech perception system does or does not remain plastic following L1 perceptual attunement" (p. 3073). That is, in order to gauge L2 perceptual acquisition, a more appropriate measure would be a three-way comparison between early learners and monolinguals of both the L1 and the L2. Accordingly, differences between early learners and Spanish monolinguals, in this case, would show plasticity in perceptual learning, even if the early learners do not perform exactly like English monolinguals.

In this study, eight English vowels were placed in a C__C context to create non-words. Subjects were then tested in an AXB paradigm on four contrasts: one easy contrast [i]–[u] and three difficult contrasts [ɪ]–[eᴵ], [ɑ]–[ʌ], [ʊ]–[oᵘ]. Overall, early learners' scores were slightly lower than English monolinguals' scores and substantially higher than Spanish monolinguals'. The Spanish monolinguals' scores were significantly lower on all four contrasts while the English monolinguals' and early learners' scores did not differ significantly. Early learners scored lower than English monolinguals on only two difficult contrasts: [ɪ]–[eᴵ] and [ʊ]–[oᵘ]. The differences between early learners and English monolinguals on two out of three difficult contrasts shows that early learners' perceptual system is not identical to that of English monolinguals; however, the early learners have obviously shown extensive perceptual learning.

Beddor and Strange (1982) investigated the role of native language experience in perceiving the oral-nasal contrast in consonants and vowels. An interesting aspect of this experiment is that it tests the same phonetic feature, [+/–nasal], in both vowels and consonants. In this study, English listeners and Hindi listeners were given identification and discrimination tasks testing perception of the oral-nasal contrast in consonants ([ba] vs. [ma]), which is phonemic in both languages, and in vowels ([ba] vs. [bã]), which is phonemic only in Hindi. Performance on the consonantal continuum did not significantly differ between groups on either the identification or the discrimination task, although the average [b]–[m] boundary for Hindi listeners was slightly more toward the oral end of the spectrum than for English listeners. In addition, both groups perceived the oral-nasal contrast in consonants categorically. Their performance on the vowel series, in contrast, differed significantly. The Hindi listeners perceived the oral-nasal distinction in vowels as categorical, while the English listeners tended to perceive it as continuous, regardless of the other features of the vowels. Although some English listeners (20 of 27) consistently divided the stimuli into two categories, they required more velar port opening to label a vowel as nasal. Furthermore, the English listeners in general were more accurate than the Hindi speakers in discriminating differences of nasality in stimuli within the oral category. The results of this study suggest that perception can be categorical for certain phonetic distinctions, in this case the oral-nasal distinction, whether they distinguish consonants or vowels, and that the same vowel parameter can be perceived as categorical by one language group and continuous by another (Beddor & Strange, 1982).

Besides segmental distinctions, adults have difficulty perceiving suprasegmental and phonotactic aspects of a second language that differ from their native language. French speakers for example do not reliably perceive contrastive differences in stress, as in English noun-verb pairs like ˈ**pro**duce vs. proˈ**duce** (Dupoux, Pallier, Sebastián-Gallés, & Mehler, 1997; Dupoux, Parlato, Frota, Hirose, & Peperkamp, 2001). This is most likely because stress in French is very regular, always appearing word finally, and thus, need not be encoded in the mental lexicon (Peperkamp, 2004).

Speakers of non-tone languages have been shown to rely on different cues than speakers of tone languages in distinguishing tones (Gandour, 1983; Gandour & Harshman, 1978). Guion and Pederson (2006) found that while native listeners of Mandarin rely on both average *F*0 and *F*0 slope to distinguish contrasting tones, naïve English listeners, rely only on average *F*0. English listeners who are advanced learners of Mandarin, however, do use F0 slope information in perceiving tones. This shows that cues that are not used in the L1 can be recruited as a result of learning a second language (Guion & Pederson, 2006).

Phonotactic properties of languages have also been shown to influence L2 speech perception (Dupoux, Kakehi, Hirose, Pallier, & Mehler, 1999; Dupoux et al., 2011; Weber & Cutler, 2006 and others). Weber and Cutler (2006) found that listeners of English and German were affected by preceding context in detecting an English word embedded in a nonsense word. The results of a wordspotting experiment showed that both groups were facilitated when the phonotactic constraints of the preceding context forced a boundary at word onset in English. For example, the English word lecture was easier to recognize when listeners heard the word *moinlecture* as opposed to *gorklecture*, since both English and German syllable structure aligns with the word edge in the first example, but not in the second. Interestingly, German learners of English seem to be sensitive to the phonotactic constraints of their L1 as well as their L2. German listeners were able to detect words with highly likely boundaries in English (e.g., *thrarshlecture*), in spite of the fact that German allows clusters like /ʃl/. On the other hand, German listeners were equally facilitated when the word aligned with a German only boundary, as *moycelecture*, while English speakers were not, since /sl/ is a licit cluster in English but not German. Thus, while German listeners gained sensitivity to L2 phonotactic constraints, they did not seem to lose or suppress sensitivity to L1 phonotactic constraints in perceiving an L2.

Listeners have also been shown to perceive illusory vowels in clusters that are not permissible in their native language (Dupoux et al., 1999; Dupoux et al., 2011). In a cross-linguistic study involving French and Japanese speakers, Japanese speakers were shown to perceive an illusory [u] in nonce VCCV stimuli, such as *ebzo*. The subjects listened to a continuum of stimuli ranging from no vowel (*ebzo*) to a full vowel (*ebuzo*). Japanese speakers, but not French speakers, reported the presence of [u] between consonants even when there was none; that is, they reported hearing [u] in both *ebuzo* and *ebzo* as well as the iterations between. In addition, Japanese speakers were unable to discriminate between stimuli like 'ebzo' and 'ebuzo' in an aural matching task. Interestingly, unlike the French speakers, they were able to distinguish between pairs based on vowel length (e.g., *ebuzo* vs. *ebu:zo*), a feature that is contrastive in Japanese (compare *tokei* 'watch' to *tookei* 'statistics'). The length of the long vowel in these stimuli was exactly double the length of the short vowel. Similar results were obtained for Brazilian Portuguese listeners, who perceived an epenthetic [i] in illicit VCCV clusters (Dupoux et al., 2011).

2.4 Models of adult and L2 speech perception

Considering the decrease in perceptual abilities of infants during the first year of life and the poor performance of adults on identification and discrimination of non-native contrasts, the logical question of the nature of this decrease in perceptual abilities arises. That is, is the decline in the ability to perceive non-native contrasts the result of sensorineural loss? This is precisely the claim of some early research (Eimas, 1975a); Werker and Tees (1984a), however, provide evidence from adult perceptual data that the decline in the ability to perceive non-native contrasts may be the result of changes in processing strategies, as opposed to neural atrophy. In this study, English-speaking subjects were tested on their ability to discriminate two non-native place-of-articulation contrasts (Hindi retroflex vs. alveolar /ʈa/–/ta/ and Thompson glottalized velar vs. glottalized uvular /k'a/–/q'a/) using a category change task as well as AX discrimination tasks. The results were that English listeners could discriminate full syllable non-English contrasts when tested in an AX procedure, even without training or familiarization trials. However, when tested again, this time with a longer inter-stimulus interval (ISI) (1500 ms instead of 500 ms), subjects were unable to reliably make the same discriminations. This implies that memory traces are available at shorter inter-stimulus intervals but decay after longer intervals; thus, "adult listeners can discriminate sounds across non-native phonetic categories in some testing conditions, but are not able to use that ability in testing conditions which have demands similar to those required in natural language processing" (Werker & Tees, 1984a, p. 1866). Since the subjects accurately distinguished between non-native contrasts, the notion of neural atrophy is not supported. Subsequent studies showing changes in perceptual attenuation, such as those mentioned in the previous section (Guion & Pederson, 2006; Weber & Cutler, 2006), also support the idea of plasticity in adult language learning.

Developmental models like NLM and WRAPSA have some implications for L2 speech perception. Recall that NLM posits that via language experience, infants develop phonetic "prototypes," and that sounds that are acoustically close to prototypes are more difficult to differentiate as compared to those around non-prototype members of a category. In other words, the perceptual space of a phonetic category is distorted so that there is less perceptual space around a prototype. This model attempts to account for developmental changes in the perception of native and foreign sounds by infants, but it may also be applied to foreign language perception by adults. That is, the magnet effects of native language prototypes can cause certain foreign-language contrasts to be less discriminable, at least in the initial stages of acquisition, specifically for those contrasts in which two L2 sounds are in close proximity to an L1 prototype.

The WRAPSA model proposes that infants organize speech input by weighting features that are most critical for lexical acquisition in the L1 and that this process becomes automatic for native-language perception (Jusczyk, 1993). Extending these claims to adult perception, the model predicts that in listening to L2 contrasts, adults will initially weight cues according to their native language, which in turn may lead to decreased perceptual acuity, due to the fact that selective attentional processes to detect and integrate phonologically relevant information are attuned to the L1.

Two of the most influential models of adult L2 speech perception are Best's Perceptual Assimilation Model (PAM) (Best, 1990, 1991; Best et al., 1988) and Flege's Speech Learning Model (SLM) (Flege, 1995). According to PAM, linguistically naïve adults perceptually assimilate L2 phonemes to L1 phoneme categories based on their perceived phonetic similarity. There are four possible assimilation patterns: (1) Two Category assimilation in which two L2 phonemes are assimilated to two different L1 categories; (2) Single Category assimilation in which two L2 phonemes are assimilated to a single L1 category; (3) Category Goodness assimilation in which two L2 phonemes are assimilated to a single L1 category but with differential goodness ratings such that one is a better fit to the L1 category than the other; and (4) Non-assimilation, which happens when L2 sounds are perceived as so different from L1 sounds that they are not able to be assimilated as speech sounds. Best claims that the easiest L2 distinctions are Two Category assimilations, while the most difficult contrasts are Single Category assimilations. Category Goodness assimilations pose intermediate levels of difficulty, and will depend on how good or bad the L2 phonemes fit to the L1 category. For example, if one L2 phoneme is a relatively good fit and the other is a bad fit to the L1 category, the contrast will be relatively easy, however if both L2 phonemes are bad fits, then the contrast will be almost as difficult as a Single Category assimilation. Non-assimilations are not perceived as speech, so they will be easy or difficult depending on the auditory salience of the acoustic differences. An example of Non-assimilation in L2 perception is the perception of Zulu clicks by American English listeners. Best et al. (1988) tested AE listeners' discrimination of 18 minimal pair contrasts of nine non-nasalized Zulu clicks of varying place of articulation and voicing. In spite of having no previous experience or training with the clicks, the AE listeners discriminated all contrasts, showing 85%–95% accuracy in all pairings except one. On a posttest questionnaire, subjects indicated that they had perceived the Zulu clicks as non-speech sounds such as "tongue clicking" or "cork popping."

Unlike Best's PAM, Flege's SLM is primarily concerned with ultimate attainment of L2 pronunciation, and therefore, focuses on L2 learners who have spoken their L2 for many years, not beginners. In addition, while both models draw on the notion of similarity between L1 and L2 phonetic categories, PAM is based on

perceived similarity while SLM phonemes are categorized as Old, New or Similar based on their acoustic distance from L1 phonetic categories. Old phones are those that are identical or almost identical to L1 categories and pose little or no difficulty to the L2 learner both in terms of perception and production. New phones are different from all L1 categories, and are initially difficult to perceive and produce; however, with experience, learners create new equivalence classes for these sounds, which results in accurate perception and production. Similar phones are assimilated to L1 categories, which is helpful in initial stages of acquisition, but since there is a mismatch between the L1 and L2 phonetic boundaries, perceptual errors and accented pronunciations may persist for these (Flege, 1995; Strange, 1992).

More recently, the notion of speech perception has been incorporated into phonological models like Optimality Theory, and some authors (Boersma 1997, 1999 and others) even propose separate OT grammars for perception. Optimality Theory (Prince & Smolensky, 1993) was first conceived as a phonological grammar made up of violable constraints that are ranked relative to each other. Two types of constraints, Faithfulness and Markedness, mediate against deviations from the input and marked forms in the output, respectively. For example, the faithfulness constraint MAX-IO states that an element that is present in the input must have a corresponding element in the output, in other words, no deletion. MAX-IO might play out differently with a markedness constraint like *CODA (i.e., no coda consonants) depending on how these two constraints are ranked relative to each other. A language that ranks MAX-IO higher than *CODA, will have coda consonants while a language with the opposite ranking will not. The tableaux in (1) and (2) illustrate how these two grammars might evaluate a form like /tap/.

(1) Realization of coda consonants

/tap/	MAX IO	*CODA
☞ a. tap		*
b. ta	*!	

(2) Coda deletion

/tap/	*CODA	MAX IO
a. tap	*!	
☞ b. ta		*

In the grammar exemplified in (1), candidate a is optimal, because it satisfies the higher ranked MAX IO, in spite of its violation of *CODA. In (2), b is the optimal candidate because *CODA is now ranked higher than MAX IO.

Based on the observation that perception does not always mirror but often precedes production in both children acquiring their first language and adults acquiring a second language, Boersma (1997, 1999) proposes a model of an OT perception grammar. Boersma makes a distinction between acoustic representations, which are rich in that they include many acoustic details about a segment, and perceptual representations, which are economical constructions that represent the rich acoustical input. An example of how an OT perceptual grammar might work is exemplified by the constraints and the tableau in (3).

(3) *Categ (vowel height: *x*): do not categorize into the vowel height class of *x* Hz (Boersma, 1997, p. 4)
*Warp (*F*1: *x*; vowel height: *y*): do not categorize an acoustic *F*1 of *x* Hz into a vowel height class of *y* Hz (Boersma, 1997, p. 4)

[530]	*Categ (other)	*Warp ([530], /740/)	*Warp ([530], /470/)	*Categ (/260/) *Categ (/470/) *Categ (/740/)
☞ a. /470/			*	*
b. /530/	*!			
c. /740/		*!		*

The tableau in (3) illustrates how a listener perceives a vowel with a first formant at 530 Hz. Assuming the categories 260 Hz (high vowel), 470 Hz (mid vowel), and 740 Hz (low vowel), a listener who hears a first formant value of 530 Hz will likely classify the vowel into the 470 Hz category. Candidate a is preferable to candidate c, because the input at 530 Hz is closer to 470 Hz than to 740 Hz, which is why the *Warp constraints are ranked as they are. The reason that candidate b fails is that the listener would essentially be creating a new category for an input of 530 Hz, instead of classifying into the closest category that already exists in the grammar. This possibility is prevented by the highest ranked constraint *Categ (other), which does not permit the creation of another category. Thus, here the acoustic representation is 530 Hz while the (economic) phonetic representation is 470 Hz.

This rigorous division between perception and production grammars follows from Boersma's (1998) Functional Phonology, which separates the roles of articulation and audition of speech sounds. This model attempts to bridge the gap between "phonetic" and "phonological" explanations of essentially the same data by formalizing the mechanisms involved in both perception and articulation as constraint-based interactions within a formal grammar.

Some interesting recent developments and future directions for research in adult perception include the effects of musical training on speech perception and neurolinguistic studies that use fMRI and other neuroimaging techniques to study speech perception. For example, American musicians studying at a university conservatory have been shown to perform significantly better than non-musicians at identifying tones in Mandarin (Gottfried, 2007). Neuroimaging studies have shown hemispheric differences in processing native and non-native language speech whereby native listeners of a language show significant left hemispheric activation while non-native listeners show participation of both hemispheres, with less overall activation (Sereno & Wang, 2007; Perani et al., 1998). Future studies in these areas could provide insight into notions of a critical/sensitive period, cortical changes resulting from language experience, and the effects of musical, perceptual and other types of training on language learning.

This chapter has focused on empirical and theoretical research related to the development of speech perception in infancy and adulthood. In the next chapter, we will see how these issues relate to the perception of coarticulated speech. Empirical analyses such as the ones we have seen in Chapter 2 provide insight into which acoustic cues listeners use to perceive certain speech sounds and the effects of cue timing, while at the same time informing experimental methodologies. Theoretical models of perceptual development account for how a listener processes acoustic information to create phonemic inventories. Studies in adult L2 perception demonstrate how listeners deal with unfamiliar speech, which may, in turn, inform models of coarticulation and sound change in the sense that losing the conditioning environment for coarticulation, which is hypothesized to be one cause of language change (Ohala, 1981, 1986, 1993, 2012), is akin to decoding unfamiliar speech in an L2.

CHAPTER 3

Coarticulation and nasalization

One of the most formidable challenges in speech recognition, whether by humans or machines, is dealing with variability in the speech code. Different instances of the "same" sound can contain enormous variation, yet listeners are for the most part unconscious of this variability. Some of this variability stems from differences in speakers, but even sounds produced by the same speaker may demonstrate significant acoustic dissimilarity due to the phonetic context in which the sound is uttered.

3.1 Context and coarticulation

One of the ways in which context plays a role in how speech sounds are perceived is in the perception of assimilated phones. An example of this is listener's perception of nasal consonants that have undergone nasal place assimilation. In many languages, the place features of final /n/ may become homorganic with the place features of the following consonant. For example, Dutch *tuin* 'garden' may be pronounced *tui*[m] if followed by a bilabial consonant as in *tiu*[m] *bank* 'garden bench.' Dutch and German speakers have been shown to overlook changes in place of articulation of /n/ in contexts that allow for assimilation of a word-final nasal, regardless of unfamiliarity with lexical items (Mitterer & Blomert, 2003). Dutch and German listeners were asked to indicate whether the /n/ in Dutch *tuin* 'garden' was pronounced canonically (i.e. as [n]) or with a change in place of articulation (in this case, as [m]). Listeners heard *tuin* and *tuim* in three conditions: in isolation, followed by *bank* 'bench' (i.e. a context that allowed for final nasal assimilation) and followed by *stoel* 'chair' (i.e. a context that did not allow for final nasal assimilation). Both Dutch and German listeners performed at near ceiling levels in the isolation and *stoel*-context conditions; however, in the *bank*-context condition, they showed a significant tendency to (mis)perceive *tuimbank* as *tuinbank*. What is interesting is that the German listeners in this study did not know Dutch; hence, it is unlikely that the results obtained are attributable to lexical inference (Mitterer & Blomert, 2003).

Even if the features of a segment do not assimilate to those of an adjacent segment, the context in which it is uttered can affect the acoustic signal, which

in turn affects perception. Such is the case in coarticulation, which is often anticipatory, although not necessarily so. The difference between assimilation and coarticulation is that assimilated segments are actually articulated with different features (different place or manner of articulation, for example), while coarticulation is the result of some feature of an adjacent segment overlapping with another (Ohala, 1981, 1986). For example, there is a tendency for [u] to have a raised *F*2 when between dental or alveolar consonants, making it more like the vowels [ʊ], [ɨ] or [y]. Here the shift in *F*2 is due to the apical constriction coarticulated with the [u], not actual changes in the position of the tongue body. That is, the tip of the tongue is slightly raised due to the adjacent coronal consonants, but the tongue body is high and back as for any other [u]. However, the interruption of the airflow by the tongue apex raises the *F*2, which in turn affects the perception of the vowel's backness in this context.

There is experimental evidence that listeners are able to disregard coarticulatory vowel fronting in the appropriate contexts. Ohala, Riordan, and Kawasaki (1978) tested listeners on perception of a synthetic continuum from /i/ to /u/ in the consonantal contexts /f_p/ and /s_t/. Subjects reported more /u/ responses in the /s_t/ condition than in the /f_p/ condition. Thus, in the /s_t/ context, speakers attributed some of the vowels' frontness to their position between two alveolar consonants (Ohala, 1986).

Listeners seem equally able to perceptually compensate for the effects of coarticulation on consonants as well, as evidenced by the perception of [s] and [ʃ] in certain contexts (Mann & Repp, 1980). One of the main acoustic differences between /s/ and /ʃ/ is that the friction noise of /ʃ/ has a lower center of gravity than that of /s/. However, when /s/ appears before a rounded vowel, the result of anticipatory lip rounding overlapping with the production of /s/ is that the frequency lowers, making it acoustically much more similar to /ʃ/. Mann and Repp (1980) found that listeners tended to judge a synthetic fricative as [ʃ] when followed by [ɑ] but then judged the same fricative as [s] when it occurred before [u]. This is because before a rounded vowel, a lower center of frequency is expected for the fricative. Thus, when the fricative is heard before a rounded vowel (in this case, [u]), the low frequency is factored out and a higher center of frequency is 'reconstructed' (Mann & Repp, 1980).

Many psycholinguistic models of speech perception, such as PAM, NLM, and SLM, are simply not equipped to deal with the role of coarticulation in speech perception, since the basic units for speech recognition in these models are segments or segmental features (Dupoux et al., 2011). In these models, the perception of one segment is essentially independent of adjacent segments, however, experimental results indicate that listeners closely track the time course of coarticulatory gestures within and across linguistic units (Beddor, 2012).

One of the first and most prominent models to take coarticulation into account is Ohala's (1981) model of sound change, which is based on listener misinterpretation of coarticulatory influences in the speech signal. As Ohala (1981, 1986, 1990a, 1990b, 1993, 1996, 2012) argues, listeners do not always accurately reconstruct the intended pronunciation of the speaker, a fact which can play a role in the initiation of sound change, as listeners "misperceive" the intended pronunciation of the speaker and this "misperception" leads to a change of norms. Ohala models how listeners deal with variable speech as three possible scenarios: correction, hypocorrection and hypercorrection. In the case of correction, a listener correctly factors out the acoustic effects of coarticulation and reconstructs the speaker's intended pronunciation. Ohala argues that this perceptual correction serves to prevent sound change. If, however, the listener fails to normalize the speech signal, the effects of coarticulation will not be factored out; instead, they will be taken at face value, and the listener will form his conception of the pronunciation of an utterance based on the acoustic signal as it is (mis)perceived. This 'hypocorrection' can then lead to sound change. Ohala speculates about two main situations in which hypocorrection might happen: (1) when the listener does not have the experience necessary to enable him to correct the speech signal and (2) when the listener fails to perceive or attend to the phonological conditioning environment that caused the change. Finally, hypercorrection occurs when a listener attributes some acoustic detail to coarticulation when it was actually part of the intended pronunciation and then factors it out, as is often the case with dissimilation of adjacent sounds. For example, Ohala postulates that the English word 'sword', which was once pronounced with a labiovelar glide /w/, lost the glide due to its labial features becoming attributed to the adjacent /o/, which were then factored out, resulting in the modern pronunciation [sɔɹd].

Ohala's model is designed to address the initiation of sound change, that is, the "mini sound change" that occurs when one listener misperceives a speaker. It is unlikely that the misperception of one listener would lead to sound change on a large scale, because (1) the listener has many opportunities to hear the utterance pronounced by a variety of speakers and (2) even if the listener did not reanalyze his perception and began pronouncing the utterance as he has (mis)perceived it, it is not expected that one speaker's innovative pronunciation would spread to great numbers of other speakers (Ohala 1986, 1993, 2012). The key aspect of misperceptions that become sound changes is that they constitute a change of norms such that "the listener forms a phonological norm that differs from that intended by the speaker" (Ohala, 1993, p. 244).

Ohala's model of sound change is supported by results of experimental research, which indicate that listeners seem to vary in how heavily they weight coarticulatory cues (Beddor, 2012). For at least some listeners, coarticulatory cues

are sufficient cues for making perceptual decisions. Beddor (2012) argues that the perception grammars of these "innovative" listeners may contribute to sound change in contexts of hypocorrection, but only if these grammars are publically manifested via interaction with other speaker/listeners.

A logical question that stems from the role of the listener in sound change is the effect of native language experience in dealing with coarticulation. Ohala's model tacitly addresses the role of L1 in speech perception by inferring that one of the possible reasons for hypocorrection is that the listener does not have the experience necessary to correct the speech signal, implying that the ability to factor out the perceptual effects of coarticulation depends on the listener's linguistic background. Other psycholinguistic models of speech perception that address coarticulation view the ability to perceptually compensate for its effects as language independent processes (Fowler & Smith, 1986; Gow, 2003; Lotto & Kluender, 1998; Mann, 1986). For example, Mann (1986) calls for distinguishing between universal and language-dependent levels of speech perception in which segments are perceived on a language-dependent level and coarticulation is dealt with on a universal level. She argues this based on Japanese listeners' apparent sensitivity to coarticulatory effects of /l/ and /r/ on the /d/–/g/ distinction despite their inability to distinguish /l/ and /r/ phonemes. In this experiment, subjects identified stimuli along a synthetic /da/–/ga/ continuum when the stimuli were preceded by natural tokens of /s/, /ʃ/, /al/ or /ar/. Because the /l/–/r/ contrast is not phonemic in Japanese, while the /s/–/ʃ/ contrast is, in the /al/–/ar/ condition, subjects were additionally asked to identify whether the stimulus contained /al/ or /ar/. The results of this task showed that while English speakers scored 100% on the identification of /al/ and /ar/, the Japanese learners showed great variability. Mann used these results to separate the Japanese listeners into two groups: subjects who scored 98% or better were called superior beginning students (n = 8), and those who scored within 15% of chance were inferior beginning students. Results of the /s/–/ʃ/ context showed that for native English and Japanese listeners alike, the /s/ context favored more /ga/ responses. In addition, for both of the Japanese groups as well as the English group, /al/ favored more /ga/ responses. Thus, despite the fact that the inferior beginning Japanese learners could not distinguish between /al/ and /ar/, they showed the same coarticulatory influence as the English listeners and superior beginning students.

Mann takes these results to imply that "native language experience has less of an influence on one aspect of speech perception, namely, the ability to untangle the acoustic consequences of articulatory interactions, than on another, namely, the ability to make phonemic categorization of speech sounds" (Mann, 1986, p. 175) and calls for two levels of speech perception, one universal and one language-dependent. According to this analysis, at the first level of perception,

listeners track articulatory movements more or less objectively, while at the second level, the speech signal is categorized into language-dependent segments that conform to phonological rules and constraints in the language. She further argues that at the universal level, the representation of speech sounds corresponds to the articulatory gestures that give rise to the speech signal.

In contrast to this, Dupoux et al. (2011) argue that coarticuatory perception cannot be based on only language independent mechanisms, because some context effects are language specific. In this follow-up to Dupoux et al. (1999), investigators tested listeners of Japanese, Brazilian Portuguese and European Portuguese on the perception of illusory vowels inside illicit consonant clusters. Japanese and Portuguese have similar restrictions on syllabic structure, and also like Japanese, Brazilian Portuguese resolves illegal consonant sequences via epenthesis. While European Portuguese has similar underlying phonotactic restrictions as Brazilian Portuguese, obstruent clusters appear in surface phonetic forms as a result of a process that deletes unstressed vowels, as in *obesidade* [obzidad] 'obesity'.

The three groups performed a vowel identification task and an ABX discrimination task. Notable differences between these stimuli and those in Dupoux et al. (1999) are that there were two vowel continua: one for /u/, the epenthetic vowel in Japanese, involving intermediate steps between /ebuzo/ and /ebzo/ and one for /i/, the epenthetic vowel in Brazilian Portuguese, involving stimuli between /ebizo/ and /ebzo/ as well as natural clusters, which did not contain a vowel or vowel remnants (like formant transitions). Participants were asked whether they heard a vowel and if so, which vowel. Looking at the native vowel continua (/u/ for Japanese listeners, /i/ for Brazilian Portuguese listeners), the results of the Japanese listeners replicate previous findings: as the vowel decreased in duration, the number of /u/ responses decreased, but even when there was no vowel, subjects responded that they heard /u/ 59% of the time. Similar results were found for Brazilian Portuguese listeners in the /i/ continuum, as /i/ was perceived 66% of the time for stimuli in which no vowel was pronounced. Comparing the two groups on the non-native vowel continua (/i/ for Japanese listeners and /u/ for BP listeners) yielded further insight into the phenomenon of illusory vowel perception. As the vowel duration shortened, the responses for that vowel declined, but at the same time, the responses for each language's own epenthetic vowel increased. Thus, at the end of the /i/ continuum, Japanese listeners gave many /u/ responses, and at the end of the /u/ continuum, Brazilian Portuguese listeners gave many /i/ responses. However, the coarticulation effect was not identical for the two groups. While the Japanese speakers reported many /u/ responses in the /i/-coarticulation condition, /i/ was still the predominate response, showing that even when all or most of the /i/ vowel had been spliced out of the stimulus, Japanese listeners were still sensitive to the effects of coarticulation. In contrast, for

the Brazilian Portuguese listeners in the /u/-coarticulation condition, the effects of coarticulation were not as strong, and /i/ responses predominated. Finally, the results of the European Portuguese listeners were similar to previous findings for French listeners in that they did not perceive illusory vowels toward the end of either vowel continuum.

The results of the ABX discrimination task corroborated those of the identification task. Japanese and Brazilian Portuguese listeners had more difficulty perceiving non-native consonant clusters and perceived clusters closer to their respective epenthetic vowels, while the European Portuguese listeners perceived non-native clusters more accurately. Again, Japanese listeners showed a stronger coarticulation effect than Brazilian Portuguese listeners, in that they perceived /i/ in the /i/-coarticulated clusters rather than their dominant epenthetic /u/. Brazilian Portuguese listeners, however, perceived their primary epenthetic vowel /i/ even in /u/-coarticulated clusters.

The findings of Dupoux et al. (2011) are significant for several reasons. First, they provide evidence that perception of coarticulatory influences cannot be entirely due to universal auditory processes, "but rather, is integrated within the language-specific process of segmental categorization" (Dupoux et al., 2011, p. 207). In addition, their results provide strong evidence that phonotactic (and perhaps general phonological) probabilities are computed based on surface distributions of segments and not on the underlying representation of words. This is evidenced in the perception differences between Brazilian and European Portuguese speakers. In this case, even when the underlying representation of words is similar in two dialects of the same language, when surface forms differ, so does perception.

In a "vector analysis" of speech perception, Fowler and Smith (1986) propose that listeners segment speech along coarticulatory rather than temporal lines. Figure 2a illustrates how coarticulated speech can overlap across time, and Figure 2b depicts two possible strategies for segmenting phones.

In (i), the listener makes divisions according to temporal cues, resulting in discrete segments that contain overlap from neighboring segments. In (ii), the

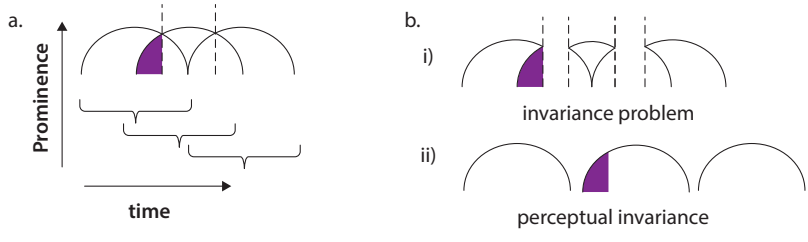

Figure 2. A schematic display of coarticulated speech (a), and two ways that it may be segmented (b) (Fowler & Smith, 1986)

listener has factored out coarticulation and 'recovered' three separate segments which are free of contextual influence. Fowler and Smith (1986) argue that listeners segment speech as in (ii). Here listeners do not base segmentation and coherence (i.e., grouping cues together into the same segment) on only temporal information. In other words, listeners not only factor out simultaneous information when segmenting, but they also ascribe that information to a successive segment, thus grouping together acoustic events that do not occur at the same time.

Two predictions of this model are that (1) acoustic effects of coarticulation will be attributed to the *influencing* segment and (2) such influences will not influence the listener's perception of the *influenced* segment (Fowler & Smith, 1986). Both of these predictions have been borne out by empirical data (Martin & Bunell, 1982; Whalen, 1982 for the first prediction and Fowler & Smith, 1986 for the second). In addition, these authors view this analysis as consistent with Ohala's views on hypocorrection, but they interpret hypocorrection as a listener's failure to "recover the natural segmentation of the produced signal" (Fowler & Smith, 1986, p. 136).

This analysis can be seen as an extension of the Direct Realist Theory (Fowler, 1986). The authors liken a vector analysis for speech perception to a similar analysis of visual perception in that the role of the acoustic signal in perception is like that of reflected light in vision. In vision, the light reflected off of objects and events serves as a proximal stimulus about those objects and events: viewers can obtain information about these from the reflected light, because in being reflected, the light takes on structure specific to that off of which it is reflected (Gregory, 1966). In a similar vein, the acoustic signal serves as information about articulation, and thus the acoustic signal is not the object of perception itself but rather a proximal stimulus, the object of perception being the distal source of the acoustic signal, which is the moving vocal tract.

If some psycholinguistic models have sidestepped coarticulation, it can be said that phonological theory has ignored it altogether. A notable exception to this is Articulatory Phonology (AP) (Browman & Goldstein, 1986, 1992), which views phonology as a set of gesture commands. As such, the notions of segment and feature are abolished, and the basic units of phonological contrast are gestures, which are abstract representations of articulatory events, and these articulatory events are modeled as dynamical systems, that is, "an equation (or set of equations) that expresses how the state of a system changes over time" (Goldstein & Fowler, 2003, p. 166). Thus, while other models of phonology conceive of the continuous motion of the vocal tract as interpolation between targets (the phonological unit), AP models the production of gestural constrictions as a dynamical system with a fixed set of parameters. Words and utterances are modeled as patterns of gestures, or *constellations*, which may overlap in time. A computational

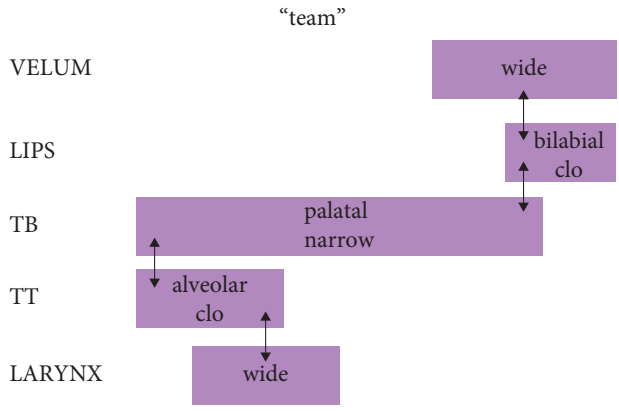

Figure 3. Gestural score for the utterance "team." TB = Tongue Body, TT = Tongue Tip (Goldstein & Fowler, 2003, p. 165)

model can then take these constellations and organize them into *gestural scores*. Figure 3 illustrates the gestural score for the word "team."

Here the rows correspond to speech organs, and the labels within the boxes represent the goal specification for that organ. For example, the specification of "wide" for the organ Larynx corresponds to the aspiration of the /t/. The arrows connect gestures that are critically phased, and the thickness of the arrow represents the bonding strength of the coordinated gestures. So the arrow connecting the Tongue Tip gesture "alveolar closure" and Larynx gesture "wide" (corresponding to aspiration) show that these gestures are critically coordinated, while the overlap of the laryngeal gesture and the Tongue Body gesture "palatal narrow" (corresponding to /i/) is incidental coarticulation. Notice that the specification of "wide" for the velum overlaps with the gesture of the tongue body (i.e., the vowel), starting from about the midway point of the vowel to the end, which represents nasalization.

There are several advantages to an AP model with regards to accounting for coarticulatory influences. First, phonological units defined as gestures provide articulatorily-based natural classes organized hierarchically by the vocal tract itself. Coarticulation is the result of different gestural commands affecting the vocal tract at the same time. In this way, invariance and variability are accounted for phonologically; when the same (invariant) consonant gesture overlaps with differing vowel gestures, the vocal tract motions will differ. In addition, the gestures and gestural organization can be categorical or gradient, and gradient changes can be perceived as categorical. This coupled with the notion that gestures can overlap partially or entirely is consistent with models of language change based on listener misperceptions (Ohala, 1981, 1993, 2012).

AP is also compatible with several of the psycholinguistic models of speech perception that have been discussed in this chapter and the last, particularly the Direct Realist model of speech perception (Fowler, 1986) and vector analysis (Fowler & Smith, 1986). Since a rift emerged between the camps of phonetics and phonology around the time of the structuralists in the early 20th century, these two interrelated fields have largely been studied independently (Ohala, 1990a). Because of this, phonological and phonetic theory have grown increasingly autonomous, especially with regards to the basic units of analysis, making it difficult to relate models of phonology, phonetics and speech perception. Ohala describes this as a lack of "common currency" among knowers of a language, producers of a language and perceivers of a language. AP bridges this important gap, since gestures are the atoms of phonology and gestures also structure the acoustic speech signal. Thus, gestures, which causally structure the acoustic signal, are, in turn, specified by them, providing a common currency for phonetics and phonology (Goldstein & Fowler, 1986).

3.2 Nasalization: Production, acoustics, and perception

Nasal coarticulation with a vowel has received its fair share of attention by phoneticians, especially after the advent of the sound spectrograph in the middle of the 20th century. Some of the first spectrographic analyses of nasalization were done by Malécot (1960), who claimed that nasal vowels are distinctive in English, particularly when a nasal consonant is followed by a voiceless obstruent (cat [kæt] vs. can't [kæ̃t]). This was based on the observation that in English the nasal murmur is almost completely elided before voiceless obstruents. In fact, crosslinguistic evidence suggests that nasal murmurs are shorter and vowel nasalization is more pronounced before voiceless consonants.

The addition of nasalization affects the acoustic structure of a vowel, and in turn, the perception of vowel quality. Much of the research done in the last 70 years has been dedicated to understanding the acoustic correlates of nasalization including how to measure these, which of these correlates affects the perception of nasalization and how nasalization and context affect the perception of vowel quality.

3.2.1 Timing and mechanics of nasalization

Cross-linguistic studies have shown that lowering the velum before articulation of nasal consonants necessarily involves some amount of overlap with the articulation of the preceding vowel, and that the length of this overlap varies across

languages (Clumeck, 1976; Cohn, 1990; Rochet & Rochet, 1991). Solé (1992, 1995, 2007) calls for a distinction between phonetic and phonological processes, arguing that the fact that cross-linguistic variability in degree of vowel nasalization exists suggests that something more than just mechanics of articulation is at play. In other words, if anticipatory nasalization were simply due to constraints on speech production, we would not expect languages to differ in the amount of time needed to lower the velum before articulation of a nasal consonant.

Solé (1992) tested this hypothesis by analyzing the nasalization behavior of speakers of Spanish and English. Using a nasograph, Solé was able to measure and track the position of the velum during speech production. Three speakers of American English and three speakers of Peninsular Spanish read words with the shape CVVC from a list at five different speech rates: (1) overarticulated, overly slow speech ("as if talking to a deaf person who was lip reading"); (2) careful, slow speech ("as if reading out loud to a formal audience in a big lecture hall"); (3) normal conversational speech; (4) fast conversational speech; (5) underarticulated, overly fast speech ("as fast as you possibly can").

In all speech rates, American English speakers lowered the velum at or before the onset of the vowel; thus, the voiced portion of the vowel was completely nasalized regardless of rate of speech. Spanish speakers, in contrast, lowered their velum roughly the same amount of time before the onset of the nasal consonant, around 100 ms prior, regardless of speech rate. This resulted in varying degrees of nasal overlap depending on the rate of speech, with slower speech having the smallest proportion of nasal overlap with the vowel and faster speech a greater proportion of overlap.

The differences between timing of nasal overlap in American English and Spanish are clear when depicted as gestural scores. Observe the scores of English "pan" and Spanish *pan* 'bread' in Figure 4.

Besides differences in vowel quality and aspiration of the initial stop, a notable difference here is the onset of the specification "wide" for the velum. According to Solé's results, this gesture begins as early as the release of the bilabial closure for English speakers, while it begins much later for Spanish speakers.

Cohn (1990) points out that measuring timing alone might be misleading when studying nasal coupling. In a crosslinguistic study of nasalization in Sudanese, American English and French, Cohn found that although vowels preceding tautosyllabic nasals were extensively nasalized, they did not show the rapid increase and plateau in amplitude of nasal flow that French contrastively nasal vowels did. Based on this, she rejected the notion of phonological anticipatory nasalization in English, but, as Solé (1995) points out, there is no reason to expect contrastively nasal vowels to be identical to noncontrastively nasalized vowels.

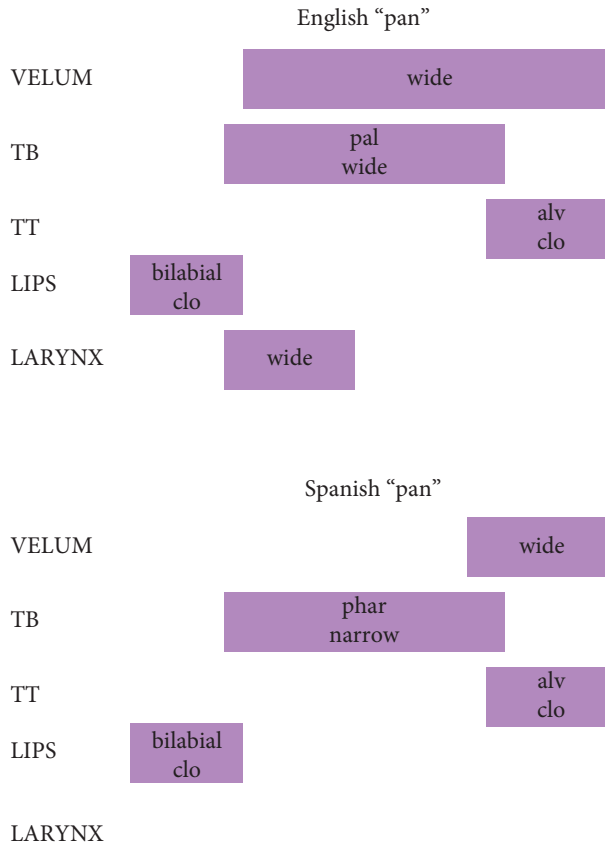

Figure 4. Gestural scores of English "pan" (above) and Spanish *pan* 'bread' (below)

In a follow-up study, Solé (1995) compared the velar port opening velocity of the American English and Spanish speakers from her original 1992 study. She found that the velocity of velar opening is unaffected by speech rate for Spanish speakers, but that it varied for English speakers, with faster speech rates showing higher velocity opening.

Solé (1992, 1995, 2007) argues that the stability of acoustic properties such as timing of anticipatory nasalization and velocity of velar port lowering found in Spanish reflects the transitional time to open the velum, while the systematic adjustment of these properties to durational variations, as found in English, indicates that the vowels are targeted as nasal. As such, nasal overlap in Spanish can be seen as a mechanical constraint on the demands of articulation and nasalization in English as phonological.

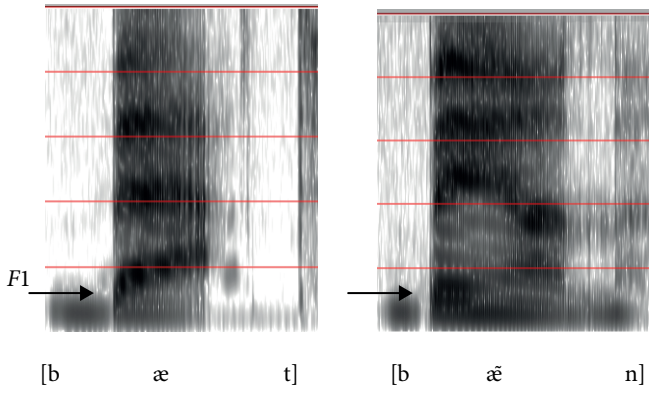

Figure 5. Spectrograms of [bæt] and [bæ̃n]

3.2.2 Acoustics correlates and perception of nasal coupling

The acoustic effects of nasalization include changes in the amplitude of oral formants, an increase in the bandwidth of the first formant by up to 60 Hz, as well as the introduction of nasal pole-zero pairs to the spectrum (Beddor, 1993; Chen, 1997; House & Stevens, 1956). Using an analog synthesizer, House and Stevens (1956) found that in order for listeners to perceive a vowel as nasal, the amplitude of the first formant (*A*1) needed to be lowered by 6–8 dB. Huffman (1990), on the other hand, found that changes in the amplitude of *F*1 relative to the amplitude of the first harmonic (*H*1) in natural speech tokens were more predictive of listeners' perception of oral vs. nasal vowels. Observe the spectrograms in Figure 5, in which the reduction of amplitude of the first oral formant in [bæ̃n] is evident by its lighter appearance in the spectrogram, especially during the second portion of the vowel.

In addition to the changes to oral formants, nasal coupling with the vowel results in the introduction of new formants, which correspond to resonances in the nasal and sinus cavities. As seen in Figure 6, two nasal peaks have been observed to amplify nasality in terms of perception: one at around 1000 Hz (Hawkins & Stevens, 1985; Maeda, 1982) and another between 250 Hz and 450 Hz (Hattori, Yamamoto, & Fujimura, 1958; Maeda, 1982). As we will see shortly, it is this lower nasal formant that plays a role in perceptual shifts of vowel quality due to its location in the region of the spectrogram associated with vowel height.

The exact acoustic consequences of nasalization vary according to speaker, vowel and phonetic context (Fant, 1960; Beddor, 2009 and others), and aspects of vowel quality, such as height, duration and context, have been shown to affect the degree of nasal coupling that is perceived. For example, low vowels require

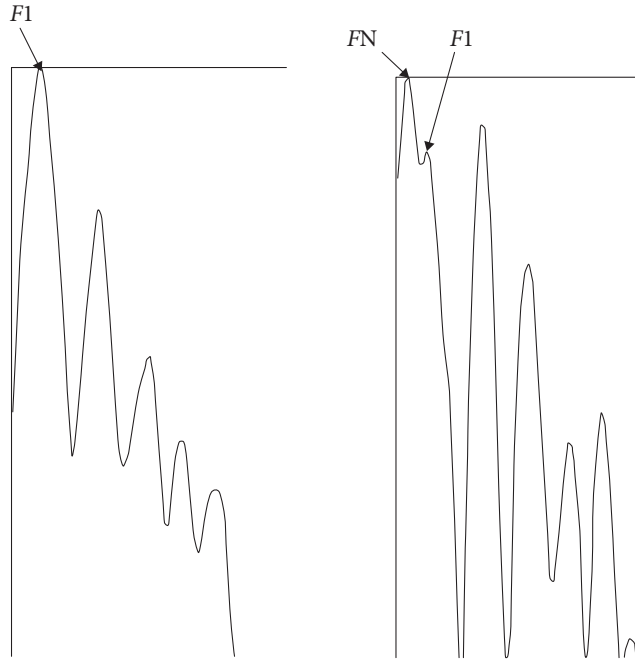

Figure 6. Comparison spectral slices of oral /ɛ/ (left) and nasal /ɛ̃/ (right) in 'bed' and 'bend', respectively

more nasal coupling to be perceived as nasal than high vowels (Abramson, Nye, Henderson, & Marshall, 1981; House & Stevens, 1956; Maeda, 1982), likely due to the fact that in natural speech, low vowels tend to be produced with a slightly lowered velum, even in oral contexts (Fritzell, 1969; Henderson, 1984; Ohala, 1981). In addition, nasal vowels are perceived as more nasal when adjacent nasal consonants are weakened (Kawasaki, 1986).

Many studies on the perception of nasal coupling examine whether there are differences in how listeners perceive the oral/nasal distinction in vowels. Cross-linguistic studies show that most listeners are able to divide oral and nasal vowels stimuli into two categories (oral vs. nasal) regardless of native language background (Beddor, 1993; Beddor & Strange, 1982; Hawkins & Stevens, 1985). In addition, listeners of languages that do not have phonemic nasal vowels demonstrate similar 50% crossover points in judging vowels as oral or nasal as listeners of languages that do have phonemic nasal vowels (Beddor & Strange, 1982; Hawkins & Stevens, 1985). However, as we saw in Section 2.3, native language background does seem to play a role in the discrimination (more so than in the identification) of oral versus nasal vowels. For example, listeners of languages with phonemic nasal vowels tend to perceive the oral-nasal distinction categorically, with

good between-category discrimination and poor within-category discrimination, while speakers of languages with only allophonic nasalization tend to show good discrimination of both between- and within-category pairs (Beddor & Strange, 1982). Context also plays a role in how nasality of vowels is perceived. For example, Fowler and Brown (2000) found that listeners use acoustic correlates of nasalization produced during the articulation of the vowel to predict whether a following consonant is nasal or not.

A number of studies have explored the effects of nasalization on perceived vowel quality, in particular vowel height, due to the presence of the lower nasal peak in the *F*1 region, which is associated with vowel height (Beddor, 1993; Chen, 1997; Krakow, Beddor, Goldstein, & Fowler, 1988). Studies show that when two peaks are close in frequency in a vowel spectrum, listeners do not perceive the individual peaks, but rather one amalgam peak (Chistovich & Lublinskaya, 1979). Because the nasal peak located at 250–450 Hz is so close to *F*1, listeners determine vowel quality based on some weighted average, or center of gravity, between the two peaks, resulting in a shift in the perceived height of nasal vowels.

Typically, high nasal vowels are perceived as lower and low nasal vowels are perceived as higher than their oral counterparts of the same height. This follows

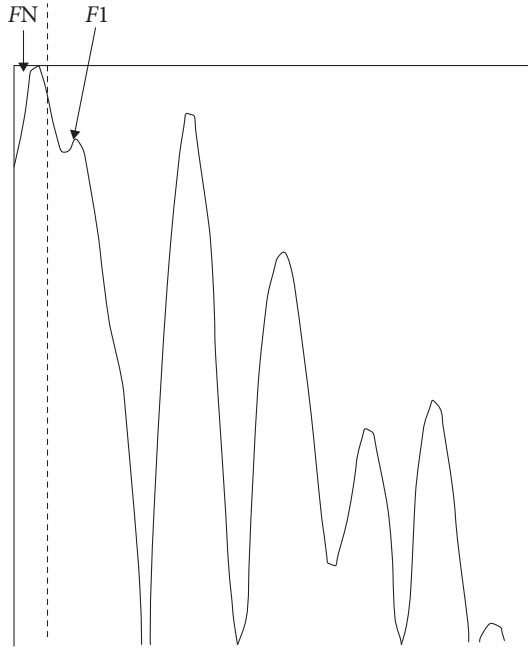

Figure 7. Spectral slice showing first formant (*F*1) and nasal formant (*F*N) where weighted average between them is indicated by dotted line

from the magnetic effects of the nasal formant, which appears at frequencies between the middle to lower extreme of the spectral region associated with vowel height. To be specific, the weighted average of a low frequency oral formant and a slightly higher frequency nasal pole results in perceiving one peak that is slightly higher in frequency than the oral formant. Because *F*1 frequency is inversely correlated to vowel height, this perceptual shift upwards results in perceiving high vowels as slightly lower. Likewise, the "center of gravity" between a high oral formant *F*1, corresponding to a low vowel, plus a mid-frequency nasal pole will result in the perception of a slightly lower peak, corresponding to a somewhat higher vowel. The perceptual effects of nasalization are different for front versus back mid vowels: mid front vowels (e.g., [e], [ɛ]) have been shown to lower, while mid back vowels (e.g., [o], [ɔ]) tend to raise (Wright, 1975). This is possibly due to the lip rounding of the back vowels, which also has an overall raising effect.

Interestingly, there is evidence that speaker/listeners are aware of the perceptual effects of nasalization on some level. Speakers of English have been shown to produce nasalized [ĩ] with a higher tongue position than oral [i], presumably to counteract the perceived lowering associated with nasalizing a high vowel (Carignan, Shosted, Shih, & Rong, 2011). These results challenge the traditional notion that oral and nasal vowels differ only in velopharyngeal position (Carignan et al., 2011).

Context has been demonstrated to be a relevant factor in the perception of nasal vowel height. Krakow et al. (1988) tested English listeners' perception of the /ɛ/–/æ/ contrast in oral ([bɛd]–[bæd]), contextual nasal ([bɛ̃nd]–[bæ̃nd]) and non-contextual nasal conditions ([bɛ̃d]–[bæ̃d]). Each condition consisted of a seven-step, synthetic vowel-height continuum from [ɛ] to [æ], and each of the nasal conditions contained five degrees of nasalization.

Comparing the results across oral and nasal conditions, listeners reported more /æ/ responses in the non-contextual nasal condition than in the oral or contextual nasal conditions, which indicates a lowering effect for only the non-contextual nasal condition. Listeners' perception of contextualized nasal vowels did not differ significantly from that of oral vowels, which shows that they accurately perceived the height of the vowel in the contextual nasal condition. At high degrees of nasalization, vowel height was perceived as lower in both non-contextualized and contextualized stimuli, although the effects were not statistically significant in the contextualized stimuli. These results indicate that English listeners are able to factor out the effects of nasalization when they can attribute the nasalization to coarticulatory influences of an adjacent nasal consonant.

Krakow et al. (1988) tentatively suggest that English listeners' experience with nasal vowels in contextual nasal conditions is what allowed them to factor out nasalization in this context, but not in the non-contextual nasal condition, since

English does not have phonemically nasal vowels. However, it is unclear whether these results support that perceptual influences of co-articulation are language-dependent. This is because English listeners were able to accurately assess the height of nasal vowels only when they were followed by tautosyllabic nasal consonants, which precisely the context in which vowels are nasalized in English. In other words, it is unclear whether English listeners were able to "cancel out" the effects of vowel nasalization due to native language experience or simply because there was an adjacent nasal consonant present to which these effects could be attributed. It is possible that listeners who do not have experience with nasal vowels in contextual nasal conditions may also be able to factor out the effects of nasal coupling when they can attribute them to an adjacent nasal consonant. Likewise, we cannot know if English listeners misperceived the height of nasal vowels in the non-contextual nasal condition due to their lack of experience with nasal vowels in this context, or if even listeners of a language with phonemic nasal vowels might also misperceive nasal vowel height when there is no following nasal consonant. In fact, diachronic changes in nasal vowel height occur precisely when the adjacent nasal is lenited or elided (Beddor, 1993, 2009; Ohala, 1981), indicating that even listeners with experience with non-contextual nasal vowels may misperceive nasal vowel height.

Because the data from English speakers is ambiguous, the best way to examine the roles of native language experience and context in the perception of nasal vowel height is to test listeners of languages with different patterns of nasalization. If the L1 does not determine whether listeners accurately perceive nasal vowel height, then crosslinguistically we should observe the same perceptual pattern found for English speakers: the height of nasal vowels will be perceived as the same as their oral counterparts only in contextual nasal conditions. However, if the Krakow et al. results are indeed due to linguistic experience, we should find that the perception of nasal vowel height varies according to the patterns of vowel nasalization in a listener's language. Spanish and Portuguese provide an excellent opportunity to test this, because some dialects of Spanish do not nasalize vowels even before tautosyllabic nasal consonants (Solé, 1992), while other dialects of Spanish have similar patterns of vowel nasalization as English (Cedergren & Sankoff, 1975; D'Introno & Sosa, 1988; López-Morales, 1981; Terrell, 1975), and Portuguese has both allophonic and (surface) contrastive nasalization (Bisol, 2014; Wetzels, 1997). Experiment 2 in Chapter 5 examines the perception of nasal vowels in different contexts by listeners of these language varieties; however, before revealing these results, it is important to review the differences in these languages with regards to nasals and nasalization in order to adequately contextualize them.

CHAPTER 4

Nasals and nasalization in Spanish and Portuguese

Spanish and Portuguese do not differ substantially in terms of inventory or distribution of nasal consonants, which is not surprising given their common ancestry.[2] Both languages have three nasal phonemes, /m/, /n/ and /ɲ/, of which the palatal /ɲ/ has the most limited distribution.

(4)		Spanish	Portuguese	
	/m/:	*mata*	*mata*	'kills'
		cama	*cama*	'bed'
	/n/:	*nada*	*nada*	'nothing'
		fino	*fino*	'fine'
	/ɲ/:	*uña*	*unha*	'(finger/toe)nail'
		ñoño	*nhonho*	'fool'

Because Spanish and Portuguese both evolved from spoken Latin, they share commonalities with regards to how nasals developed. In both Spanish and Portuguese, Latin geminate -mm- simplified to -m- (*flamma* > Spanish *llama*, Portuguese *chama* 'flame'). In addition, /n/ was the only nasal to survive word finally from Latin in both languages; final /m/ was deleted (*iam* > Spanish *ya*, Portuguese *já* 'already'; *sum* > *so* > Spanish *soy*, Portuguese *sou* '(I) am'), with the exception of a few monosyllables (*quem* > Spanish *quien*, Portuguese *quem* 'who'; *tam* > Spanish *tan*, Portuguese *tão* 'so') (Penny, 2002).

The most significant differences in (consonantal) phonemic inventory between the two languages stem from the development of the palatal nasal, corresponding to the graphemes ñ in Spanish and nh in Portuguese. Most instances of the palatal nasal in Spanish developed from one of three Latin origins: (1) geminate -nn- (*annu* > *año* 'year'; *damnu* > *dannu* > *daño* 'damage'), (2) the sequence -gn- (*pugnu* > *puño* 'fist'), or (3) /n/ followed by a palatal glide (*vīnea* > *vinia* > *viña* 'vineyard'). Latin sequences -gn- and -nj- also became palatal nasals in Portuguese (*pugnu*>*punho* 'fist' and *vīnea* > *vinia* > *vinha* 'vineyard,' respectively). However, in Portuguese, geminate -nn-, like -mm-, was also simplified with no

2. Unless otherwise stated, Portuguese refers specifically to Brazilian Portuguese.

further modification (*annu* > *ano* 'year'; *damnu* > *dannu* > *dano* 'damage'). In addition, diphthongs containing nasal [ĩ] (usually as a result of nasalization and deletion, see Section 4.3 for further discussion of this) gave rise to a palatal nasal in Portuguese (*vinum* > *vĩo* > *vinho* 'wine'; *reginam* > **ragina* > *raĩa* > *rainha* 'queen,' cf. Spanish *vino* 'wine' and *reina* 'queen'). Finally, unlike Spanish, Portuguese lost many instances of intervocalic /n/ and /l/ (*luna* > Portuguese *lua*, Spanish *luna* 'moon'), some of which were later recovered (*unum* > *uno* > *uũ* > *um* 'one') (Colina & Díaz-Campos, 2006; Williams, 1961).

In Modern Spanish and Portuguese, bilabial /m/ and coronal /n/ occur in both word-initial and word-medial positions, while palatal /ɲ/ occurs most often word medially and rarely word initially. In fact, Brazilian Portuguese seems to have even less tolerance for word-initial /ɲ/, as evidenced by the fact that an epenthetic [i] is often inserted at the beginning of words like *nhoque* [ĩˈɲɔki] 'gnocchi'. In spite of these similarities, there is substantial crosslinguistic and crossdialectal variation in the phonetic realization of nasals and in patterns of vowel nasalization between these languages.

4.1 Dialectal variation in Spanish

The most common of the three nasals in Spanish is the coronal /n/, which also has the most extensive distribution in most dialects of Spanish. In syllable-initial position, whether word medial or word final, /n/ is pronounced as an alveolar nasal, but in syllable-final position, there is considerable allophonic and dialectal variation. Here we will examine neutralization of nasals, nasal place assimilation and nasal velarization/absorption, all of which have been implicated in a more general tendency toward nasal lenition across dialects of Spanish (López-Morales, 1981; Piñeros, 2006 and others).

Variationist perspectives divide American Spanish into two macrozones: *tierras altas* 'high lands,' which consist of mountainous regions as well as interior cities (e.g., Mexico City, Lima, etc.), and *tierras bajas* 'low lands' regions, which include coastal regions of the continent and the islands. Spanish varieties in *tierras altas* as well as north-central Spain tend to be more conservative in terms of phonological variation and changes as compared to *tierras bajas*, which have been said to share many characteristics with Andalusia, Spain. Specifically, 'conservative' dialects of Spanish tend to preserve final consonants with their features intact (e.g. /taksi/ [ˈtak.si] 'taxi'; /kaspa/ [ˈkas.pa] 'dandruff'; /boton/ [bo.ˈton] 'button'), while so-called 'radical' dialects in *tierras bajas* tend to delete, weaken or neutralize consonants in coda position (e.g. /taksi/ [ˈta.si] 'taxi'; /kaspa/ [ˈkah.pa] 'dandruff'; /boton/ [bo.ˈtõŋ]~[bo.ˈtõ] 'button') (Colina, 2009; Hualde, 2005; Piñeros,

2006).[3] Such variations in the realization of codas depend on dialectal as well as sociolinguistic factors such as sex, class and situational formality (Colina, 2009; Hualde, 2005).

4.1.1 Nasal place assimilation

In conservative dialects of Spanish, syllable-final /n/, both word medially as well as word finally, agrees in place of articulation with a following consonant. Thus, /n/ becomes bilabial before /p/ and /b/, labiodental before /f/, dental before /t/ and /d/, etc.[4] In word-final position before a vowel, /n/ will be re-syllabified in initial position and remain [n]. Observe the following data:

(2) Nasal place assimilation

a.	*un beso*	[um.ˈbe.so]	'a kiss'
b.	*confiar*	[koɱ.ˈfi͡aɾ]	'confide'
c.	*onza*	[ˈon̪.θa][5]	'ounce'
d.	*entre*	[ˈen̪.t̪re]	'between'
e.	*un yunke*	[uɲ.ˈʝuŋ.ke]	'a yolk'
f.	*conjugar*	[koŋ.xu.ˈɣaɾ]	'conjugate'
g.	*un árbol*	[un.ˈaɾ.βol]	'a tree'
h.	*pan#*	[ˈpan]	'bread'

Nasal place assimilation has been described under different phonological paradigms. Standard descriptions of Spanish often resort to devices such as alpha notation or coindexing in order to express the total place assimilation of final /n/ in a simplified way. This kind of linear, rule-based derivation is exemplified in (5).

(5) Final nasal assimilation

$[+\text{nasal}, +\text{coronal}] \rightarrow [\text{place}_i] \ / \ _____]_\sigma \ [\text{C}, \text{place}_i]$

Autosegmental accounts of nasal place assimilation involve feature sharing between the nasal from the following consonant. Goldsmith (1976), for example, proposes the rule in (6).

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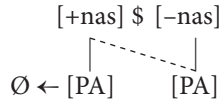
3. Although is should be noted that coda deletion in taxi and other high frequency words also occurs in conservative dialects.

4. Note that Spanish coronal stops are not alveolar but dental /t̪/ and /d̪/.

5. In Peninsular Spanish, orthographic <z> is pronounced /θ/; in Latin-America, it is pronounced [s]. In both cases, the final /n/ assimilates to the following consonant.

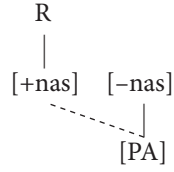
(6)



Here, $ stands for syllable boundary, and PA refers to all features related to place of articulation. The vertical lines indicate association with place of articulation features, and the rule indicated by the arrow states that the nasal becomes disassociated from its own PA features. The dashed line represents the nasal's reassociation with the PA features of the following segment's PA features.

Harris (1984a, 1984b) and Hualde (1989a, 1989b) formulate an autosegmental rule of feature spreading that does not reference the syllable boundary *per se*, but rather targets nasals in the rhyme, since a nasal must be homorganic with the following consonant whether it is syllable final (e.g., *tran.si.ción* 'transition') or not (e.g., *trans.for.ma.ción* 'transformation'). This type of rule is formulated as in (7).

(7)



An important distinction between Goldsmith (1976) and Harris (1984a) is that Harris conceives of the loss of coda nasal PA features as a separate rule, different from but related to feature spreading involved in place assimilation. Thus, all nasals in the rhyme lose their PA features and are subsequently treated as underspecified by the phonology: they obtain PA features either from a following consonant (e.g., *álbum precioso* [ˈal.βum.pɾe.ˈθi͡o.so] 'cute album', *harem grande* [aˈɾeŋ.ˈgɾan̪.d̪e] 'large harem') or by assignment of default unmarked features (as in [ˈal.βun] and [aˈɾen]).

Analyses of nasal place assimilation in Optimality Theory typically conceive assimilation as the result of licensing restrictions on coda place features, a constraint known as Coda Condition (Colina, 2009), or as a restriction on adjacent consonants with different places of articulation, a constraint termed Agree (place) (Bacović, 2001; Piñeros, 2006; Lloret & Mascaró, 2006). Colina (2009) proposes that nasals and laterals satisfy Coda Condition by assimilating to the place features of the following consonant. In this way, place features are licensed through the onset. This is depicted in the tableau in (8).

(8) Coda Condition: A coda cannot license place features.
Have Place: All segments must have place features.
Ident (place): The place features in the input must match those of the output.

Input: /man.ga/ 'sleeve'	Have Place	CodaCond	Ident (place)
☞ a. [ˈmaŋ.ga]			*
b. [ˈman.ga]		*!	
c. [ˈmaN.ga]	*!		*

Candidate a is not entirely faithful to the input, but it is optimal, because it violates only the lowest ranked constraint Ident (place). While candidate b is the most faithful to the input, it fails because it violates Coda Condition by licensing place features in the coda. Candidate c satisfies Coda Condition by being placeless, but it too fails, because it violates the highest ranked constraint, Have Place. Note that the dotted line between Have Place and Coda Condition indicates that it is unclear how these two constraints are ranked with respect to each other; crucially, however, they are both ranked above Ident (place).

Regressive place assimilation of nasals may have a basis in speech perception. According to studies by Ohala (1990b), Kurowski and Blumstein (1987) and others, in a sequence of nasal + stop consonant, the perception of the stop consonant's place features predominates the perception of nasal features, due to the fact that stops have more robust, and therefore salient, PA cues than nasals. This is also why progressive assimilation in nasal + stop sequences is unattested crosslinguistically (Herrera, 2002). Herrera (2002) found that Spanish listeners perceived nasal place features more accurately in homorganic NC sequences than in isolation (nasal murmur only), in non-homorganic NC sequences, and surprisingly, in NV sequences. This would suggest that when a coda nasal benefits from the place features of an adjacent consonant, it is even more perceptible than when it is in onset position.

4.1.2 Neutralization

The distribution of /ɲ/ and /m/ has been described as "defective" in that neither occurs in absolute final position (Navarro Tomás, 1999), with only [n] appearing word finally in conservative dialects (excluding assimilation contexts) due to neutralization.[6] For example, typically loanwords that end in orthographic <m> are pronounced with final [n] almost without exception (e.g., *harem* 'harem', also

6. Note that dialectal variation exists even within otherwise conservative dialects. For example, final nasals are often realized as [m] (e.g., [pam] for *pan* 'bread') in the Cuaca-Valle area of Colombia and the Yucatan region of Mexico (Canfield, 1981) and as [ŋ] (e.g., [paŋ]) in Galicia, Spain (Ramsammy, 2011). This is most likely due to contact with indigenous languages in Colombia and Mexico and Portuguese and/or Galician in Galicia.

spelled *harén*, both pronounced [a.ˈɾen]) (Navarro Tomas, 1999). In some cases, the plural form retains the [m] (e.g., sg. *álbum* [ˈal.βun] 'album', pl. *álbumes* [ˈal.βu.mes] 'albums'), while in other cases, the plural form appears with [n] as well (e.g., *harenes* [a.ˈɾe.nes] 'harems'). It is likely that this variation is due to the level to which the loanword has been incorporated into Spanish phonology, with more recent loans retaining [m] in plural forms (see Bonet [2006] for discussion of differential OT constraint rankings for native words vs. loans). Likewise, the production of final [m] in *cultismos* like *referéndu*[m] 'referendum' and *currículu*[m] 'curriculum' should not be considered as evidence in contra neutralization, because these are learned forms that have not entered into the phonology, and these often coexist with derivational forms like *referendo* and *currículo*, especially when plural, e.g., *referendos*, *currículos*.

A classic and oft cited account of neutralization is Harris (1984a), which is couched in autosegmental phonology and lexical phonology. In this account, nasals are underspecified for place features and receive features via language-particular and universal markedness rules, also known as redundancy rules. In this way, a nasal that is specified [–coronal] will receive the unmarked features [+anterior, –back] via universal markedness rules, because these are the unmarked values for non-coronal consonants. The coronal nasal is considered maximally unmarked in terms of place of articulation, and therefore receives all of its place features via redundancy rules. Thus, the phonemes /m/ and /ɲ/ are specified only for features that are marked: /m/ is specified [–coronal] and is assigned remaining default features [+anterior, –back] while /ɲ/ is specified [–coronal, –anterior] and is assigned the default feature [–back]. According to this account, neutralization is the result of the application of an autosegmental rule that deletes nasal place of articulation features in word-final position and redundancy rules that specify unmarked values for place features.[7] Accordingly, in a loanword like /ˈal.βum/, the nasal loses its place features, because it is word final, and then the resulting nasal, now unspecified for place features, is assigned the unmarked PA features, which results in the form [ˈal.βun].

Optimality Theory accounts of nasal neutralization draw on the fact that only coronal consonants are possible word finally in Spanish (Colina, 2009; Piñeros, 2006). Thus, word-final neutralization of nasals is attributed to a group of markedness constraints that mediate against place features. Under the umbrella of Place Hierarchy, the constraints *Coronal, *Labial and *Dorsal are organized into a fixed hierarchy according to which dorsal is the most marked place

7. The rule actually specifies that PA features are lost at the end of a lexical category, which, in the examples presented here, is the end of the word. See Harris (1984a) for more details on the difference.

had only two nasals, 136 (98.55%) of them had /m/ and /n/. Piñeros explains that these nasals benefit from relative articulatory economy as well as relative perceptibility as compared to palatal and velar nasals: labial and coronal nasals' forward place of articulation leaves a longer space in the oral cavity, which he claims makes the oral anti-resonances during the articulation of the nasal more perceptible, while nasals with a more posterior place of articulation leave less space in the oral cavity in which anti-resonances are generated. Thus, a perceptual hierarchy for nasals would rank labial as the most perceptible, followed by coronal, palatal then dorsal, which is at odds with the articulatory hierarchy coronal > labial > dorsal > palatal. Accordingly, whether a language's third nasal is [ɲ] or [ŋ] depends on whether that language's grammar prioritizes economy in articulation or discernibility of perception.

However, experimental studies have shown that the primary cue for nasal place of articulation is not nasal antiresonances, but adjacent formant transitions. Malécot (1956), for example, examined the perception of nasal consonants [m], [n], and [ŋ] by speakers of English and found they perceived nasal murmurs in isolation much less accurately than when an adjacent vowel was present. In addition, when a nasal resonance was presented with conflicting formant transitions, listeners used the transitions as the primary cue for the PA features of the nasal. For example, when the resonance of [m] was spliced and connected to coronal transitions, listeners perceived [n]. When the nasal appeared in a syllable with the appropriate formant transitions, onset nasals were generally perceived more accurately than coda nasals, with the exception of [ŋ], which was perceived more accurately in coda. Furthermore, there was a general trend in which [m] was the most accurately perceived nasal, followed by [n] and then [ŋ]. These last results are somewhat in accord with Piñeros (2011), however, these authors' reasoning for the increased saliency of some nasals as compared to others is still at odds.

Another important aspect of nasal place perception is the role of a listener's native language. Recall that Marckwardt's (1946) study (Section 2.2.2) showed that Spanish speakers perceived final [n] most accurately as compared to final [m] and [ŋ]. Herrera (2002) also found that Spanish speakers perceived [n] more accurately than [m] or [ɲ] when identifying nasal murmurs in isolation, although even [n] was only correctly identified 40% of the time, followed by [m] (20%) and [ɲ] (15%). In contrast, onset nasals in NV sequences were perceived much more accurately, with [m] leading (87%), followed by [n] (83%) and [ɲ] (53%) (Marckwardt, 1946). Thus, there is some evidence that universal tendencies in markedness and perception are tempered by a listener's native language. On the other hand, while Malécot assumes that the velar nasal is more perceptible (to English listeners) in coda rather than onset due to familiarity with [ŋ] syllable finally, it turns out that crosslinguistically, [ŋ] rarely occurs in onset. This is likely because

it is the most vowel-like of the nasals (Ohala, 1975; Widdison, 1997), and in terms of sonority, less sonorous (i.e., less vowel-like) consonants are preferred in onset position (Clements, 1988, 1990). Thus, it is unclear exactly how/if one's native language influences the perception of nasal place contrasts in different contexts.

4.1.3 Velarization and absorption

One of the most studied aspects of dialectal variation found in the production of nasal consonants in Spanish is velarization and absorption (or elision) of final /n/. This process is productive in Andalusia, Spain, Guatemala, Veracruz, Mexico, and in most of *tierras bajas* as well as in Galician Spanish. While velarization data are often presented in a cut-and-dried fashion, especially in phonological analyses, there is actually considerable variation in terms of the contexts in which nasals are velarized (Cedergren & Sankoff, 1975; D'Introno & Sosa, 1988; López-Morales, 1981; Terrell, 1975). Depending on geographic and social factors, syllable-final nasals may be velarized only word finally, word medially but only before [–cont] consonants, or in all syllable-final contexts. These patterns are illustrated in (10)–(12):

(10) Velarization of word-final /n/ only
[ũŋ.ˈaɾ.βol] *un árbol* 'a tree'
[kõŋ.ˈpe.ðɾo] *con Pedro* 'with Pedro'
[ũŋ.ˈʲuŋ.ke] *un yunke* 'a yolk'

[kõm.ˈfi͡aɾ] *confiar* 'to confide'
[ˈkõŋ.ʲju.xe] *conyuge* 'spouse'
[kõm.ˈpa.ðɾe] *compadre* 'buddy'

(11) Velarization of word-medial /n/ before [+cont]
[kõŋ.ˈfi͡aɾ] *confiar* 'to confide'
[ˈkõŋ.ʲju.xe] *conyuge* 'spouse'
[ˈkãŋ.so] *canso* '(I) tire'
[õŋ.ˈra.ðo] *honrado* 'honored'

[kõm.ˈpa.ðɾe] *compadre* 'buddy'
[ˈkãn̪.t̪o] *canto* '(I) sing'

(12) Velarization or absorption of final /n/ in all contexts
[ũŋ.ˈaɾ.βol] ~ [ũ.ˈaɾ.βol]
[kõŋ.ˈpe.ðɾo] ~ [kõ.ˈpe.ðɾo]
[ũŋ.ˈʲũŋ.ke] ~ [ũ.ˈʲũ.ke]
[kõŋ.ˈfi͡aɾ] ~ [kõ.ˈfi͡aɾ]
[ˈkãŋ.so] ~ [ˈkã.so]

[õŋ.ˈra.ðo] ~ [õ.ˈra.ðo]
[kõŋ.ˈpa.ðɾe]~ [kõ.ˈpa.ðɾe]
[ˈkãŋ.t̪o]~ [ˈkã.t̪o]

It is important to note that even among radical dialects, there is considerable variation in the rates of velarization and absorption of final /n/. For example, in Puerto Rico, although final nasals are often velarized, assimilation of final /n/ before a consonant is still quite common (López-Morales, 1981). In contrast, assimilation is much less common in Caracas, where velarization is preferred even word medially (D'Introno & Sosa, 1988). López-Morales (1981) found for Puerto Rican Spanish that although velarization of final /n/ is common before vowels and pauses (at rates of 26.6% and 69.3%, respectively), it was found in only 13% of preconsonantal contexts, with assimilation preferred over 80% of the time in that context. Absorption was found most often before a pause, but even then in only 8.1% of the cases. In Caracan Spanish, D'Introno & Sosa (1988) found velarization of final /n/ in similarly high rates across preconsonantal, prevocalic and prepausal contexts (76.5%, 94.6% and 92.1%, respectively). Terrell (1975), in contrast, found almost no velarization of word-medial, syllable-final /n/ in Cuban Spanish, but many cases of absorption, especially before [+continuant] consonants. However, several social factors, including age, sex, socio-economic status, and urban vs. rural origin have been identified as predictors for velarization and absorption (Cedergren & Sankoff, 1975). For example, Guitart (1973, 1976) claims that in educated Spanish of Havana, all nasals become velar (not necessarily absorbed) before consonants, regardless of that consonant's place of articulation. This however is not supported by electropalatographic studies of Cuban Spanish (Kochetov & Colantoni, 2011), which find categorical assimilation of nasals before coronal consonants in Cuban Spanish. Thus, there is some inconsistency with regards to experimental studies performed with Cuban Spanish speakers, especially when comparing impressionistic data to instrumental data.

In general, vowel nasalization is more extreme in radical dialects than other dialects, and absorption of final nasals is preceded by nasalization of the preceding vowel (Cedergren & Sankoff, 1975; D'Introno & Sosa, 1988; López-Morales, 1981; Terrell, 1975). Terrell (1975) describes the stages of word-medial, syllable-final nasal weakening as (1) nasal assimilation with following consonant; (2) preceding vowel nasalization; and (3) absorption of nasal consonant. It is interesting to note that diachronic analyses show a similar path to absorption of final nasals in French (Azra, 2000) and Galician (Colina & Díaz-Campos, 2006). Under Terrell's analysis, conservative dialects of Spanish could be described as being in stage (1): nasal assimilation with following consonant, but no nasalization of

preceding vowel yet. Radical dialects, on the other hand, can be seen as being at various points between stages (2) and (3).

Velarization has been problematic for many phonological analyses, particularly due to the fact that coronal has been argued by so many linguists to be the least marked place of articulation, leading Paradis and Prunet (1994) to propose the Weak Coronal Hypothesis, which states that there is only one universal default articulator, and that it is coronal. Why then do radical dialects of Spanish seem to treat the velar as the default nasal?

Harris (1984a) gives a lacking explanation for this, claiming that [n] is the default nasal for conservative dialects of Spanish while [ŋ] is the default for radical dialects. The problem with this analysis is that he himself argues that universal rules of markedness dictate that [ŋ] is in fact the most marked of the nasals, and must lexically specify all its features, so it is unclear why one dialect would opt for the universally least marked nasal as its default while another would opt for the most marked as its default.

Baković (2001), in contrast, argues that velarization is really debuccalization, which results in a placeless nasal that sounds velar due to the acoustic similarity between nasal vowels and velar nasals (preceded by nasal vowels). Accordingly, in radical dialects, Coda Condition would be satisfied by eliminating place features in the coda. Presumably, this constraint would be ranked higher than Have Place and Ident (place), as depicted in (13).

(13)

Input: /man.ga/ 'sleeve'	CodaCond	Have Place	Ident (place)
a. [ˈmaŋ.ga]	*!		*
b. [ˈman.ga]	*!		
☞ c. [ˈmaN.ga]		*	*

A critical difference between the notion of Coda Condition as applied in tableau (8) and here is that Baković (2001) does not consider Coda Condition to be a licensing condition, but rather an outright prohibition of place features in the coda. This is why candidate a fails in tableau (13): assimilation here is considered to violate Coda Condition because the coda nasal may not have place features at all, whereas in (8), it satisfied this constraint by "licensing" place features through the following onset.

The central assumption for Baković's argument is that the velar nasal is actually placeless. Why then has it been described by so many linguists as velar? As Widdison (1997) points out, the reason [ŋ] is so much less salient than its labial and coronal counterparts is that the resonance chamber is so much shorter (due to its backness) that the dampening effect occurs at much higher frequencies, well

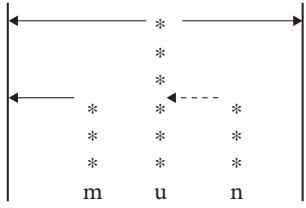
above the range where other nasal murmurs are realized. "Given the negligible effect the tiny secondary chamber exerts on the velar nasal, the auditory impression is that of a sound wave enhanced by a single resonating cavity, much like that of a vowel" (Widdison, 1997, p. 143). In other words, a nasalized vowel plus [ŋ] sounds very much like a nasal vowel with no following consonant.

However, electropalatalographic analyses of "velarizing" dialects do not support that final velar nasals are placeless (Kochetov & Colantoni, 2011; Ramsammy, 2013). Ramsammy (2013), for example, confirms that word-final velarized nasals in Galician Spanish are consistently realized with dorso-velar contact in both prevocalic (ditén#a-) and prepausal (e.g., ditén##) contexts. Kochetov and Colantoni (2011) find that Cuban Spanish speakers variably realize final /n/ as [ŋ] or as [ũɥ], the latter of which is most likely congruent with Baković's conception of a placeless nasal. Nonetheless, it is obvious that velarization does entail a coda nasal with place features at least some of the time for the dialects studied instrumentally. At the very least, the observations regarding the perceptual similarity between velar nasals and nasalized vowels calls into question many of the descriptive analyses of velarization and absorption that have been presented for Spanish, since most of these (especially the more dated ones, which also happen to be the most cited) rely on impressionistic judgments made by investigators.

Now we are still left with the conundrum of how the velar nasal can be the "default" in some dialects and yet have the most marked place of articulation. In point of fact, Spanish is not the only language that presents this dilemma. Cross-linguistically the velar sometimes appears to be the "weak" nasal, for example, when assimilation targets only velar nasals in coda position, or when nasals reduce synchronically or diachronically to [ŋ] (Paradis & Prunet, 1993). Paradis and Prunet suggest that what is happening in these cases is that the nasal has lenited to the point of losing its place of articulation node and then shares the feature [+dorsal] with the preceding vowel in order to be pronounced. This would also explain why vowels are typically nasalized in contexts of velarization.

Piñeros (2006) capitalizes on the notion of velarization as the result of sharing the vowel's place of articulation by proposing that final nasal velarization and nasal absorption have the same cause: a constraint that requires nasal consonants to be aligned with the left edge of the syllable, which he calls Align-C (nasal). Nasal consonants in onset obviously satisfy this constraint, while coda nasals do not, and Piñeros argues that a nasal with linked structures to a vowel would at least partially satisfy Align-C (nasal). This is because the vocalic nucleus is more sonorous than the consonants in the onset and coda and can therefore be aligned with the right or left edge of the syllable. This is illustrated in (14).

(14)



The sonority grid represented here assesses segment-to-syllable alignment (Piñeros, 2006). The highest mark on the sonority grid for each segment is where alignment is checked. The first nasal in the syllable is in compliance with Align-C (nasal), as is the vowel, due to its greater sonority, which allows it to reach over the less sonorous segments to align with either syllable edge; however, the post-nuclear nasal consonant violates Align-C (nasal), because it is blocked from aligning with the left edge of the syllable by the sonority column of the nucleus. Thus, nasal velarization and absorption are the result of the interaction of Align-C (nasal) with faithfulness constraints prohibiting deletion of features, Max (feature), or segments Max (seg). When Align-C (nasal) outranks Max (feature), as in (15), the optimal output is absorption of the final nasal. Here, Max (seg) prevents total deletion of the final nasal.

(15) Align-C (nasal): Every nasal consonant must be aligned with the left edge of a syllable.
Max (feature): The features of input segments must be preserved in the output.
Max (seg): Every segment in the input must have a correspondent in the output.

Input: /kon.fun.d̪en/ 'they confuse'	Max (seg)	Align-C (Nasal)	Max (Feature)
a. [kon.fun.d̪en]		*!**	
b. [koŋ.fuṉ.d̪en]		*!**	
☞ c. [kõ.fũ.d̪ẽ]			***
d. [ko.fu.d̪e]	*!**		***

Note that each asterisk corresponds to a violation of one of the three nasal consonants in the input. This tableau depicts a grammar that prefers nasal absorption. Candidate a fails because the nasals cannot be aligned with the left edge of the syllable. Candidate b fails for the same reason; place agreement does not resolve the conflict, because, even though the nasals are homorganic with the adjacent consonants, they still cannot be aligned with the left edge of their own

respective syllables, violating dominant Align-C (nasal). Candidate d fails because it violates the dominant Max (seg) by completely deleting the nasal segment. Therefore, candidate c is the optimal output because, although it violates the faithfulness constraint Max (feature), unlike the other candidates, it satisfies dominant Align-C (nasal) and Max (seg).

Piñeros argues that velarization is a gradient process, and hypothesizes a spectrum of lenition that spans from no lenition at all to vowel nasalization, nasalization and velarization, nasalization followed by a nasal fricative, nasalization followed by a placeless nasal, and total absorption: Vn]σ > Ṽn]σ > Ṽŋ]σ > Ṽɣ̃]σ > Ṽɴ]σ > Ṽ]σ. The various realizations of final nasal consonants can be explained by Align-C (nasal) gradually gaining precedence over Max (feature) (Piñeros, 2006). According to Feature Class Theory (Padgett, 1996a, 1996b), features are organized according to class membership and constraints can refer to these classes. When Align-C (nasal) completely outranks Max (feature), as in the tableau in (15), the result is absorption of the nasal in coda position. However, breaking down Max (feature) according to the relevant feature classes Major (e.g. [sonorant, consonantal, etc.]), Stricture (e.g. [continuant, high, low, etc.]), Place and Nasality results in a set of more specific faithfulness constraints. In this way, velarization is obtained when Align-C (nasal), outranks only Max (place), as in (16) (Piñeros, 2006).

(16) Nasal velarization

Input: /en.la.se/ 'link'	Max (seg)	Max (nas)	Max (maj)	Max (strict)	Align-C (nasal)	Max (place)
a. [en.la.se]					*!	
b. [ẽn.la.se]					---!	
☞ c. [ẽŋ.la.se]					--	*
d. [e.la.se]	*!	*	*	*		*

Candidate a fails (in radical dialects that prefer velarization) because it violates the higher-ranking Align-C (nasal), and d fails due to the total loss of the nasal segment, which violates dominant Max (seg). Both candidates b and c only partially satisfy Align-C (nasal). Dashes in the cells indicate gradient satisfaction of a particular constraint. For example, candidate b partially satisfies Align-C (nasal) by becoming linked to the previous vowel via nasalization; however, candidate c satisfies this constraint to a higher degree (shown by fewer dashes) by also sharing place of articulation features with the preceding vowel.

4.2 Brazilian Portuguese

One of the most striking differences between Portuguese and Spanish phonology is the occurrence of nasal vowels and diphthongs in Portuguese. These developed from spoken Latin via regressive nasalization and deletion (i.e., nasal absorption) and, in fewer cases, progressive nasalization. Typically, when absorption of coda nasals resulted in two adjacent vowels that were identical or close in the vowel space, they fused, as in *lanam* > *lãa* > *lã* 'wool'; when the resulting adjacent vowels were farther away from each other, a nasal diphthong was formed, e.g., *manum* > *mão* 'hand.' In some cases, particularly in rising diphthongs, nasalization was lost, as in *luna* > *lũa* > *lua* 'moon' and *bonam* > *bõa* > *boa* 'good fem.' (cf. *bonum* > *bõo* > *bom* [bõ] 'good masc.'). Progressive assimilation also resulted in nasal vowels and diphthongs, especially following /m/, e.g., *matrem* > *mae* > *mãe* 'mother' and *meam* > *mia* > *mĩa* > *minha* 'my.' Note the subsequent nasal reinsertion in the latter example, just as in the previously mentioned examples *vinum* > *vĩo* > *vinho* 'wine' and *reginam* > **ragina* > *raĩa* > *rainha* 'queen.' While many cases of nasal reinsertion resulted in a palatal nasal, this was not always the case, e.g., *unum* > *uno* > *ũu* > *um* 'one masc.' and *unam* > *una* > *ũa* > *uma* 'one fem.' Of note is that the resulting PA features of the reinserted nasal are the same as those of the nasal vowel, palatal in the case of nasal [ĩ], and labial in the case of nasal [ũ], which is reminiscent of the idea of linked structures in VN sequences discussed for Spanish velarization in 4.1.3 (Colina & Díaz-Campos, 2006; Williams, 1961). This notion will be discussed further with regards to Modern Portuguese later in this chapter.

The phonemic inventory of Modern Brazilian Portuguese (henceforth BP) vowels includes seven oral vowels and five nasal vowels, as illustrated below.

(17)

	a. Oral vowels			b. Nasalized vowels		
+high	i		u	ĩ		ũ
–high, –low	e		o	ẽ	ɐ̃	õ
+low[8]	ɛ	a	ɔ			

Analyses of BP nasal/nasalized vowels are quite varied: some treat all nasalized vowels as phonemically oral vowels that undergo nasalization (Quicoli, 1990, 1995; Reed & Leite, 1947), while others posit phonemically nasal vowels (Hall, 1943). Still others propose that nasal vowels are phonological diphthongs (Parkinson,

8. Analysis of ɛ and ɔ as low vowels is by no means uncontroversial. I have chosen to follow Quicoli (1990, 1995), Hall (1943), Reed & Leite (1947) and Head (1965) in assuming three vowel heights defined in terms of tongue-body features. See Redenbarger (1981) and Lopez (1979) for alternative analyses.

1983). In this section, nasalization data in BP will be presented, and various analyses that have attempted to account for these patterns will be discussed.

4.2.1 Stress-induced nasalization

While contrastive nasalization is the cause of much controversy among BP linguists, stress-induced vowel nasalization is not highly disputed. Quicoli (1995) offers perhaps the most thorough analysis of stress-induced nasalization, including the myriad of morphophonological variations that result from such nasalization. Observe the following data adapted from Quicoli (1995).

(18) a. [ˈfĩ.nu] – [a.fi.ˈna.du] *fino* 'fine' – *afinado* 'sharpened'
b. [ˈpẽ.nɐ] – [pe.ˈna.ʃu] *pena* 'feather' – *penacho* 'headpiece'
c. [ˈfũ.mu] – [fu.ˈma.sɐ] *fumo* 'tobacco' – *fumaça* 'smoke'
d. [ˈgõ.mɐ] – [go.ˈma.du] *goma* 'starch' – *gomado* 'starched'
e. [ˈgrẽ.mɐ] – [gra.ˈma.du] *grama* 'grass' – *gramado* 'turf'

Note that low vowels raise when nasalized, thus /a/ becomes nasal [ɐ̃]. Here we can see that nasalization in an open syllable is stress dependent such that stressed vowels before onset nasals are nasalized while unstressed vowels are not.

Because this type of nasalization is dependent upon stress, interesting patterns of vowel nasalization and raising emerge as the gamut of morphophonological forms are considered. Observe the following (Quicoli, 1995).

(19) a. [ba.ˈnɐ̃.nɐ] – [ba.na.ˈnaw] – [ba.na.ˈna.dɐ]
banana – bananal – bananada
'banana' – 'banana grove' – 'banana paste'
b. [ˈʃɐ̃.mɐ] – [ʃa.ˈma.dɐ]
chama – chamada
'(he) calls' – 'roll call'
c. [bɾi.ˈtɐ̃.ni.ɐ] – [bɾi.ˈtɐ̃.ni.ku]
Britânia – britânico
'Britain' – 'British'
d. [fa.ɾa.ˈɔ] – [fa.ɾa.ˈõ.ni.ku]
faraó – faraônico
'pharaoh' – 'pharaoh-style'
e. [ˈɐ̃.ni.mu]–[a.ˈni.mi.ku]
ânimo – anímico
'animus' – 'pertaining to the spirit'

While prefixes never affect stress in BP, some suffixes do. In (19a) and (19b), the suffixes -al and -ada are stressed while the suffix -ico, in examples (19c)–(19e), is

not. Quicoli (1995) designates suffixes that are stressed, such as those in (19a) and (19b), as Type I, while unstressed suffixes as in (19c)–(19e) are Type II. Notice that Type II suffixes, although not stressed themselves, may affect the stress of the root. These suffixes require that stress fall on the syllable immediately preceding the suffix (i.e. the last syllable of the root) (Quicoli, 1995). Thus, in examples (19c) and (19d), stress seems unaffected because it was already on the last syllable of the root, but in example (19e), stress is shifted to the antepenultimate syllable.

Following work by Halle and Vergnaud (1987), Quicoli (1995) accounts for the differences in main stress obtained in (19) by postulating cyclic, noncyclic and stress domain affixes in BP. Accordingly, prefixes in BP are considered non-cyclic, because they do not affect stress. Type I and II suffixes are cyclic, as is the BP Main Stress Rule (MSR). The difference between Type I and Type II suffixes is that Type II suffixes are lexically marked as exceptional with respect to the stress rule (i.e. they cause the MSR to assign stress to the syllable immediately preceding the suffix), while Type I suffixes receive stress through normal application of the MSR or by being lexically marked to bear stress (Quicoli, 1995). The following illustrates how the cyclic application of MSR works with Type I and Type II suffixes.

(20)

Type I	Type II	
[[fo*ɾ*m]al]	[[sifili(t)]iko]	
ˈo	ˈi	cycle 1: MSR
o ˈa	i ˈi	cycle 2: MSR & Stress Erasure
[fo*ɾ*.ˈmaw] *formal* 'formal'	[si.fi.ˈli.ti.ku] *sifilítico* 'syphilitic'	

The question remains, how do cyclic rules and suffixes interact with stress-induced nasalization? If nasalization were cyclic, we would not expect to see the alternation between nasalized and oral vowels in (19), as nasalization would be assigned twice in the derivation.

(21)

[[banan]al]	
ˈa	cycle 1: MSR
ˈẽ	Nasalization & Ṽ Raising
ẽ ˈa	cycle 2: MSR & Stress Erasure
---	Nasalization & V Raising
*[bʌ̃naˈnaw]	

As shown by the failed derivation in (21), Nasalization and Nasal Vowel Raising cannot be cyclic rules. Quicoli proposes that Nasalization and Nasal Vowel Raising

apply in the noncyclic stratum. In this way, both rules apply only once, after all cycles have transpired and main stress is assigned. The following derivations exemplify how these rules interact with Type I and Type II suffixes.

(22)

Type I		Type II	
[[banan] al]		[[faraɔ(n)]iko]	
			Cyclic Stratum
ˈa		ˈɔ	cycle 1: MSR
a	ˈa	vacuous app	cycle 2: MSR & Stress Erasure
			Noncyclic Stratum
---		ˈɔ̃	Nasalization
---		ˈõ	Ṽ Raising
[ba.naˈnaw]		[fa.ɾa.ˈõ.ni.ku]	

The derivation in (22) accounts for the observations in (19), but an unexpected pattern arises with the diminutive suffix *-(z)inho*, the superlative suffix *-íssimo*, the adverbial suffix *-mente* and a few others. Despite the fact that these suffixes are stressed, stressed-induced Nasalization and Nasal Vowel Raising still apply in the root.

(23)
a. [ba.nẽ.ˈnĩ.ɲɐ] *bananinha* 'little banana'
b. [ba.kẽ.ˈni.si.mɐ] *bacaníssima* 'very elegant'
c. [ba.kẽ.na.ˈmẽ.t͡ʃi] *bacanamente* 'elegantly'

These suffixes cannot be cyclic, like Type I and Type II suffixes, because forms such as *[ba.na.ˈnĩ.ɲɐ] and *[ba.kɐ.ˈni.si.mɐ] would be result; yet they cannot be non-cyclic either, because, unlike prefixes, they affect word stress. Quicoli (1995) analyzes suffixes like these as stress domain suffixes. In this way, both the root and the suffix are assigned stress in the first cycle. Observe the derivation in (24).

(24)

[banan]	[iɲa]	[bakana]	[mente]	
				Cyclic Stratum
ˈa	ˈi	ˈa	ˈe	cycle 1: MSR
[ba.ˈna.ˈni.ɲa]		[ba.ˈka.na.ˈmen.te]		Noncyclic Stratum
ˈã	ˈĩ	ˈã	ˈẽ	Nasalization
ˈɐ̃		ˈɐ̃		Ṽ Raising
[ba.nẽ.ˈnĩ.ɲɐ]		[ba.kẽ.na.ˈmẽ.t͡ʃi]		

A late non-cyclic rule of Conflation would eliminate all but the rightmost stress to produce the correct forms [ba.nẽ.ˈnĩ.ɲɐ] and [ba.kẽ.na.ˈmẽ.t͡ʃi] (Quicoli, 1995). Another striking example of the interaction of the MSR and Nasalization is the following minimal pair pointed out in Quicoli (1990).

(25) a. [ka.ˈmĩ.ɲɐ] *caminha* ‘(he) walks’
b. [kẽ.ˈmĩ.ɲɐ] *caminha* ‘little bed’

This difference can be attributed to the internal structure of the words, as in (26) (Quicoli, 1995).

(26)

a.	b.	
[kamiɲ + a]	[kam] [iɲa]	
		Cyclic Stratum
ˈi	ˈa ˈi	cycle 1: MSR
[ka.ˈmiɲ + a]	[ˈka.ˈmi.ɲa]	Noncyclic Stratum
ˈĩ	ˈã ˈĩ	Nasalization
---	ˈẽ	Ṽ Raising
[ka.ˈmĩ.ɲɐ]	[kẽ.ˈmĩ.ɲɐ]	

In spite of the fact that stress-induced nasalization is rather straightforward and uncontroversial in the literature on BP nasalization, it yields very interesting results when considered in the context of morphophonology. These are important, because they demonstrate that BP listeners make phonological distinctions between nasalized and non-nasalized vowels in open and closed syllables (as we will see in the next section) and in similar words with different morphological structure. This is significant for questions related to perception of nasal vowels in certain contexts, because we know that BP listeners perceptually distinguish between minimal pairs such as [ka.ˈmĩ.ɲɐ] and [kẽ.ˈmĩ.ɲɐ], which differ only in terms of vowel nasalization (and the accompanying raising).

4.2.2 Nasal(ized) vowels in closed syllables

Nasal vowels in closed syllables have been the subject of controversy among BP linguists. In this case, stress is irrelevant to the vowel’s nasality. At issue is the phonemic status of the nasal vowel and whether it is targeted as oral and then nasalized, or whether it is targeted as nasal. Part of the problem is that word finally and before [+cont] consonants, typically no nasal consonant is pronounced, as seen in (28), while before stops there may be a reduced, homorganic nasal transition (Bisol, 2014), as illustrated in (27).

(27)	a.	[ˈfũ$^{(n)}$.du]	*fundo* 'bottom'
	b.	[põ$^{(n)}$.te.ˈa.du]	*ponteado* 'stitched'
	c.	[ˈpĩ$^{(ŋ)}$.gu]	*pingo* 'drop'
	d.	[ˈkɐ̃$^{(m)}$.pɐ]	*campa* 'gravestone'
(28)	a.	[lɐ̃]	*lã* 'wool'
	b.	[sĩ]	*sim* 'yes'
	c.	[ˈkɐ̃.sɐ]	*cansa* '(he) tires'
	d.	[ɐ̃.ˈfi.biu]	*anfíbio* 'amphibian'

While phonology typically does not differentiate between nasalized and nasal vowels, Medeiros (2011) calls for such a distinction based on acoustic analyses of pairs like *pampa* 'grasslands' and *pama* (nonce form). She found that nasal vowels (which, for her would be those in (27) and (28) above) are more nasal than stress-induced nasalized vowels, such as those discussed in Section 4.2.1, as measured by nasal and oral airflow.

In terms of phonological analysis, there are basically three main types of arguments for nasal vowels in this context: those that posit a nasal vowel followed by a nasal consonant that is lenited or deleted (e.g., Reed & Leite, 1947; Trager, 1943a, 1943b), those that propose monophonemic representations of [+nasal] vowels (Hall, 1943; Hammarström, 1962), and a somewhat intermediate approach in which nasal vowels are considered to be biphonemic in nature, but not necessarily followed by a nasal consonant (Alameida, 1976; Battisti, 1998; Bisol, 1999, 2014; Câmara, 1953; Parkinson, 1983;Wetzels, 1997, etc.)

Based on analyses of the Paulista dialect, spoken in São Paulo, Reed and Leite (1947) propose that nasalized vowels are a sequence of an oral vowel followed by a nasal consonant that is reduced. They identify three nasal phonemes /m/, /n/, and /ɲ/, which surface as [m], [n] and [ɲ] corresponding to graphemes m, n and nh, respectively, in syllable-initial position. When describing the distribution of syllable-final nasals, however, this analysis becomes rather convoluted. According to Reed and Leite, reduced nasal off-glides [^{m}] and [^{n}] that (may) appear syllable finally correspond to the phonemes /m/ and /n/, respectively, and the phoneme /ɲ/ becomes [^{ŋ}] before /k/ and /g/. However, there is no evidence that reduced nasal segments that appear before homorganic stops are phonemically distinct. It would seem more logical to suppose that final nasals that are homorganic with following stops involve place assimilation of one nasal phoneme, as we saw in the analysis of Spanish nasals in coda.

There is another important observation, however, that stems from this study. In word final position, reduced palatal and velar nasals appear after front vowels ([sĩ^{ɲ}] *sim* 'yes') and back vowels (e.g., [ũ^{ŋ}] *um* 'a'), respectively. This is evocative of the notion of nasals sharing place features with vowels, as seen in 3.1. Here,

the reduced nasal segment could be argued to share coronal PA features with front vowels and dorsal PA features with back vowels. Note that the idea of vowels sharing place features besides [dorsal] is not unprecedented. One example of spreading of nuclear Coronal and Labial articulators is the case of Fula, in which Coronal and Labial features spread from the vowel to the glide in onset as in *yi*, *ye*, *wo*, *wu* (Paradis & Prunet, 1993). Another example can be seen in French, where a latent [m] can only be preceded by a round vowel, as in [parˈfœ̃] *parfum* 'perfume'/[parfyˈme] *parfumer* 'to perfume', and a latent [ɲ] can only be preceded by a front vowel, as in [ˈpɛ̃] *peint* '(he) paints'/[ˈpɛɲ] *peignent* '(they) paint' (Paradis & Prunet, 1993).

On the opposite side of the spectrum, the existence of minimal pairs like *lã* 'wool' and *lá* 'there' have prompted other linguists (Hammarström, 1962; Head, 1965 and others) to analyze nasalization in this context as part of the lexical representation of the vowel. For example, the difference between *lá* 'there' and *lã* 'wool' would be that the vowel in the latter is specified as [+nasal]. Hammarström (1962) notes the communicative irrelevance of the nasal off-glides in coda position, since they only appear before stops and even then, as a reduced transition segment that is homorganic with the following consonant. (Although recall that Reed and Leite (1947) find transitional segments word finally as well, as in ([sĩⁿe] *sim é* 'yes it is'.) Acoustic analyses of these transitional nasal segments show that they are significantly shorter than onset nasals (Almeida, 1976; Medeiros, 2011).

Perhaps the most problematic aspect of proposals that posit phonemic nasal vowels is that they cannot account for why syllables with nasal vowels are treated by the phonology as closed syllables (Bisol, 1999, 2014; Câmara, 1953; Wetzels, 1997). Câmara (1953) noted this in the distribution of the allophones of /ɾ/ in BP as well as in patterns of Vowel Fusion. Intervocalically, /ɾ/ is pronounced [ɾ] (e.g., [ˈpa.ɾɐ] *para* 'for'), while word initially and after closed syllables [x] is produced (e.g., [ˈxa.tu] *rato* 'rat', and [iz.xa.ˈɛw] *Israel* 'Israel'). After nasal vowels, /ɾ/ is pronounced as [x] (e.g., [ˈõ.xɐ] *honra* 'honor'), supporting the analysis of these syllables as closed.

Further evidence that these syllables should be analyzed as closed is the fact that Vowel Fusion does not occur with a combination of a final nasal vowel followed by an initial oral vowel (Wetzels, 1997). Vowel Fusion refers to the phenomenon in which word-final vowels contract with following initial vowels. The data in (29) are taken from Wetzels (1997).[9]

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9. Due to final vowel raising, only /i, a, u/ are found in unstressed final position. Word initial [ĩ] in *encabulado* and [i] in *escuro* stem from an optional rule of word initial unstressed vowel raising.

(29) a. [a + a] menina alegre menin[a]legre 'joyful girl'
b. [i + i] leque escuro > lequ[i]scuro 'dark fan'
c. [u + u] vejo usinas > vej[u]sinas 'I see factories'
d. [i + ĩ] assistí encabulado > assist[ĩ]ncabulado 'I watched shyly'

In BP, Vowel Fusion applies to sequences of identical vowels, typically when both are unstressed but also, less often, when the first is stressed and the second is unstressed. (29d) shows that nasalization does not impede Fusion when it is the second vowel that is nasalized. However, when the nasal vowel is word-final, fusion rarely applies.

(30) [ã + a] irmã adoravel *irm[ã]doravel 'adorable sister'
[ĩ + i] cupim inofensivo *cup[ĩ]nofensivo 'harmless termite'
[ũ + u] jejum urgente *jej[ũ]rgente 'urgent fasting'

In addition to Fusion between identical final and initial vowels, word-final oral vowels can diphthongize with following word-initial vowels that are different; however, this is blocked when the word-final vowel is nasal.

Accounts that take a more intermediary approach analyze nasal vowels as a sequence of an oral vowel plus some nasal element (Alameida, 1976; Câmara, 1953; Parkinson, 1983; Wetzels, 1997), which finds support in acoustic analyses showing that nasal vowels have longer durations than nasalized vowels (Medeiros, 2011). These are often dubbed VN analyses, but the exact nature of the nasal mora varies in different accounts. Parkinson (1983) claims that nasal monophthongs are phonological diphthongs, and proposes a phonemic representation that includes two vocalic segments, the first one oral and the second nasal, i.e., [Ṽ] < /V+Ṽ/. A similar argument has been made for plural -ns sequences in Galician, a closely-related language spoken in northwestern Spain (Colina & Simonet, 2014). Parkinson cites support for this type of representation from spectral data showing that nasal vowels and diphthongs show medial-final or final nasality. In actuality, the fact that nasal vowels are more nasal toward the middle and end would support any analysis that posits a nasal element, vocalic or not; however, the major obstacle for Parkinson's proposal is that, like the proposals for phonemic nasal vowels, it cannot account for why syllables with nasal diphthongs would act like closed syllables while those with oral diphthongs do not (e.g., [ma.ˈdej.ɾɐ], not *[ma.ˈdej.xɐ] for *madeira* 'wood').

Other VN accounts analyze nasal vowels as an oral vowel plus some kind of nasal mora (Alameida, 1976; Câmara, 1953; Wetzels, 1997). Câmara (1953) proposed a nasal archiphoneme N that appears in syllable codas, although Alameida (1976) criticizes Câmara for this because an archiphoneme complicates the inventory of nasal consonants (/m/, /n/, /ɲ/ and a fourth segment N). In a similar vein,

Wetzels (1997) hypothesizes the lexical representation of nasal vowels as an oral vowel plus a nasal mora that is consonantal in nature. In this way, syllables with a nasal vowel have heavy rhymes, which is consistent with rules of Stress Placement in BP. When the penultimate syllable contains a nasal vowel, it can never be skipped over in stress assignment, which is consistent with the stress assignment in other types of heavy rhymes. Because the mora is considered consonantal, this analysis is also able to account for the distribution of /ɾ/ with regards to nasal vowels as well as the inability of nasal vowels to participate in Vowel Fusion.

An interesting aspect of Wetzel's analysis is that he is able to incorporate data related to nasalization before the palatal /ɲ/. As illustrated in (31), vowels are nasalized before the palatal nasal, regardless of stress.

(31) [ˈlĩɲo] *linho* 'linen'
[pẽˈɲasku] *penhasco* 'cliff'
[veɾˈgõɲɐ] *vergonha* 'shame'
[veɾgõˈɲozu] *vergonhoso* 'shameful'

Wetzels (1997) claims that /ɲ/ in BP is a phonological geminate as evidenced by several factors. First, while [m] and [n] can be preceded by a branching rhyme (e.g., *alma* 'soul', *adorno* 'adornment') /ɲ/ cannot. Furthermore, /ɲ/ does not occur word initially except in loanwords (Wetzels, 1997), and as previously mentioned, epenthesis is often attested in these forms [ĩˈɲɔki]. Interestingly, the other palatal sonorant, /ʎ/, displays the same distribution. /ʎ/ does not occur in initial position, with the obvious exception of clitics *lhe* 'to him/her/you', *lhes* 'to them/you' *lha* (< lhe + a) and *lho* (< lhe + o), which are rarely used in colloquial BP. In loanwords, such as *lhama* 'llama', borrowed from Spanish *llama*, an epenthetic [i] is often produced initially [iˈʎẽmɐ]. According to Wetzels, the reason that a palatal sonorant cannot occur after a branching rhyme is that its own first half is occupying the second position of that rhyme, rendering it maximally filled. In this way, nasalization of a vowel before /ɲ/ follows from the nasal portion of the geminate occupying the coda of the same syllable.

After examining the distribution of nasal consonants and the patterns of vowel nasalization in Spanish and BP, and considering the close genetic relationship of the two, it seems natural to propose that the same basic process is involved in both, namely lenition of final nasal consonants. In terms of Terrell's (1975) steps of nasal weakening – (1) nasal place assimilation; (2) vowel nasalization; (3) nasal deletion/absorption – conservative dialects of Spanish could be said to be in stage (1), while radical dialects of Spanish are between stages (2) and (3), and BP is firmly in stage (3).

Goodin-Mayeda (2015) analyses nasalization and other processes that affect codas in BP as the result of a group of constraints on codas based on sonority.

due to epenthesis, making candidate e the winner with its minimal violation of UNIFORMITY.

This analysis still holds even in cases in which a transitional nasal segment is attested in BP. I would argue that transition segments that are homorganic with the following consonants are indeed just that, transition segments. Recall that Solé (1992) distinguishes between phonetic and phonological processes, arguing that overlap between velar aperture and a preceding vowel before a nasal consonant is not part of the programming instructions in Spanish, but rather due to mechanical demands of lowering the velum in time for the nasal. The same could be said for producing an oral stop after a nasal vowel. A homorganic transitional segment in this context is not evidence for a nasal consonant in the input but rather is an articulatory artifact due to the fact that oral articulators are moving at the same time as the velum is closing. Thus, a transitional nasal segment might occur if the oral articulators come together before the velum has entirely closed.

The truly interesting aspect of comparing Spanish and Portuguese nasals and nasalization is that these offer insights into synchronic variation and diachronic shifts with regards to nasal lenition, so that it is possible to imagine that at one time, Brazilian Portuguese nasals looked much like those of today's Cuban Spanish. It is for this reason that this particular language pairing is so felicitous for examining phonetic and phonological aspects of nasals and nasalization. Chapter 5 will present data from two perceptual experiments testing crosslinguistic perception of nasal consonant place and nasal vowel height in different contexts.

CHAPTER 5

Studies on the perception of nasals and nasalization in Spanish and Portuguese

This chapter presents the results of two studies on the perception of nasal place of articulation and nasal vowel height by speakers of different varieties of Spanish and Portuguese in order to examine predictions of some of the psycholinguistic and phonological models of speech perception and coarticulation that have been discussed in previous chapters.

5.1 Experiment 1: Nasal place perception

As we saw in Section 4.1.2, some linguists have argued that the perceptibility of nasal place of articulation is related to the amount of space in the vocal tract, with more forward-articulated nasals being more perceptible than those that are more back (Piñeros, 2011; Widdieson, 1997). Recall, for example, Piñeros' (2011) perceptual hierarchy for nasals with its ranking labial > coronal > palatal > dorsal. This hierarchy is somewhat in accordance with data-driven studies that test accuracy of nasal place perception; however, these studies also show that listeners rely more heavily on formant transitions rather than nasal antiresonances in order to make judgments on nasal PA. In addition, due to the fact that formant transitions are more perceptible in onset position, it has been shown that nasal place is perceived more accurately in onset than in coda (Herrera, 2002; Malécot, 1956; Ohde, Haley & Barnes, 2006 and others).

What remains to be seen is the role of a listener's native language and how that interacts with these other factors. Most studies test the perception of nasals that are phonemic in a listener's native language, which makes sense given the decline of perceptual abilities in the first year of life (Kuhl, 1992; Sundara et al., 2008; Werker & Tees, 1984a, 1984b and others). However, it is possible that some nasals are simply more salient than others, regardless of native language experience and/or syllable position. Recall that native language is not necessarily deterministic in how all sounds are perceived. For example, both Best's Perceptual Assimilation Model and Flege's Speech Learning Model predict that non-native sounds that are very unlike native sounds will actually be perceived more accurately than

non-native sounds that are close to (but not exactly like) an L1 category, because the more distinct the foreign sound is, the less likely it is to be assimilated to an L1 sound.

Experiment 1 tests the perception of four nasals with differing places of articulation (bilabial [m], alveolar [n], palatal [ɲ] and velar [ŋ]) by speakers of Spanish and English. Both languages have all four sounds, but not all are phonemic. As we have seen, Spanish has three nasal phonemes /m, n, ɲ /, with the velar nasal occurring in assimilation contexts, while English has /m, n, ŋ/ with a palatal (or palatalized) nasal in words like can<u>y</u>on. The goal is to examine the role of a listener's L1 in the perception of nasal place and how this interacts with syllable position. Recall that Malécot (1956) assumes that English listeners perceive the velar nasal more accurately in coda, because the velar nasal only occurs in coda in English; however, the velar nasal is crosslinguistically rare in onset position, because it is the most vowel-like of the nasals (Ohala, 1975; Widdison, 1997). This experiment seeks to address the following questions:

(RQ1) Are some nasal consonants easier to perceive regardless of native language and/or syllable position?
(RQ2) Are some nasal consonants easier to perceive in some syllable positions than others?
(RQ3) Does a listener's L1 phonemic inventory determine the nasals that are most accurately perceived?

5.1.1 Methodology and procedure

The stimuli for Experiment 1 consisted of tokens of [m], [n], [ɲ] and [ŋ] naturally spoken in isolation and in CV and VC contexts. The speaker is a phonetically-trained, bilingual female who produced each CV and VC syllable with English-like vowels for the English stimuli and Spanish-like vowels for the Spanish stimuli. The murmurs were produced in isolation, but then also cut so as to include no formant transitions whatsoever (from releasing the articulation, for example), and all were 600 ms long.

Subjects were tested using a forced-choice identification task in which they listened to the stimuli and chose among the four options, which were presented in orthographically appropriate ways for each language and included example words (e.g., ny as in *canyon*, ng *como manga*). The advantage of using a forced-choice task is that the limited (and experimentally fixed) set of possible responses presents a lighter memory load for listeners (Beddor & Gottfried, 1995). While an open-set identification may provide more information on how listeners perceive the stimuli, the general practice has been to provide subjects with linguistic labels,

the assumption being that any auditory sensitivities that might not be evident with limited response choices are only relevant insofar as they enable listeners to choose the appropriate response (Beddor & Gottfried, 1995).

The program Praat was used to run the experiment and record responses. For each condition, Praat was programmed to counterbalance the stimuli across five blocks so that each subject heard each stimulus a total of five times during the experiment. In addition, the order of the CV and VC sets was counterbalanced across subjects. The isolated murmur condition was always presented last, as it was the most difficult. No feedback regarding the subjects' performance was given during or after the experiment.

Subjects were tested in groups of 1–3 in a quiet environment. The experimenter gave instructions in the subjects' native language indicating that they would hear a series of syllables and that they should focus on the consonant they heard. They were instructed to use a mouse to click on the appropriate response indicating which consonant they thought best fit what they heard. The pace of the experiment was determined by how quickly the subject chose each response, but the entire experiment generally took between 15 and 20 minutes. Each stimulus played automatically after the previous response had been chosen, with a period of 0.7 sec of silence preceding each token. Responses were coded and recorded in table format by Praat.

The experiment was administered to monolingual speakers of English (n = 12; 8 female, 4 male) ages 19–38 (average 25 years) and functional monolingual speakers of Mexican Spanish (n = 12; 6 female, 6 male) ages 35–56 (average 44 years). All subjects were native speakers of their prospective languages and reported no hearing impairments. Because the subjects were recruited in the Houston, TX area, it was difficult to find true monolinguals of Spanish who had absolutely no exposure to English; however, the Spanish subjects all arrived to the U.S. after age 18 (range 18–37, mean 26.5) and were unable to respond in English to simple questions like 'What day was it yesterday?' and 'What is your favorite kind of food?' In addition, they live and work in an area in which they use Spanish for basically all of their daily activities.

Accuracy scores were analyzed using mixed ANOVA (repeated measures, general linear model) with two within-subjects factors (nasal place: m, n, ɲ and ŋ; context: isolation, onset, coda) and one between-subjects factor (English vs. Spanish listener).

5.1.2 Results

Tables 1 and 2 contain the proportion that subjects reported hearing each nasal. In each table, Row 1 indicates stimuli while Column 1 indicates responses. For example, in Table 1a, when presented with [m] in isolation, English subjects reported hearing [m] 78% of the time. Likewise, when presented with the [ŋ] murmur, they reported hearing [ŋ] only 12%. In fact, for English listeners, regardless of which nasal murmur was played, [m] and [n] were the most often selected.

Statistical analysis indicates that context was significant, $F(2, 44) = 61.34$, $p < .01$, as were nasal place, $F(3, 66) = 78.20$, $p < .01$, and language, $F(1, 22) = 973.55$, $p < .01$. Furthermore, interactions were found for context*language, $F(2, 44) = 10.09$, $p < .01$, context*nasal place, $F(6, 132) = 20.73$, $p < .01$, and context*nasal place*language, $F(6, 132) = 2.36$, $p = .03$.

The results of Experiment 1 support that the perception of nasals is influenced by place of articulation, the context in which the nasal appears (in isolation

Table 1. English listeners' perception of nasal consonants in three contexts: (a) isolation, (b) CV and (c) VC. Values shown are percentages.

(a) murmur

	m	n	ɲ	ŋ
m	78	45	45	48
n	8	42	27	30
ɲ	8	8	13	10
ŋ	5	5	15	12

(b) onset

	m	n	ɲ	ŋ
m	91	16	4	32
n	3	56	29	26
ɲ	3	16	56	23
ŋ	2	12	10	19

(c) coda

	m	n	ɲ	ŋ
m	58	7	10	11
n	14	46	35	19
ɲ	14	31	32	21
ŋ	14	17	23	49

Table 2. Spanish listeners' perception of nasal consonants in three contexts: (a) isolation, (b) CV and (c) VC. Values shown are percentages.

(a) murmur

	m	n	ɲ	ŋ
m	82	60	50	55
n	8	25	27	28
ɲ	2	7	3	2
ŋ	8	8	20	15

(b) onset

	m	n	ɲ	ŋ
m	93	12	1	17
n	1	76	1	17
ɲ	0	3	93	23
ŋ	6	8	4	43

(c) coda

	m	n	ɲ	ŋ
m	64	4	2	9
n	16	68	25	25
ɲ	7	9	41	11
ŋ	13	18	32	55

vs. onset vs. coda) as well as the phonemic inventory of the listener's L1. Listeners of both Spanish and English had higher accuracy for the perception of [m] in almost all contexts, the only exception being Spanish speakers' slightly more accurate perception of [n] in coda vs. other conditions. In addition, nasals in the onset were generally perceived more accurately than in the coda, and these in turn were more accurate than murmurs. The only exceptions to this generalization are isolated [m] and coda [ŋ]. For listeners of both languages, judgments of [m] in isolation were more accurate than in coda, but less accurate than onsets. In fact, there was a tendency for all nasal murmurs to be judged as [m], regardless of place of articulation. Note that non-[m] nasals in isolation were judged to be [m] over 50% of the time by Spanish listeners and nearly as often by English listeners. Differences between the language groups appear in the perception of [ɲ] and [ŋ] in the onset, which Spanish listeners perceived more accurately than English listeners.

These results are significant to the notion of perception of coarticulated speech, because they show that a listener's native language background interacts with other phonological factors, such as syllable position and phoneme inventory, in how accurately sounds are perceived. Given that accurate perception of coarticulation involves detecting the conditioning environment in order to factor it out, a logical question is whether some conditioning environments are more easily detected by listeners of some languages versus others. Experiment 1 speaks to the ability of listeners to perceive sounds in different syllable positions and provides evidence that perception in some environments is more likely to destabilize depending on the listener's L1. In other words, a listener's native language experience may affect his ability to reconstruct the intended pronunciation of the speaker. These results especially bear upon cases of coarticulation that are due to the effects of a coda, and which may involve the subsequent loss of said coda, such as vowel nasalization.

5.2 Experiment 2: Perception of nasal vowel height

Experiment 2 examines the role of native language experience in compensating for the acoustic effects of nasal coarticulation with the preceding vowel in different contexts. Recall that Krakow et al. (1988) found that English listeners were able to perceptually compensate for the effects of nasal coupling only when there was an adjacent coda nasal to which nasalization could be attributed (e.g., b[ẽ]nd); when there was no nasal present (e.g., b[ẽ]d), the acoustic effects of nasalization were perceived as a difference in vowel height. What remains unclear is whether English speakers were able to "undo" the effects of nasalization in the nasal context because that is the context in which nasalization occurs in English, or whether a speaker of any language would compensate for nasalization in this (and only this) context.

The patterns of nasalization in Spanish and Portuguese present an opportunity to investigate whether the ability of listeners to perceptually compensate for the effects of nasal coupling is related to their native language or not. If so, one would expect context effects on the perception of nasal vowel height to be affected by cross-linguistic differences. Note that the question is not whether listeners perceive the difference between oral and nasal vowels, as this distinction seems to be one that speakers of all languages are able to make (Beddor & Strange, 1982). The question here is whether or not listeners are able to accurately perceive the *height* of a nasal vowel as compared to its oral counterpart in certain contexts. Experiment 2 seeks to address the following questions:

(RQ4) Does native language background play a role in the perception of nasal vowel height?

(RQ5) What, if any, interaction exists between native language and context effects (i.e., whether or not there is a nasal vowel present) on the perception of nasal vowel height?

The first question has been peripherally addressed by John Ohala's work on diachronic shifts in nasal vowels, but it has not been directly tested until Goodin-Mayeda (2011, 2012), and is further examined here. The second question is touched upon in Krakow et al. (1988), but their main focus was the role of context on the perception of nasal vowels, and not necessarily the interaction of this with language experience.

Experiment 2 directly investigates the role of language experience in the perception of nasal vowel height by testing listeners of Peninsular Spanish (PS), Cuban Spanish (CS) and Brazilian Portuguese on their perception of nasal vowel height in different contexts. Recall that BP has both contrastive and allophonic vowel nasalization (Wetzels, 1997), while PS vowels are subject to very little nasal overlap with the vowel regardless of context (Solé, 1992). CS has been described as somewhere between these two extremes: vowels are nasalized before tautosyllabic nasal consonants, and these nasal consonants may be subsequently lenited or deleted (Terrell, 1975). Given these patterns of production, specific predictions can be made with regards to BP, PS and CS listeners' perception of nasal vowels in certain contexts, which are illustrated in Figure 8.

Language-dependent

	Accurately perceive contextual nasal vowel height	**Accurately perceive non-contextual nasal vowel height**
Peninsular Spanish	X	X
Cuban Spanish	✓	X?
Brazilian Portuguese	✓	✓

Non-language dependent

	Accurately perceive contextual nasal vowel height	**Accurately perceive non-contextual nasal vowel height**
Peninsular Spanish	✓	X
Cuban Spanish	✓	X
Brazilian Portuguese	✓	X

Figure 8. Comparing predictions of hypotheses that coarticulatory effects on perception of nasal vowel height are language dependent vs. non-language dependent

If a listener's L1 determines the types of coarticulatory influences that can be compensated for, then BP listeners should accurately perceive contextual and non-contextual nasal vowels while PS listeners should demonstrate a perceptual shift in vowel height in both nasal conditions. CS listeners would be expected to accurately perceive nasal vowel height in the contextual nasal condition and probably not in the non-contextual nasal context; however, given that deletion of final nasals is attested in Cuban Spanish, it is possible that like Portuguese listeners, they may compensate for nasalization in the latter condition as well. In contrast, if native language experience does not interact with context effects for perceiving nasal vowel height, then listeners of all three varieties should demonstrate the same pattern of perception, which, incidentally, is the same pattern that Krakow et al. (1988) found for English listeners.

5.2.1 Methodology and procedure

Experiment 2 is modeled after the experiment in Krakow et al. (1988), but because the subjects of this study are from different language backgrounds, non-words that conform to the phonotactic constraints of Spanish and Portuguese were chosen to test the /u/–/o/ distinction. The stimuli for this experiment consisted of synthesized non-words in which vowel height and degree of nasalization were manipulated. The choice of synthetic or modified natural speech tokens (or hybrid tokens) depends largely on the type of distinction being examined, and each comes with its own advantages and disadvantages (Werker & Lalond, 1988). Editing techniques such as deletion, vowel shortening/lengthening, transposition of portions of the acoustic signal, etc. are easily achieved using computer software, and can be used to modify natural speech tokens. Unfortunately, natural speech tokens, unlike synthetic tokens, do not permit total control over variations in the acoustic signal. While synthetic stimuli may not succeed in representing some perceptually relevant properties of the signal (Beddor & Gottfried, 1995), a positive correlation between perception of natural and synthetic speech has been attested (Werker & Lalond, 1988; Yamada & Tohkura, 1992).

Stimuli were synthesized using the program Synthworks®, a KLATT-based parallel speech synthesizer with up to 48 manipulable parameters. The interface of the program is a spreadsheet in which frequency/amplitude values for each parameter, listed in rows, can be manually entered across columns, which correspond to time. Natural tokens of the endpoints [gos], [gus], [gõns], and [gũns] were recorded by the researcher and provided the basis for synthesizing the basic shape of the stimuli, including parameters such as segment length, formant transitions for

Table 3. *F*1, *F*2 and *F*3 frequency values for seven vowel configurations

Stimulus	*F*1 (Hz)	*F*2 (Hz)	*F*3 (Hz)
1 [u] endpoint	225	750	2800
2	262	750	2800
3	300	750	2800
4	337	750	2800
5	375	750	2800
6	412	750	2800
7 [o] endpoint	450	750	2800

initial /g/, voicing cues, total utterance length, etc.[10] The fundamental frequency (*F*0) of all utterances began at 100 Hz and fell linearly to 85 Hz over the first 200 ms of the utterance and remained there until the end of the utterance. The vowels were generated by first synthesizing the endpoints /u/ and /o/ and then manipulating the first formant (*F*1) in equal increments to synthesize five intermediate shapes in order to ultimately create a seven-step vowel continuum from /u/ to /o/, as shown in Table 3. While naturally spoken [u] and [o] also differ (minimally) in terms of *F*2, which corresponds to vowel backness, this was not manipulated in order to ensure that subjects made judgments based only on vowel height, and not backness. The seven vowel heights were used in the three synthetic continua: oral [gos]–[gus], contextual nasal [gõns]–[gũns] and non-contextual nasal [gõs]–[gũs].

For the two nasal conditions, five degrees of nasalization were synthesized in order to account for possible speaker variability in the degree to which vowels must be nasalized to be perceived as nasal and also to permit examination of this effect on perceived vowel height. As we have seen previously, the primary cues for nasal coupling are reduction in amplitude of the first formant, an increase in *F*1 bandwidth of up to 60 Hz as well as the introduction of two nasal peaks, one at around 1000 Hz and another between 250 and 450 Hz (Chen, 1997). Nasalization was synthesized by introducing nasal poles and zeros at varying frequencies, depending on the level of nasalization. In addition, the amplitude of the first oral formant was decreased in the nasal continua. Nasality judgments have been correlated with the magnitude of *F*1 amplitude reduction (Chen, 1997; House & Stevens, 1956), thus, the lower and middle degrees of nasalization were synthesized with a 6 dB, 7 dB and 8 dB reduction in *F*1 amplitude, respectively, and the highest two degrees of nasalization were synthesized with a reduction in *F*1

10. It should be noted that while Spanish allows syllables of the shape CVns (e.g., cons-tan-te), there are very few, if any, monosyllabic words with this shape. A possible exception is the pronunciation [tons] for 'entonces.'

Table 4. Frequency of nasal pole-zero pairs and amount of amplitude reduction of oral *F*1 (AF1) for all degrees of nasalization

Degree of nasalization	fnp (Hz)	fnz (Hz)	AF1 (dB)
a	250	310	–5
b	265	365	–6
c	290	420	–7
d	305	475	–8
e	315	528	–8

Table 5. Three stimulus sets synthesized for identification task

Stimulus sets	
1	Oral condition [gus]–[gos] 7 tokens (seven vowel heights)
2	Contextual nasal condition [gũns]–[gõns] 35 tokens (seven vowel heights X five degrees of nasal coupling)
3	Non-contextual nasal condition [gũs]–[gõs] 35 tokens (seven vowel heights X five degrees of nasal coupling)

amplitude of 8 dB. Finally, the amplitude of the nasal pole increased from 30 dB to 60 dB over the duration of the (nasal) vowel.

The stimuli consisted of 77 utterances in all, divided into three sets as shown in Table 5. After all stimuli were synthesized, they were converted into wav audio file format.

Subjects were administered a forced-choice identification task via Praat in which they listened to the stimuli and chose whether they heard 'o' or 'u.' For each stimulus set, Praat was programmed to counterbalance the stimuli across ten blocks so that each subject heard each stimulus a total of 10 times during the experiment. In addition, the order of the stimulus sets was counterbalanced across subjects. Subjects heard the stimuli over headphones, and used the mouse to click on the computer screen to indicate whether the vowel they had heard was [u] or [o]. To avoid the possible confound of orthography, the entire word was not spelled out. These response choices posed no difficulty for the subjects, because both Spanish and Portuguese orthography are somewhat more phonetically reliable as compared to languages like English, for example, and because [u] and [o] correspond to orthographic 'u' and 'o' respectively in both languages. Each condition was preceded by a short practice session. The stimuli for the practice sessions contained the endpoint vowel heights as well as the next closest vowel height for each set, in other words, vowel heights 1, 2, 6, and 7 from the seven-step vowel

continuum. No feedback regarding the subjects' performance was given after the practice sessions.

Subjects were tested in groups of 1–4 in a quiet environment. The experimenter read instructions in the subjects' native language which indicated that they would hear a series of "invented" words. They were instructed to use a mouse to click on the 'u' button or 'o' button on the screen to indicate which vowel they thought best fit the vowel they heard. The experiment generally took between 50 and 70 minutes, and each stimulus played automatically after the previous response had been chosen, with a period of 0.7 sec of silence preceding each token. Responses were coded and recorded in table format by Praat.

The experiment was administered to 40 total participants, and data were analyzed for 36 listeners of Peninsular Spanish (n = 12; 7 female, 5 male) ages 18–33, Cuban Spanish (n = 12; 8 female, 4 male) ages 19–50, and Brazilian Portuguese (n = 12; 8 female, 4 male) ages 20–44. Two PS listeners were excluded, because they reported being Catalan dominant, and two CS listeners were excluded due to investigator error resulting in loss of some or all data. All subjects were native speakers of their respective languages and reported no hearing impairments.

It has been noted that as models of speech perception become more refined in terms of their predictions regarding the influence of linguistic experience, so must the descriptions of the relevant languages become increasingly detailed (Beddor & Gottfried, 1995). At the same time, the possible influence of dialectal variation must also be considered. As languages spoken in many parts of the world and often in contexts of language contact, Spanish and Portuguese are subject to a wide range of dialectal variation. For this reason, an effort was made to ensure that the subject groups were as homogenous as possible in terms of their linguistic background, both L1 and L2. The Peninsular Spanish (PS) subjects were all listeners from Barcelona, Spain. While it may seem strange to choose to conduct an experiment of this sort in an area of bilingualism, the Solé (1992) experiment measuring Spanish speakers' production of vowels before tautosyllabic nasal consonants was carried out in this region, so we can be relatively sure of the subjects' linguistic input, at least in terms of nasal vowels in Spanish, because production data from the same region (and actually, the same lab) as the current study exist. In addition, although Catalan phonology is very different from that of Spanish, experience with Catalan should pose no problems for studies involving nasalization of vowels since, at least in this respect, Catalan and Spanish are quite similar (Wheeler, 2005). However, to err on the side of caution, as previously mentioned, only subjects who reported being Spanish dominant were considered.

The CS subjects were recruited in Houston, TX. All subjects moved to the US after age 20 with a mean length of residence of five years (range 5 months – 12 years). Subjects reported using Spanish on a regular basis in their daily lives, and

judged their English proficiency to be advanced (n = 5), intermediate (n = 2), basic (n = 3) and none (n = 1).

BP subjects were Brazilians living in Los Angeles and had learned English as adults. All reported using Portuguese on a regular basis. Subjects were from Porto Alegre (n = 2), São Paulo (n = 3), Rio de Janeiro (n = 3), Minas Gerais (n = 2), Recife (n = 1) and Maceió (n = 1).

A regression analysis was used to calculate the 50% crossover point from /u/ to /o/ for each subject's judgments in each continuum. Crossover values were based on a scale of 1 to 7 corresponding to the seven synthesized vowel heights. 1 corresponds to the /u/ endpoint ($F1$ frequency of 225 Hz) and 7 corresponds to the /o/ endpoint ($F1$ frequency of 450). Thus, at these endpoints, subjects reported hearing [u] and [o], respectively, most of the time, and the regression calculated the point at which they chose [u] half of the time and [o] half of the time, usually somewhere between tokens 3 and 5. The idea is that dissimilar crossover values between continua correspond to perceived vowel height differences. These crossover values were then used for statistical analyses. A repeated-measures, mixed ANOVA (general linear model) was performed with one within-subjects factor (condition: oral vs. contextual nasal V vs. non-contextual nasal V) and one between-subjects factor (L1: PS, CS or BP).

5.2.2 Results

Statistical analyses indicate a significant effect of condition, $F(1.62, 53.55) = 9.25$, $p < .01$, and language group, $F(2, 33) = 7.37$, $p < .01$, on the perception of vowel height, but there was no interaction between the two, $F(3.25, 53.55) = 1.56$, $p = .21$. Note that Greenhouse-Geisser F values were used to calculate significance for the within-subjects factor, because sphericity could not be assumed ($p = .01$). Table 6 illustrates the 50% crossover values for all three language groups.

Figure 9 illustrates PS speakers' rate of /u/ responses in each condition. Here the percentage of /u/ responses as a function of vowel height collapsed over degree

Table 6. Mean crossover values for PS, CS and BP listeners corresponding to the identification results of the [gus]–[gos], [gũns]–[gõns] and [gũs]–[gõs] continua

Crossover values across language groups			
Condition	PS	CS	BP
[gus]–[gos]	3.90	4.12	3.77
[gũns]–[gõns]	4.71	4.79	3.86
[gũs]–[gõs]	4.53	4.87	3.95

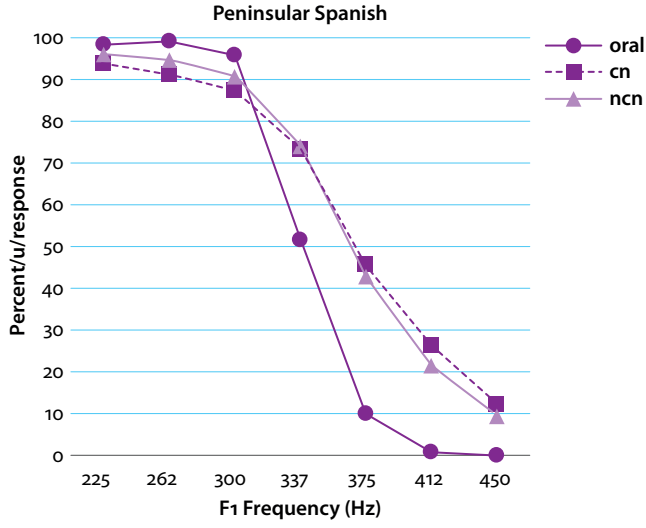

Figure 9. PS identification responses to the oral [gVs], contextual nasal [gṼns], and non-contextual nasal [gṼs] continua for Spanish speakers

of nasalization (for nasal stimuli) is shown. The figure represents the pooled responses of all 12 subjects.

As the figure illustrates, PS speakers reported significantly more /u/ responses in both nasal conditions as compared to the oral condition. This corresponds to a perceptual raising effect such that PS speakers interpreted the coarticulatory effects of nasalization as a difference in vowel height. Thus, PS speakers were unable to perceptually factor out the effects of nasalization in either nasal condition, regardless of whether there was an adjacent nasal consonant to which nasalization could be attributed.

The CS results are shown in Figure 10. Contrary to either hypothesis proposed in Figure 8, CS listeners also reported more /u/ responses in both nasal conditions than in the oral condition. Post hoc Bonferroni correction confirms that BP listeners performed significantly differently from PS and CS groups (p = .03 and p < .01, respectively), but PS and CS groups did not significantly differ from each other (p = 1.00).

Figure 11 depicts the results of the BP group. In contrast to both the PS and the CS results, the number of /u/ responses for the BP listeners was not significantly different in any context. These results support that in both nasal contexts, BP listeners were able to perceptually "undo" the effects of nasalization in order to accurately assess the height of the vowel. Figure 11 shows the rate of /u/ responses across vowel height frequency values in each condition for BP listeners.

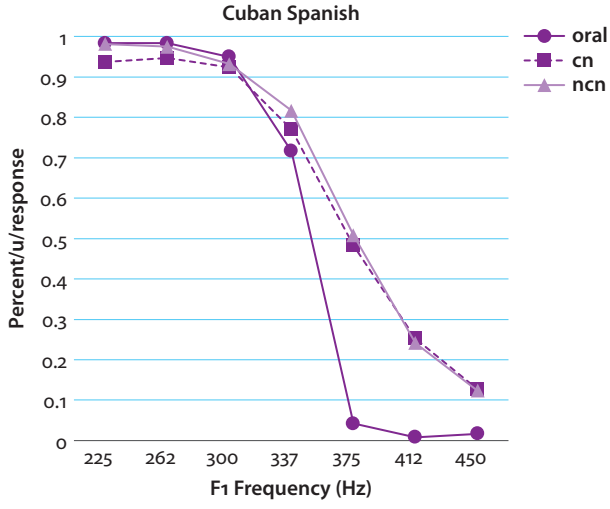

Figure 10. CS identification responses to the oral [gVs], contextual nasal [gṼns], and non-contextual nasal [gṼs] continua for Spanish speakers

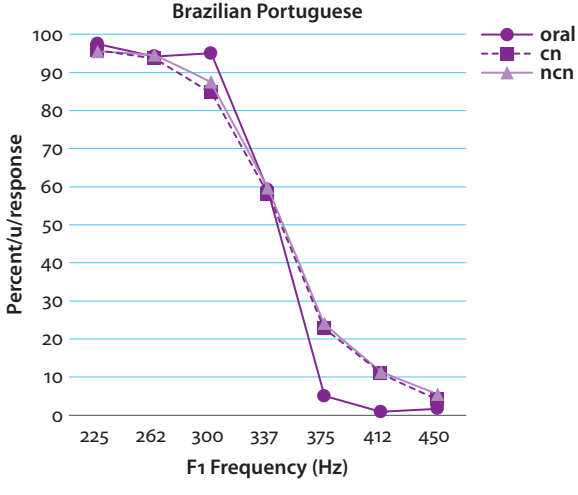

Figure 11. BP identification responses to the oral [gVs], contextual nasal [gṼns], and non-contextual nasal [gṼs] continua for BP speakers

In order to examine the effect of degree of nasalization, a repeated-measures ANOVA adding level of nasalization as a within-subjects factor was performed. Results reveal that degree of nasalization is also a significant factor $F(1.45, 2.90) = 22.29$, $p < .01$ and that there is an interaction with language group $F(2.90, 47.88) = 3.48$, $p = .02$ (Again, note the use of Greenhouse-Geisser F values due to lack of

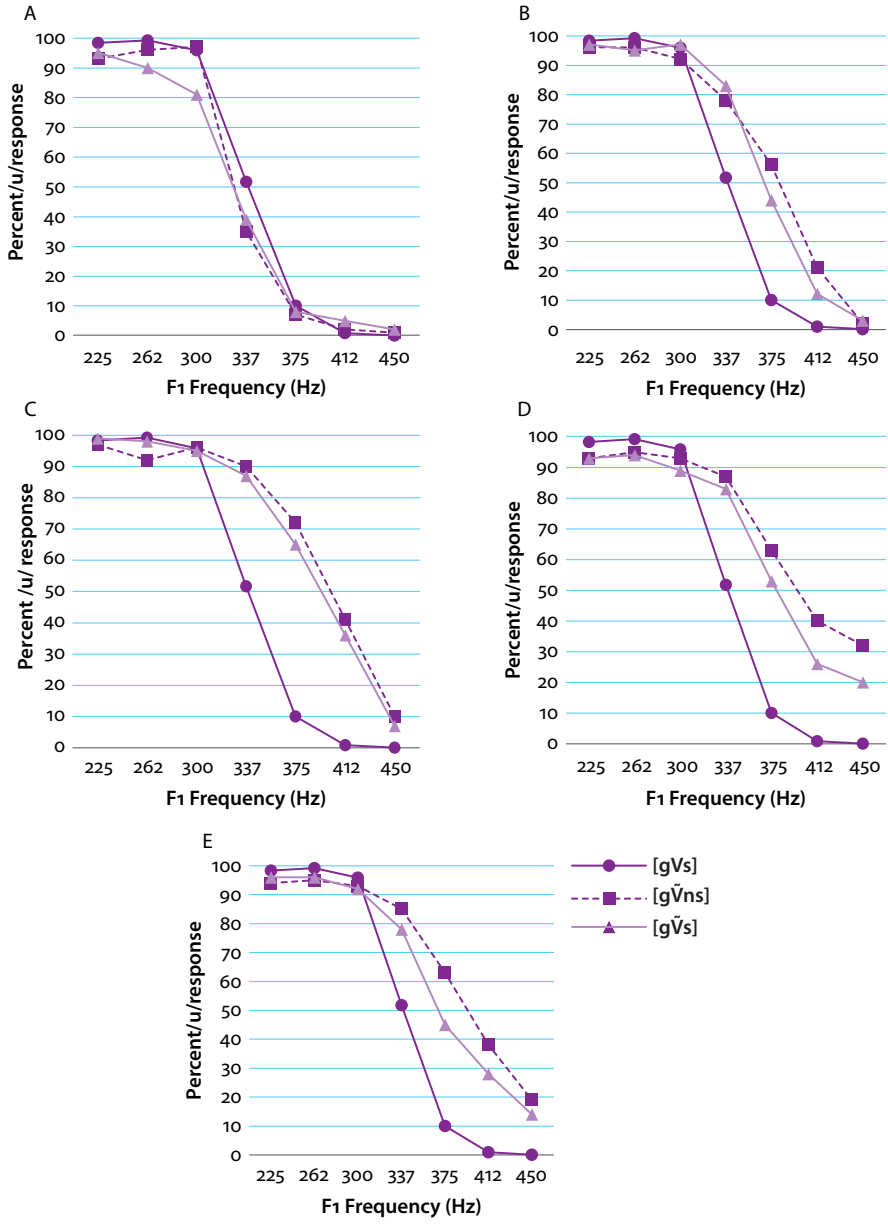

Figure 12. PS listener identification functions for contextual [gṼns] and non-contextual [gṼs] nasal continua over five degrees of nasal coupling from slight (level A) to heavy (level E) for PS group. The identification results for the oral [gVs] condition have been reproduced in each graph for purposes of comparison

sphericity in the data). Figure 12 shows PS listener identification results for all five degrees of nasal coupling (A–E where A is the lowest degree of coupling and E is the highest degree) compared to oral judgments. Here the results of the oral continuum have been replicated in each graph for purposes of comparison. As the figure illustrates, nasalization is perceived as a difference in vowel height in both nasal contexts across all degrees of nasalization.

Again, a post hoc Bonferroni correction reveals that the BP group results were significantly different from the PS and CS results (p =.02 and .02, respectively), but that PS and CS groups did not significantly differ from each other (p = 1.00).

To summarize the results shown here, PS and CS speakers perceived /u/ more often in both nasal conditions, which indicates that they were unable to perceptually factor out the coarticulatory effects of nasalization on the vowel regardless of whether such effects were attributable to an adjacent tautosyllabic nasal consonant. BP speakers, however, were able to factor out the coarticulatory effects of nasalization in both nasal conditions across all levels of nasalization. This was somewhat surprising in the case of the Cuban Spanish speakers, since nasalization in closed syllables ending in a nasal is supposedly more prominent in this dialect, and even nasal consonant deletion has been attested. It was hypothesized that CS speakers would accurately perceive nasal vowel height at least in the CN context, if not both nasal contexts; however, this was not borne out by the data.

5.3 Discussion

As we saw in 5.1.2, the general trends in perception of nasal place of articulation were similar for English and Spanish listeners. These results corroborate what Malécot (1956) found for English listeners, but only partially confirm Herrera's (2002) result for Spanish listeners. Like Herrera, Spanish listeners here perceived [m] in onset more accurately than the other nasals; however, unlike Herrera, who found that Spanish listeners perceived [n] most accurately in isolation, Spanish listeners here perceived [m] most accurately in this condition. This could be due to differences in methodology: while Herrera spliced the murmurs from naturally spoken words, the murmurs here were produced in isolation and were articulated for a longer duration than a typical nasal in situ.

There were some notable differences between English and Spanish listeners that deserve attention. First, while both English and Spanish listeners perceived [m] most accurately in the murmur and onset conditions, Spanish listeners perceived [n] most accurately in the coda (although only marginally so compared to [m]). Furthermore, Spanish listeners' perception of coda [n] was more accurate than that of English listeners. This follows from simple frequency effects, since [n]

occurs more commonly in absolute final position in Spanish, since it is the only nasal allowed in that position (in Mexican Spanish, at least).

In addition, Spanish listeners' judgments of [ɲ] and [ŋ] in onset were more accurate than the English listeners'. This is not surprising in the case of [ɲ], since the palatal nasal is phonemic in Spanish and not English, but how do we explain that Spanish listeners perceived onset [ŋ] with 43% accuracy, while English listeners' judgments were at chance levels, especially given that this sound is phonemic in English? I suggest that this result is due precisely to the fact that [ŋ] is not contrastive in Spanish. Following the Perceptual Assimilation Model (Best, 1995), if [ŋ] were assimilated as a Category Goodness fit, it would be relatively perceptible. Recall that PAM predicts that contrasts in which one sound is a good match to an L1 category and another is a poor match to an L1 category are the most easily perceived (besides those sounds that are not assimilated at all, as in the case of English listener perception of Zulu clicks). In this case, [m], [n], and [ɲ] are good/perfect matches to Spanish /m/, /n/ and /ɲ/, respectively, and thus [ŋ] may stand out as a poor match to any of these three. For this reason, its perceptibility in onset can be explained by its "otherness."

Likewise, the depressed accuracy of onset [ɲ] by English listeners (as well as onset [n], for that matter) could be explained by Single Category assimilation of both [n] and [ɲ] to English /n/, since these are articulatorily (and presumably perceptually) close to each other. In Single Category assimilations, two phones are equally good fits to one L1 category, and this type of assimilation results in the poorest discrimination of all. Note that there is evidence for this kind of confusion between /n/ and /ɲ/ in synchronic and diachronic facts of Spanish. Many instances of /ɲ/ in Modern Spanish stem from the sequence n+palatal in Latin (*vīnea* > *vinia* > *viña* 'vineyard'). In addition, it has been observed that Spanish speakers in the US tend to produce /ɲ/ as [n] especially before front vowels (Schwegler & Kempf, 2009), as in [ˈni.njo] for *niño* 'child'.

Finally, both groups performed similarly on coda [ŋ], despite the fact that this is phonemic only in English. There are two possible explanations for this. First, as just argued, it is possible that [ŋ] distinguished itself among the other coda nasals for the Spanish listeners by not easily assimilating to any L1 category; however, it is also possible that the particular Spanish listeners tested here have had enough exposure to English words, especially gerunds ending in -ing, that they recognize [ŋ] in coda.

Let us now focus on the perception of nasal vowel height by listeners of Peninsular Spanish, Cuban Spanish and Brazilian Portuguese. The results indicate that both Spanish groups resolved the coarticulatory effects of nasalization in both nasal contexts as differences in vowel height; that is, they were unable to perceptually compensate for nasalization even when there was a nasal consonant present. The

BP group, however, was able to do so in both nasal contexts. Interestingly, the PS and BP groups performed as expected under a language dependent hypothesis; however, the CS listeners performed in line with the PS listeners, which was not anticipated under either language dependent or non-language dependent hypotheses. The fact that CS listeners were not able to perceptually compensate for the acoustic effects of nasalization points to the possibility that perhaps the degree to which CS speakers nasalize (in terms of velar aperture, not frequency) is not as significant as anecdotal observation might suggest, and certainly not as significant as the degree of velar aperture involved in BP nasal vowels. As we have seen, Kochetov and Colantoni (2011) show via electropalatography that Cuban speakers assimilate preconsonantal nasal consonants more than impressionistic analyses have suggested, however Kochetov and Colantoni only tested three Cuban speakers. Therefore, further instrumental analysis of the velopharyngeal dynamics of CS nasalization is warranted.

Harmonizing the results of Experiment 2 with those of Krakow et al. (1988) supports the hypothesis that a listener's native language background, as well as the context in which a nasal vowel is produced, are both important in determining how a listener perceptually compensates for the acoustic effects of nasalization. Specifically, a listener's native language influences the contexts in which she will be able to tease apart the effects of coarticulation, at least in the case of nasal coupling.

Why then did Mann (1986) find that Japanese listeners were still sensitive to coarticulatory effects of /l/ and /ɹ/ on the /d/–/g/ distinction, in spite of their lack of experience with these sounds and their inability to distinguish between stimuli containing /l/ and /ɹ/? There are several possible reasons for the disparity between Mann's conclusions and those reported here. First, it is possible that some effects of coarticulation are simply different than others, and thus affect perception differently. This may seem like an ad hoc explanation, but let us consider the nature of nasal coupling. While coarticulatory influences on consonants tend to occur at segment edges, the coarticulatory effects of nasalization are realized simultaneously during the articulation of the vowel. For example, the coarticulatory effects of /l/ on adjacent stop consonant place of articulation consist of neighboring formant transitions of /l/ affecting the perception of formant transitions related to the stop. In contrast, nasalization involves, among other acoustic consequences, the addition of nasal poles and zeros *simultaneously* during the articulation of the vowel. Indeed crosslinguistic instrumental measurements of nasal(ized) vowels show that nasalization can be present from the onset of the vowel (Solé, 1992), and is present at least during the latter half of the vowel's duration (Almeida, 1976; Medeiros, 2011; Solé, 1992). In this way, the task of segmenting phones along coarticulatory lines (as we saw in a vector analysis [Goldstein & Fowler, 1986]) for the näive listener who does not have experience with the acoustic effects of nasalization may

be more difficult, as the spectral consequences of nasalization may be less salient as compared to changes in formant transitions at the segment edge.

It is also possible that Japanese speakers' experience with other, similar sounds in their language aided them in untangling the effects of coarticulation of /l/. Both Japanese and English speakers demonstrated a similar shift in perception of the /da/–/ga/ continuum when preceded by /s/ as opposed to /ʃ/ (Mann, 1986). Both /l/ and /s/ are similar in that they are produced with the tongue in a relatively forward position, while /ɹ/ and /ʃ/ are produced with a more retracted tongue position. It is possible that Japanese speakers' sensitivity to the coarticulatory effects of /l/ are due to their experience with the coarticulatory effects of /s/. In this way, they are not factoring out entirely novel acoustic effects of coarticulation.

The effects of nasalization, in contrast, are unlikely to be similar to coarticulatory effects of other sounds in Spanish. For this reason, Spanish speakers could not make use of knowledge of similar coarticulatory influences of other sounds to compensate for the acoustic effects of nasal coupling in either nasal condition. The same is true for English speakers in the noncontextual nasal condition found by Krakow et al. (1988). Because the coarticulatory effects under consideration in the current study are acoustic effects associated only with nasalization (and not coarticulation with other types of segments), the results of this study provide stronger evidence that a speaker's native language determines the types of coarticulatory influences she is able to perceptually factor out and in which contexts. Therefore, the results of this study are not compatible with Mann (1986).

The current results are however compatible with other models of perception and coarticulation, such as vector analysis (Goldstein & Fowler, 1986) and Ohala's models of perception and language change. Recall that vector analysis supposes that listeners perceive speech along coarticulatory lines rather than temporal lines. Here Portuguese listeners succeeded in perceiving nasal vowels along coarticulatory lines in the contextual nasal condition. Spanish listeners, on the other hand, failed to recover the natural segmentation of the produced signal, resulting in inaccurate parsing of the speech signal. It is not clear how the results of the non-contextual nasal condition fit into this model, since there is no nasal segment present. That is, the model does not specifically address the notion of loss of conditioning environment, since technically that would no longer be a case of coarticulation. Vector analysis does however specify that listeners may fail to segment speech along coarticulatory lines if they fail to notice the conditioning environment or its role in coarticulation, which is consistent with Ohala's conception of hypocorrection. Recall from Section 3.1 that hypocorrection refers to a situation in which, for whatever reason, a listener is unable to factor out the effects of coarticulation, which results in a failure to reconstruct the speaker's intended

pronunciation, i.e., listener misperception. This study provides evidence that a listener's L1 may be a key factor in how they deal with coarticulated speech.

The role of the L1 in perceiving the effects of coarticulation has implications for the study of sound change. Ohala (1993) notes the parallels between phonetic variation and sound change. For example, allophonic nasalization of vowels before nasal consonants is the environment which most often results in distinctively nasal vowels via sound change (Ohala, 1993). In this process, an essentially oral vowel that precedes a nasal consonant eventually becomes nasalized. Then the nasal consonant that conditioned the vowel nasalization becomes weakened and is deleted (or absorbed). This, as we have seen, is exactly how Portuguese came to have distinctively nasal vowels (Latin *lanam* > BP *lã* 'wool'), and it seems to be in progress synchronically in radical dialects of Spanish. Sound change occurs when listeners "misperceive" the intended pronunciation of the speaker and this "misperception" leads to a change of norms (Ohala, 1981, 1986, 1990, 1993, 1996). Thus, if CS speakers do not parse nasalization from the vowel and attribute it of the following nasal, which seems to be the case based on the current results, then Cuban Spanish (and perhaps all radical dialects of Spanish) is particularly susceptible to vowel quality changes in nasal contexts.

Ohala also notes that sound changes that are the result of coarticulatory influences, be they assimilations or other effects resulting from speech production, often occur with the simultaneous loss of the conditioning environment. A change of this nature has already occurred in Portuguese. Low vowels /ɛ/, /a/ and /ɔ/ raise to mid [ẽ], [ɐ̃] and [õ], respectively, when they become nasal. An example of this is *tempo* 'time', which is pronounced with [ẽ] but derives from open [ɛ] in Vulgar Latin (cf. Spanish *tiempo*). Presumably, the loss/absorption of the nasal consonant that conditioned the nasalization of /ɛ/ resulted in the reanalysis of the height of the vowel in Portuguese.

How does hypocorrection come about in the first place? That is, how is it that listeners come to stop parsing the effects of coarticulation from the affected segment? Ohala (1993) points out that hypocorrection results when a listener does not notice the conditioning environment for some reason. This may be due to weakening of the conditioning segment, and/or, as our results support, the listener's lack of experience with coarticulation in such environments. Beddor (2012) proposes that a listener's perception grammar includes weighting of the coarticulatory cues for a given percept. In other words, listeners' attention to coarticulatory cues can be selective, and they may differ in the perceptual weight they assign to a given unit that carries cues. For example, listeners have been shown to use coarticulatory nasalization as a cue in judging the following consonant as oral or nasal (Fowler & Brown, 2000). While some listeners weigh the presence of the nasal consonant more heavily, regardless of the nasality of the preceding vowel,

other listeners weigh the nasality of the vowel more heavily. Beddor claims that listener variation in perceptual cue weighting follows from the fact that such cues *can* differ in production. Therefore, novel cue weighting by some listeners could result in hypocorrection (Beddor, 2012).

As is so often the case with scientific inquiry, we are left with as many new questions as we attempted to address from the outset. For example, given the patterns of perception attested by CS listeners here, do speakers of Cuban Spanish nasalize to the degree that impressionistic analyses have claimed for so many decades? Instrumental analyses of Cuban Spanish, or another velarizing/nasalizing dialect, would be valuable to ascertain the aerodynamics of velopharyngeal aperture in pre-nasal contexts. Specifically, it would be useful to know more about the timing of velum lowering and the level of airflow through the nasal cavity as compared to other Spanish dialects. Also, given the fact that CS listeners did not parse nasalization from the vowel in Experiment 2, instrumental analyses comparing the production of oral and nasal vowels in Cuban and/or other varieties of Spanish might indicate whether these are indeed more susceptible to vowel quality changes in nasal contexts.

Further insights into the perception of nasal place of articulation could be made by testing listeners of languages with different phonemic inventories and phonotactic restrictions. For example, comparing the results of Spanish and English listeners to those of Vietnamese listeners would be useful, since Vietnamese has four nasal phonemes /m/, /n/, /ɲ/ and /ŋ/, all of which occur in both onset and coda positions (Thompson, 1965). In addition, it would be interesting to test other language pairings with regard to nasal vowel height. A particularly interesting pairing might be Portuguese, Galician and Spanish, since Galician has also been argued to fit within the continuum of nasalization between Portuguese and Spanish (Colina & Simonet, 2014). In fact, given the fact that Galician has nasal vowels much like Portuguese (Sampson, 1999) as well as velarization (Colina & Díaz-Campos, 2006), it may serve as a more revealing comparison to Spanish and Portuguese than Cuban Spanish. In order to represent more of the continuum, it would also be interesting to test listeners of a contact variety like Galician Spanish.

CHAPTER 6

Summary and conclusions

In this book, I have set out to explore nasals and nasalization in Spanish and Portuguese from the standpoint of speech perception, including how the notion of speech perception is integrated into models of phonetics, phonology and language change. In Chapter 1, the polemical nature of integrating phonetic considerations into phonological theory was introduced. While some phonologists reject any relevance of phonetics to the phonological grammar (Buckley, 2000; Hale & Reiss, 1998; Kaye, 1989), it has been maintained throughout this volume, in accordance with other linguists (Browman & Goldstein, 1986, 1992; Hyman, 2001; Ohala, 1990 and others), that issues in phonetics and perception are essential to the study of phonology. As we saw in Chapter 2, perception plays a role in how sounds are classified and phonologized. Infants are able to perceive any speech contrast in any language until about 6 months of age, and as this ability diminishes over the next 6 months, they become more attuned to the distinctions that are important in their own language (Aslin, Pisoni & Jusczyk, 1983; Jusczyk, 1993, 1994; Kuhl, 1991, 1993; Werker & Tees, 1984a, 1984b). Models such as Motor Theory (Liberman & Mattingly, 1985), Work Recognition and Phonetic Structure Acquisition (WRAPSA) (Jusczyk, 1993), Native Language Magnet (Kuhl, 1987, 1991), and Direct Realist Theory (Best, 1995; Fowler, 1986) consider how speech perception develops during this period. Once these L1 categories have been formed, adult listeners tend to perceive new, unfamiliar phones either within their existing L1 repertoire or with reference to it (Best, 1990, 1991; Flege, 1995; Kuhl, 1987, 1991). As we saw, two of the leading models of adult/L2 perception are the Perceptual Assimilation Model (PAM) (Best, 1990, 1991) and the Speech Learning Model (SLM) (Flege, 1995).

In Chapter 3, the relevance of context and coarticulation to speech perception and, in turn, phonology and language change was discussed. We saw that a vector analysis (Fowler & Smith, 1986) of speech perception attempts to account for how listeners segment speech. On one hand, the fact that speech is continuous leads to coarticulation, especially at segment peripheries, which affects the acoustic signal. On the other hand, listeners seem to factor out the acoustic consequences of coarticulation under certain circumstances, thereby normalizing the speech signal. Vector analysis claims that this is because listeners segment speech along coarticulatory lines rather than temporally. Ohala refers to this as "correcting"

in order to recover the speaker's intended pronunciation. Listeners may not always correct the input, however, which could result in hypo- or hyper-correction, both of which potentially lead to language change (Ohala, 1981, 1986, 1990, 1993, 1996, 2012).

In spite of decades of disengagement between the fields of phonetics and phonology, several current models of phonological theory explicitly incorporate phonetic elements, including perception, into their paradigms. For example, Optimality Theoretic (Prince & Smolensky, 1993) constraints are grounded in phonetic concerns such as economy of articulation and perceptual ease. Indeed, OT has been proposed as a model for perception, that is, an OT perception grammar regarded as different and separate from the production grammar (Boersma, 1997, 1999 and others). Articulatory Phonology (Goldstein & Fowler, 2003), on the other hand, conceives the units of a phonological grammar as a set of gestural commands, which makes it particularly useful for modeling coarticulation.

The empirical analyses reported in Chapter 5 also contribute to the first research objective of this book. The results of Experiment 1, which examined nasal place perception by English and Spanish listeners, support models such as the NLM (Kuhl, 1987, 1991) and PAM (Best, 1990, 1991), in that Spanish listeners perceived /ŋ/, which is not phonemic in Spanish, better than even the English listeners in some contexts, likely because the velar nasal is a bad fit for any of the nasals that exist in Spanish. In this way, NLM would view the salience of /ŋ/ as stemming from it being so unlike the prototype of any of the existing nasals and therefore, impervious to the magnetic effects of any prototype, while PAM would view the velar nasal as a poor Category Goodness fit to any nasal phoneme, thus, increasing its perceptibility. Likewise, English listeners had difficulty discriminating the palatal nasal, which is not phonemic in English, from /n/, because these are articulatorily and acoustically similar, thus, English listeners parsed them into the same category.

The results of Experiment 2, along with other empirical results, show that rigorous phonetic analyses are imperative to the study of a language's (or language variety's) phonological grammar. For example, we saw in Chapter 4 that most phonological accounts of nasalization in Cuban Spanish place this variety in the middle or toward the farther extreme on a continuum of nasal lenition (Guitart, 1973, 1976; Piñeros, 2006; Terrell, 1975). Piñeros depicts the continuum of nasal lenition as gradual: Vn]σ > Ṽn]σ > Ṽŋ]σ > Ṽɣ̃]σ > Ṽɴ]σ > Ṽ]σ. Here, the first step is no lenition at all, then nasalization of the preceding vowel with the features of the nasal left intact, then nasalization and velarization, followed by nasalization plus loss of manner and/or place features, and finally total nasal absorption. Researchers (Guitart, 1973, 1976; Piñeros, 2006; Terrell, 1975) have described Cuban Spanish as being anywhere between the third and last stages of lenition,

depending on sociolinguistic and other factors; however, this is not borne out by instrumental research (Kochetov & Colantoni, 2011).

Recall that Baković (2001) argues that what has traditionally been dubbed velarization is actually debuccalization, the result of which is a placeless nasal. The crux of his OT analysis of coda nasals lies in the argument that this placeless nasal does not violate the highly ranked Coda Condition, because there are no place features to be licensed. However, as we have seen, electropalatographic analyses of Cuban Spanish demonstrate categorical assimilation of nasals before coronal consonants and vacillation between [ŋ] and [ũ̞] elsewhere (Kochetov & Colantoni, 2011). Therefore, velarization, at least in Cuban and other varieties of Spanish (Ramsammy, 2013), does entail a coda nasal with place features at least some of the time. Likewise, Piñeros' (2006) OT analysis of nasal lenition as an alignment issue hinges on the coda nasal gaining access to the left edge of the syllable via vowel nasalization. However, the fact that Cuban Spanish speakers performed similarly to Peninsular Spanish speakers in Experiment 2 alludes to the possibility that nasalization is not as robust in this dialect as previously thought. Hence, accurate phonological analyses depend on accurate, current phonetic investigation in order to adequately account for the data.

Another research objective of this volume has been to investigate the roles of language experience and context in the perception of coarticulated sounds, and how these influence nasals and nasalization in Spanish and Portuguese. Results of the studies presented in Chapter 5 support that native language experience indeed plays a role in the way in which a listener compensates for the acoustic effects of coarticulation. These results are in accord with various models of perception and coarticulation, particularly vector analysis (Goldstein & Fowler, 1986) and Ohala's (1981, 1986, 1990b, 1993, 2012) model of hypocorrection and language change. Given the fact that Cuban speakers did not perceptually factor out the effects of nasalization and that there is instrumental evidence of changes in coda nasal place and manner of articulation (Kochetov & Colantoni, 2011), one could predict possible vowel changes in nasal contexts, if Cuban listeners commit hypocorrection, which could then lead to language change.

In addition to the research goals outlined earlier, this book has also served to highlight Spanish and Portuguese as a particularly auspicious language pairing to study phonological aspects of synchronic and diachronic variation. Given the intimate relationship between Spanish and Portuguese as well as the array of dialectal variation in each, crosslinguistic inquiry into these languages can be especially revelatory. Future directions in the study of Spanish and Portuguese phonology might consider the production and perception of nasals and nasalization in European versus Brazilian dialects of Portuguese as well as in situations of contact between Spanish, Portuguese and/or Galician.

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of articulation and coronal is the least (Prince & Smolensky, 1993; DeLacy, 2002). Like most markedness constraints, Place Hierarchy is grounded in phonetic principles whereby dorsal is the least economic place of articulation and coronal is the most economic. This follows from the observation that alveolar (i.e., coronal) articulations involve the least amount of effort given the flexibility of the tongue apex and blade (i.e., corona). In contrast, the tongue body is relatively immobile, making dorsal articulations the most costly. The intermediate labial involves movements of the lips and mandible, making it less economic than coronal, but not as costly as dorsal.

In Piñeros' (2006) account, neutralization is the result of Place Hierarchy's ranking relative to two other constraints, Ident (place) and Have Place. As the tableau in (9) illustrates, the relatively low ranking of Ident (place) leaves coda nasals susceptible to neutralization due to the higher ranking Place Hierarchy, while the preferential ranking of Have Place ensures that final nasals will be produced with some place of articulation.

(9)

/al.bum/	Have Place	Place Hierarchy			Ident (place)
		*D	*L	*C	
a. [al.βum]			*!		
☞ b. [al.βun]				*	*
c. [al.βuɲ]		*!		*	*
d. [al.βuŋ]		*!			*
e. [al.βuɴ]	*!			*	*

While candidate a is the most faithful, it fails because faithfulness is outranked by Place Hierarchy. Candidate e fails, due to the precedence of Have Place. Candidates c and d are neither faithful, nor the best alternatives in terms of place hierarchy, which leaves candidate b as the optimal output.

A common thread among these and other accounts of nasal neutralization is the notion that coda (as opposed to onset) nasals dispreferentially lose or fail to faithfully realize place features. This is consistent with the fact that place features are difficult to perceive in coda position, especially in the case of nasal consonants. Piñeros (2011) accounts for crosslinguistic nasal phoneme inventories by proposing an interaction of the abovementioned Place Hierarchy constraints on articulation and another hierarchy based on perception, both presumed to be universal. Of note is the tendency for languages to include /m/ and /n/ in their phonemic inventories and then either /ŋ/ or /ɲ/, if they have a third. Piñeros examined the inventories of 451 languages and found that of the 138 languages that

According to the Sonority Dispersion Principle (Clements, 1988, 1990), the most harmonious codas are those in which the sonority falls uniformly and shallowly out of the peak. Languages satisfy these and other conditions on the coda in different ways and to varying degrees. Indeed crosslinguistically some of the most common phonological processes (synchronically and diachronically) are those that result in either open syllables or more sonorous codas. Sonority is addressed in OT by constraints (or sub-constraints) based on manner features. Given the importance of sonority in determining the types of segments allowed in coda, McCarthy (2002) and others have argued that *Coda can be broken down into sub-constraints according to manner of articulation (*Coda(obstruent), *Coda(nasal), *Coda(liquid), etc.). Goodin-Mayeda (2015) argues that the interaction of these constraints on codas with faithfulness constraints is responsible for what in other accounts might seem like unrelated phenomena in BP, such as epenthesis after obstruents (as in *ad*[i]*vogado* 'lawyer'), gliding of /l/ (*fa*[w]*ta* 'lack'), and debuccalization of /ɾ/ (*ca*[x]*ta* or *ca*[h]*ta* 'letter'). In the case of nasalization, the relevant constraints are *Coda(Nasal), Max-IO, Dep-IO and Uniformity. The tableau in (32) illustrates this.

(32) *Coda(nasal): No nasals in coda position.
Max-IO: All elements in the input must have a corresponding element in the output.
Dep-IO: All elements in the output must have a corresponding element in the input.
Uniformity: Input and output elements must stand in a one-to-one correspondence relationship with each other (McCarthy & Prince 1995).

/kaNsa/	*Coda(Nas)	Max-IO	Dep-IO	Uniform
a. [kan.sɐ]	*!			
b. [kɐ̃n.sɐ]	*!			*
c. [ka.sɐ]		*!		*
d. [ka.ni.sɐ]			*!	
☞ e. [kɐ̃.sɐ]				*

Because coda nasals still have a correspondent in the output, by "leaving" their nasality with the preceding vowel rather than eliding entirely, they do not violate Max-IO. However, the fact that the feature [nasal] is expressed on a different segment violates the constraint Uniformity. Thus, candidates a and b fail, because a nasal appears in the coda, while c fails because the nasal segment is deleted entirely. Candidate d fails in this case, because it violates the higher-ranked Dep-IO

Nasality, whether part of a consonant or vowel, has certain phonetic and phonological characteristics that lead to outcomes seen time and again in languages with and without common ancestries. Spanish and Portuguese constitute a particularly fruitful language pairing for studying phonological aspects of synchronic and diachronic variation, given their intimate relationship as well as the array of dialectal variation in each. This research monograph offers a comprehensive exploration of nasals and nasalization in Spanish and Portuguese with a special focus on the role of perception in order to provide insight into how perception informs models of phonetics, phonology and language change. Of interest to researchers and advanced students alike, this volume integrates phonetic and phonological models of speech perception and production, and discusses these with regards to original empirical research on the perception of nasal place features and vowel nasalization by listeners of Peninsular Spanish, Cuban Spanish and Brazilian Portuguese.

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